Advanced Macroeconomics II Notes

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Lecture 1

Difference Equation

1.1 Difference operators

$$\text{1-st order difference:} \qquad \Delta y_t = y_t - y_{t-1}$$

$$\text{2-nd order difference:} \qquad \Delta^2 y_t = \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = \Delta y_t - \Delta y_{t-1} = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2})$$
$$= y_t - 2y_{t-1} + y_{t-2}.$$

1.2 Period

Period: an interval of time $\Rightarrow y_t$ is the y for t^{th} interval of time, not for a point of time.
* t is a period concept, not a point-in-time concept.

$$\Delta y_t = y_t - y_{t-1}$$

is the change from $(t-1)^{\text{th}}$ time interval to t^{th} interval, not a derivative at a point of time t.
* Δy_t is a change, not a derivative or growth rate at time t.

$$\text{Derivative} = \dot{y}_t = \frac{\Delta y_t}{y_{t-1}} = \frac{y_t - y_{t-1}}{y_{t-1}} = \frac{y_t}{y_{t-1}} - 1 \approx \log\left(\frac{y_t}{y_{t-1}}\right)$$

$$\text{when } \frac{y_t - y_{t-1}}{y_{t-1}} \text{ is small, } \log\left(1 + \frac{y_t}{y_{t-1}} - 1\right) \approx \frac{y_t}{y_{t-1}} - 1 = \frac{y_t - y_{t-1}}{y_{t-1}}.$$

1.3 Solve a first-order difference equation

Solve a first-order difference equation $\Longleftrightarrow$ rewrite y_t as a function of t.
Example:

$$y_t + a y_{t-1} = C.$$

1.3.1 Brute-force iterative method

$$y_t = -a y_{t-1} + C$$
$$y_{t-1} = -a y_{t-2} + C$$

Hence
$$y_t = a^2 y_{t-2} - aC + C, \qquad y_{t-2} = -ay_{t-3} + C.$$
Continuing,
$$y_t = -a^3 y_{t-3} + a^2 C - aC + C - \cdots$$
and in general,
$$y_t = (-a)^t y_0 + C \sum_{j=0}^{t-1} (-a)^j.$$
If $a \neq -1$,
$$\sum_{j=0}^{t-1} (-a)^j = \frac{1-(-a)^t}{1+a}.$$
So
$$y_t = (-a)^t \left(y_0 - \frac{C}{1+a} \right) + \frac{C}{1+a}, \qquad a \neq -1.$$
Equivalently,
$$y_t = A(-a)^t + \frac{C}{1+a}, \qquad a \neq -1.$$
If $a = -1$,
$$\sum_{j=0}^{t-1} (-a)^j = \sum_{j=0}^{t-1} 1 = t \qquad \Rightarrow \qquad y_t = y_0 + Ct.$$
Thus
$$y_t = \begin{cases} A(-a)^t + \dfrac{C}{1+a}, & a \neq -1, \\ Ct + y_0, & a = -1, \end{cases} \qquad \text{where } A = y_0 - \frac{C}{1+a}.$$

1.3.2 General method

Since the system is *linear*, we can separate:

1) the part driven by constant C,
2) the part comes from the dynamics.

Why $y_t = y_t^p + y_t^c$ is valid?

Define linear operator
$$L(\cdot) = (\cdot)_t + a(\cdot)_{t-1}.$$
Then
$$y_t + ay_{t-1} = C \qquad \Rightarrow \qquad L(y) = C.$$
If y^p is any particular solution such that
$$L(y^p) = C,$$
and if y^c is any solution of the homogeneous equation
$$L(y^c) = 0 \qquad \text{(complementary solution)},$$

then by linearity,

$$L(y^p + y^c) = L(y^p) + L(y^c) = C + 0 = C.$$

Therefore,

$$y^p + y^c$$

solves the equation

$$y_t + ay_{t-1} = C.$$

$$\boxed{\text{All solutions = one particular solution + all homogeneous solutions.}}$$

Now we solve the difference equation with general method.

Complementary solution

Since exponential sequences are eigenfunctions of the lag operator, try

$$y_t = Ab^t.$$

Then for the homogeneous equation,

$$Ab^t + aAb^{t-1} = 0 \quad \Rightarrow \quad b + a = 0 \quad \Rightarrow \quad b = -a.$$

Hence

$$y_t^c = A(-a)^t.$$

Particular solution

Since the forcing term is a constant C, we guess

$$y_t^p = k.$$

Then

$$k + ak = C \quad \Rightarrow \quad k = \frac{C}{1+a}, \qquad a \neq -1.$$

If $a = -1$,

$$y_t - y_{t-1} = C \quad \Rightarrow \quad y_t^p = Ct.$$

Therefore,

$$y_t = y_t^c + y_t^p = \begin{cases} A(-a)^t + \dfrac{C}{1+a}, & a \neq -1, \\ A + Ct, & a = -1. \end{cases}$$

Using the initial condition:
If $a \neq -1$,

$$y_0 = A(-a)^0 + \frac{C}{1+a} = A + \frac{C}{1+a} \quad \Rightarrow \quad A = y_0 - \frac{C}{1+a}.$$

If $a = -1$,

$$y_0 = A \quad \Rightarrow \quad A = y_0.$$

1.4 Dynamic stability of equilibrium

Consider

$$y_t + a y_{t-1} = C.$$

The steady state is defined by

$$y_t = y_{t-1} = \bar{y},$$

so

$$\bar{y} + a\bar{y} = C \quad \Rightarrow \quad \bar{y} = \frac{C}{1+a} \qquad (a \neq -1).$$

Subtract steady state from the original equation:

$$y_t - \bar{y} + a(y_{t-1} - \bar{y}) = C - \bar{y} - a\bar{y}.$$

Define deviation from steady state as

$$\tilde{y}_t = y_t - \bar{y}.$$

Then

$$\tilde{y}_t + a\tilde{y}_{t-1} = 0 \quad \Rightarrow \quad \tilde{y}_t = (-a)\tilde{y}_{t-1}.$$

Let

$$b = -a.$$

Then

$$\tilde{y}_t = b\tilde{y}_{t-1} \quad \Rightarrow \quad \tilde{y}_t = b^t \tilde{y}_0.$$

* $b = -a$ controls convergence.

$$\begin{cases} |b| < 1 \Rightarrow b^t \to 0,\ \tilde{y}_t \to 0,\ y_t \to \bar{y} & \text{(convergence)}, \\ |b| > 1 \Rightarrow b^t \to \infty,\ \tilde{y}_t \to \infty & \text{(divergence)}. \end{cases}$$

1.5 Sign of b and oscillation / monotonicity

Sign of b also controls oscillation / monotonicity.

$$\begin{cases} b > 0: & b^t \text{ keeps the same sign for all } t \Rightarrow \tilde{y}_t \text{ keeps the same sign for all } t, \\ & \text{path is non-oscillatory (monotone, towards / away from } \bar{y}\text{)}. \\ b < 0: & b^t \text{ alternates sign } \Rightarrow \tilde{y}_t \text{ flips sign each period} \\ & \Rightarrow \text{oscillate around } \bar{y}. \end{cases}$$

Further:

$$\begin{cases} b > 1, & y_t \text{ monotonically diverges from } \bar{y}, \\ b = 1, & y_t \text{ keeps the same distance away from } \bar{y}, \\ 0 < b < 1, & y_t \text{ monotonically converge to } \bar{y}, \\ b = 0, & y_t \text{ stays at steady state } \bar{y} \text{ always}, \\ -1 < b < 0, & y_t \text{ oscillate around and converge to } \bar{y}, \\ b = -1, & y_t \text{ oscillate around } \bar{y} \text{ and keeps same absolute distance}, \\ b < -1, & y_t \text{ oscillate around and diverge from } \bar{y}. \end{cases}$$

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1.6 The role of A

Since the general solution can be written as

$$y_t = \bar{y} + Ab^t \quad \Rightarrow \quad A = y_0 - \bar{y}$$

(initial deviation).

Scale effect:

$$|A| \uparrow \Rightarrow \text{ start farther from equilibrium} \Rightarrow \text{bigger movements along path.}$$

Mirror effect:

$$A > 0: \text{ start above } \bar{y}; \qquad A < 0: \text{ start below } \bar{y}.$$

1.7 Example: Inventory model

Example: Inventory model.

$$Q_{d,t} = \alpha - \beta p_t \qquad \text{demand}$$

$$Q_{s,t} = -\gamma + \delta p_t \qquad \text{supply}$$

$$p_{t+1} = p_t - \sigma \left(Q_{s,t} - Q_{d,t}\right)$$

$$\alpha, \beta, \gamma, \delta, \sigma > 0 \qquad \text{solve for } p_t\text{'s path}$$

Then

$$Q_{s,t} - Q_{d,t} = -(\alpha + \gamma) + (\beta + \delta) p_t.$$

Plug in:

$$p_{t+1} = \left[1 - \sigma(\beta + \delta)\right] p_t + \sigma(\alpha + \gamma).$$

At steady state,

$$p_{t+1} = p_t = \bar{p} \quad \Rightarrow \quad \bar{p} = \frac{\alpha + \gamma}{\beta + \delta}.$$

Thus, the deviation dynamics is

$$p_t - \bar{p} = \left[1 - \sigma(\beta + \delta)\right](p_{t-1} - \bar{p}).$$

Stability condition tells us:

$$\left|1 - \sigma(\beta + \delta)\right| < 1 \quad \Rightarrow \quad 0 < \sigma(\beta + \delta) < 2,$$

system converges.

Oscillation condition tells us:

$$1 - \sigma(\beta + \delta) < 0 \quad \Rightarrow \quad \sigma(\beta + \delta) > 1$$

creates oscillation.

* Economic intuition: if price adjustment is too aggressive $\sigma \uparrow\uparrow$, creates oscillation and divergence.

If supply or demand elasticity in terms of price too high

$$\beta \uparrow\uparrow / \delta \uparrow\uparrow$$

$\Rightarrow$ creates oscillation or even divergence.

1.8 Higher-order difference equations

2-nd order linear difference equation with constant coefficients & constant term:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = C.$$

1.8.1 Particular solution y^p: the intertemporal equilibrium level

Try a constant:

$$y^p = k \quad \Rightarrow \quad k(1 + a_1 + a_2) = C \quad \Rightarrow \quad k = \frac{C}{1 + a_1 + a_2}, \qquad 1 + a_1 + a_2 \neq 0.$$

Try a first-degree polynomial:

$$y^p = kt.$$

Then

$$k(t+2) + a_1 k(t+1) + a_2 kt = C$$
$$\Rightarrow \quad k\big[(1 + a_1 + a_2)t + (2 + a_1)\big] = C.$$

Impose

$$1 + a_1 + a_2 = 0$$

then

$$k = \frac{C}{2 + a_1}, \qquad 2 + a_1 \neq 0.$$

Try a second-degree polynomial:

$$y^p = kt^2.$$

Then

$$k(t+2)^2 + a_1 k(t+1)^2 + a_2 kt^2 = C.$$

The notes impose

$$\begin{cases} 1 + a_1 + a_2 = 0, \\ 2 + a_1 = 0, \end{cases} \qquad \Rightarrow \qquad \begin{cases} a_1 = -2, \\ a_2 = 1. \end{cases}$$

Substituting:

$$k(t^2 + 4t + 4) - 2k(t^2 + 2t + 1) + kt^2 = C$$
$$\Rightarrow \quad 2k = C \quad \Rightarrow \quad k = \frac{C}{2}.$$

Therefore

$$y^p = \begin{cases} \dfrac{C}{1 + a_1 + a_2}, & 1 + a_1 + a_2 \neq 0, \\ \dfrac{C}{2 + a_1}\, t, & 1 + a_1 + a_2 = 0,\ a_1 \neq -2, \\ \dfrac{C}{2}\, t^2, & a_1 = -2,\ a_2 = 1. \end{cases}$$

1.8.2 Complementary solution y^c: deviation from the equilibrium in each period

Look at the homogeneous equation:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = 0.$$

Try

$$y_t = Ab^t.$$

Then

$$b^2 + a_1 b + a_2 = 0.$$

Two distinct real roots

If

$$a_1^2 - 4a_2 > 0,$$

then

$$y^c = A_1 b_1^t + A_2 b_2^t$$

where

$$b_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \qquad b_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}.$$

Repeated real root

If

$$a_1^2 - 4a_2 = 0,$$

then

$$b = b_1 = b_2 = -\frac{a_1}{2},$$

and

$$y^c = A_3 b^t + A_4 t b^t.$$

Complex roots

If

$$a_1^2 - 4a_2 < 0,$$

then

$$b_{1,2} = h \pm vi, \qquad h = -\frac{a_1}{2}, \qquad v = \frac{\sqrt{4a_2 - a_1^2}}{2}.$$

So

$$y^c = A_5 (h + vi)^t + A_6 (h - vi)^t.$$

Using De Moivre's Theorem:

$$(h \pm vi)^t = R^t \big(\cos(\theta t) \pm i \sin(\theta t)\big)$$

where

$$R = \sqrt{h^2 + v^2} = \sqrt{a_2}, \qquad \cos\theta = \frac{h}{R} = -\frac{a_1}{2\sqrt{a_2}}, \qquad \sin\theta = \frac{v}{R} = \sqrt{1 - \frac{a_1^2}{4a_2}}.$$

Hence

$$y^c = R^t\left(A_5 \cos\theta t + A_6 \sin\theta t\right),$$

where

$$R = \sqrt{a_2}, \qquad \theta \text{ is defined as } \begin{cases} \cos\theta = -\dfrac{a_1}{2\sqrt{a_2}}, \\ \sin\theta = \sqrt{1 - \dfrac{a_1^2}{4a_2}}. \end{cases}$$

1.9 Convergence under higher-order difference equation

1° Distinct root: b_{dom} is the root with highest absolute value. Convergence requires

$$|b_{\text{dom}}| < 1.$$

2° Repeated root:

$$|b| < 1 \Rightarrow \text{converges.}$$

3° Complex root:

$$R > 1: \text{ explosive oscillations}, \qquad R < 1: \text{ damped oscillations.}$$

1.10 Simultaneous difference equation (state-space form)

Let

$$x_t = y_{t+1}.$$

Then

$$\begin{cases} x_{t+1} + a_1 x_t + a_2 y_t = C, \\ x_t = y_{t+1}. \end{cases}$$

Therefore,

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Generalize:

$$X_{t+1} = AX_t + B$$

where

$$X_t = (x_{1t}, x_{2t}, \dots, x_{nt})$$

is state space.

A is a constant matrix with a_{ij} as elements $(i, j = 1, \dots, n)$,

B is an array with $B_i,\ i = 1, \dots, n$ as elements.

In this example,

$$X_{t+1} = \begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix}, \qquad A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Everything about dynamics and stability is controlled by eigenvalues of A.

Ignore constant term B to study deviation from steady state:

$$X_{t+1} = AX_t.$$

Try solution:

$$X_t = vb^t.$$

Plug in:

$$vb^{t+1} = A(vb^t) \quad \Rightarrow \quad bv = Av.$$

Here b is an eigenvalue of A, with eigenvector v.
A nonzero v exists iff

$$\det(A - bI) = 0.$$

1.11 Characteristic equation

Characteristic equation:

$$p(b) = |A - bI| = 0.$$

For a 2×2 matrix:

$$p(b) = b^2 - Tb + D = 0,$$

where

$$T = \mathrm{trace}(A), \qquad D = \det(A).$$

Also,

$$T = b_1 + b_2, \qquad D = b_1 b_2,$$

so

$$p(b) = b^2 - (b_1 + b_2)b + b_1 b_2 = (b - b_1)(b - b_2).$$

Eigenvalues are the solution of the characteristic equation: b_1, b_2.
Back to the example, since

$$A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix},$$

we have

$$T = \mathrm{tr}(A) = -a_1, \qquad D = \det(A) = a_2.$$

Thus characteristic equation:

$$b^2 + a_1 b + a_2 = 0.$$

Eigenvalues:

$$\frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}, \qquad \text{if } a_1^2 > 4a_2.$$

1.12 Stability triangle on the (T, D) plane

1°

$$\Delta = T^2 - 4D.$$

$$\begin{cases} \Delta < 0: & \text{complex roots } \Rightarrow \text{above red curve,} \\ \Delta > 0: & \text{real roots } \Rightarrow \text{below red curve.} \end{cases}$$

2°

$$p(1) = 0, \quad p(-1) = 0$$

are the black curves.

$$p(1) = 1 - T + D, \qquad \text{right to } p(1) = 0 \Rightarrow p(1) < 0.$$

$$p(-1) = 1 + T + D, \qquad \text{left to } p(-1) = 0 \Rightarrow p(-1) < 0.$$

3°

$$D = 1 \text{ and } |b| = 1 \text{ for complex roots}$$

is the green curve.

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Why D controls $|b|$ in complex case?
If roots are complex conjugates

$$b_{1,2} = re^{\pm i\theta} \quad \Rightarrow \quad D = b_1 b_2 = r^2 \quad \Rightarrow \quad r = \sqrt{D}.$$

Stable ($r < 1$) implies

$$D < 1.$$

Thus, above green curve $|b| > 1$, below green curve $|b| < 1$.
* Conditions for two stable roots (both inside unit circle)

$$D < 1 \quad \text{and} \quad p(1) > 0 \quad \text{and} \quad p(-1) > 0$$
$$\Rightarrow \quad D < 1, \quad 1 - T + D > 0, \quad 1 + T + D > 0$$
$$\Rightarrow \quad D < 1 \quad \text{and} \quad |T| < 1 + D.$$

$$\begin{cases} \text{complex roots with modulus } < 1, & \text{if } \Delta < 0 \ (T^2 < 4D), \\ \text{real roots both between } -1 \text{ and } 1, & \text{if } \Delta > 0 \ (T^2 > 4D). \end{cases}$$

* Conditions for one stable root and one unstable root.

$$p(1) < 0 \ \& \ p(-1) > 0 \qquad \text{or} \qquad p(1) > 0 \ \& \ p(-1) < 0.$$

That is,

$$\begin{cases} 1 - T + D < 0, \\ 1 + T + D > 0, \end{cases} \qquad \text{or} \qquad \begin{cases} 1 - T + D > 0, \\ 1 + T + D < 0. \end{cases}$$

Thus,

$$|T| > |1 + D|.$$

$$p(1)\ \&\ p(-1) \text{ has opposite signs } \Rightarrow \text{There is a root in the interval } (-1, 1).$$

Since

$$p(1) < 0,\ p(b) \longrightarrow +\infty \text{ as } b \longrightarrow +\infty$$

or

$$p(-1) < 0,\ p(b) \longrightarrow +\infty \text{ as } b \longrightarrow -\infty,$$

there is another root outside the interval $[-1, 1]$.

* Conditions for explosive and unstable roots (both roots outside unit circle or complex with modulus > 1)

Four cases:

1° $p(1) < 0,\ p(-1) < 0,\ D < 0,\ |T| > 1$

2° $p(1) > 0,\ p(-1) > 0,\ D > 1,\ T < -2$

3° $\Delta < 0$ and $D > 1$

4° $p(1) > 0,\ p(-1) > 0,\ D > 1,\ T > 2$

1.13 Region-by-region summary

Region	Conditions	Roots
1	$p(1) < 0\ \&\ p(-1) > 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1, 1), \\ 1 \text{ unstable root in } (1, +\infty) \end{cases}$
2	$p(1) < 0\ \&\ p(-1) < 0$	explosive: $\begin{cases} 1 \text{ unstable root in } (1, +\infty), \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
3	$p(1) > 0\ \&\ p(-1) < 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1, 1), \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
4	$p(1) > 0\ \&\ p(-1) > 0\ \&$ $\Delta > 0\ \&\ D > 1\ (T < -2)$	explosive: 2 real unstable roots in $(-\infty, -1)$
5	$p(1) > 0\ \&\ p(-1) > 0\ \&$ $\Delta < 0\ \&\ D > 1$	explosive: complex conjugates with modulus $r = \sqrt{D} > 1,$ oscillates & diverges
6	$p(1) > 0\ \&\ p(-1) > 0\ \&$ $\Delta < 0\ \&\ D < 1\ (-2 < T < 2)$	stable: complex conjugates with modulus $r = \sqrt{D} < 1,$ oscillates & converges
7	$\Delta > 0\ \&\ D < 1\ \&\ p(1) > 0\ \&\ p(-1) > 0$	stable: 2 real roots $\lvert b_1\rvert < 1,\ \lvert b_2\rvert < 1$
$7a$	$D > 0\ \&\ T < 0$	$-1 < b_2 \leq b_1 < 0$ (same sign, and $T = b_1 + b_2$ negative) decay but can flip sign
$7b$	$D < 0$	$-1 < b_2 < 0 < b_1 < 1$ (opposite sign, $D = b_1 b_2$)decay but can flip sign
$7c$	$D > 0\ \&\ T > 0$	$0 < b_2 \leq b_1 < 1$ (same sign, and $T = b_1 + b_2$ positive) monotone converge
8	$\Delta > 0,\ D > 1,\ T > 2$ $p(1) > 0,\ p(-1) > 0$	explosive: 2 real roots $b_1 > 1,\ b_2 > 1$

1.14 Blanchard and Kahn (BK) conditions

Predetermined variable: initial value is given, states like k, h, b (i.e. capital, hour, bond).
Jump variable: initial value not given, can jump to satisfy equilibrium, like c, p.

$$\text{Stable roots} = \#\text{ predetermined variables} \Rightarrow \text{Saddle path (Determinacy)}$$

$$\text{Stable roots} < \#\text{ predetermined variables} \Rightarrow \text{Source (Explosive, no convergent equilibrium)}$$

$$\text{Stable roots} > \#\text{ predetermined variables} \Rightarrow \text{Sink (Indeterminacy)}$$

1.15 Why?

Linearize around steady state, we write deviations as a first-order system:

$$X_{t+1} = AX_t.$$

If A has eigenpairs (b_i, v_i), the general solution can be written as sum of modes:

$$X_t = \sum_i \alpha_i b_i^t v_i.$$

If

$$|b_i| < 1,$$

mode dies out $\Rightarrow$ stable.
If

$$|b_i| > 1,$$

mode blows up $\Rightarrow$ unstable / explosive.

$$\text{A convergent equilibrium path must impose: } \alpha_i = 0 \quad \text{for every } |b_i| > 1,$$

which is a set of restrictions, one restriction per unstable root.
Now partition variables into

$$\{\text{predetermined states } s_t \text{ (given } s_0\text{) with dimension } n_s,$$

$$\text{jump variables } j_t \text{ (free } j_0\text{) with dimension } n_j\}.$$

$$X_t = (s_t, j_t).$$

Initial conditions pin down s_0, but can choose j_0.
Let

$$n_u = \#\text{ unstable roots}, \qquad n_\ell = \#\text{ stable roots}.$$

$$n_u + n_\ell = n_s + n_j.$$

To satisfy n_u restrictions, we have n_j free choices to satisfy them. Thus

$$n_u = n_j \qquad \Longleftrightarrow \qquad n_\ell = n_s.$$

1.15.1 Determinacy

$$n_\ell = n_s \quad (n_u = n_j):$$

jump variables jump to land on saddle path.

1.15.2 No convergent equilibrium

$$n_\ell < n_s \quad (n_u > n_j):$$

too many unstable modes to eliminate, but too few jump variables to adjust.

1.15.3 Indeterminacy

$$n_\ell > n_s \quad (n_u < n_j):$$

more jump freedom needed to kill unstable modes (multiple equilibria)

Lecture 2

Log Linearization

Business cycle is fluctuation / deviation around trend. Log linearization: take logs to linearize the system and then take deviation from the trend / steady state.

* Can not work with discontinuous functions, eg: zero lower bound, default, work choice,etc.

2.1 Taylor expansions

Taylor expansion of order k of function f around a is to derive the best approximation of f around a by a polynomial of order k.

Suppose f is $n+1$ times differentiable in an interval around a

$$x \in (a-\varepsilon,\, a+\varepsilon),$$

then

$$f(x) = f(a) + \sum_{i=1}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i + \underbrace{o\big((x-a)^n\big)}_{\text{Peano remainder terms.}}.$$

eg:

$$\text{1st order:} \qquad f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a)^1 = f(a) + f'(a)(x-a)$$

$$\text{2nd order:} \qquad f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a)^1 + \frac{f''(a)}{2!}(x-a)^2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$$

2.2 Taylor expansion on two dimensions

Suppose f is second-order continuous (C^2) in a circle around (x_0, y_0) that contains (x_0+h, y_0+k).

For a small move (h,k) around (x_0,y_0), how to approximate changes in f.

$$f(x_0+h, y_0+k) = f(x_0,y_0) + \Big[f_1(x_0,y_0)h + f_2(x_0,y_0)k\Big] + \frac{1}{2}\Big[f_{11}(x,y)h^2 + 2f_{12}(x,y)hk + f_{22}(x,y)k^2\Big]$$

where

$$(x,y) = (x_0+ch, y_0+ck) \qquad \text{for some } c \in (0,1).$$

A vector format:

$$f(z_0 + \Delta) = f(z_0) + \nabla f(z_0)^T \Delta + \frac{1}{2}\Delta^T H_f(z_0 + c\Delta)\Delta$$

$$\text{where } z_0 = (x_0, y_0), \qquad \Delta = (h, k)$$

2.3 Log linearization

$$\text{percentage deviation:} \qquad \hat{x}_t = \log X_t - \underbrace{\log \bar{X}}_{Steadystate}$$

Since $Y_t \to 0$ when Y_t is sufficiently small, we have

$$e^{Y_t} = e^0 + e^0(Y_t - 0) = 1 + Y_t \qquad \Rightarrow \qquad Y_t \approx \log(1 + Y_t).$$

$$\hat{x}_t = \log \frac{X_t}{\bar{X}} = \log\left(1 + \frac{X_t - \bar{X}}{\bar{X}}\right) \approx \frac{X_t - \bar{X}}{\bar{X}}.$$

$\hat{x}_t \times 100\%$ is approximately the percentage deviation w.r.t. S.S.

$$X_t = \bar{X}\left(\frac{X_t}{\bar{X}}\right) = \bar{X}e^{\log(X_t/\bar{X})} = \bar{X}e^{\hat{x}_t}$$

Taking first-order Taylor approximation:

$$\bar{X}e^{\hat{x}_t} \approx \bar{X}e^0 + \bar{X}e^0(\hat{x}_t - 0) = \bar{X} + \bar{X}\hat{x}_t = (1 + \hat{x}_t)\bar{X}$$

$$\Rightarrow \qquad X_t = \overbrace{\bar{X}}^{\text{Steady state}} + \overbrace{\bar{X}\hat{x}_t}^{\text{deviation from steady state}}$$

2.4 Examples

2.4.1 Linear equation

$$Y_t = aX_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t) \qquad \Rightarrow \qquad Y_t \approx a\bar{X}(1 + \hat{x}_t) = a\bar{X} + a\bar{X}\hat{x}_t$$

Since

$$Y_t = aX_t,\ \forall t \qquad \Rightarrow \qquad \bar{Y} = a\bar{X},$$

we get

$$Y_t \approx \bar{Y} + \bar{Y}\hat{x}_t \qquad \Rightarrow \qquad Y_t \approx \bar{Y} + \bar{Y}\hat{y}_t$$

hence

$$\hat{x}_t = \hat{y}_t.$$

2.4.2 Product

$$Z_t = X_tY_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t), \qquad Y_t \approx \bar{Y}(1 + \hat{y}_t)$$

$$\Rightarrow \quad Z_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t)(1 + \hat{y}_t) = \bar{X}\bar{Y}\Big(1 + \hat{x}_t + \hat{y}_t + \underbrace{\hat{x}_t\hat{y}_t}_{\approx 0}\Big)$$

So

$$Z_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t + \hat{y}_t).$$

Since

$$Z_t = X_tY_t,\ \forall t \qquad \Rightarrow \qquad \bar{Z} = \bar{X}\bar{Y},$$

we get

$$Z_t \approx \bar{Z} + \bar{Z}(\hat{x}_t + \hat{y}_t) \qquad \Rightarrow \qquad Z_t \approx \bar{Z} + \bar{Z}\hat{z}_t$$

hence

$$\hat{z}_t = \hat{x}_t + \hat{y}_t.$$

2.4.3 Constant product

Log linearize equation

$$X_t Y_t = C \qquad \Rightarrow \qquad \bar{X}\bar{Y} = C.$$

By (2),

$$X_t Y_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t + \hat{y}_t) \qquad \Rightarrow \qquad C \approx C(1 + \hat{x}_t + \hat{y}_t)$$

so

$$\hat{x}_t + \hat{y}_t = 0.$$

2.4.4 Ratio

$$\frac{X_t}{Y_t} = C \qquad \Rightarrow \qquad \frac{\bar{X}}{\bar{Y}} = C.$$

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}(1 + \hat{x}_t)}{\bar{Y}(1 + \hat{y}_t)}$$

Use

$$(1 + \hat{y}_t)(1 - \hat{y}_t) = 1 + \hat{y}_t - \hat{y}_t - \hat{y}_t\hat{y}_t \qquad \Rightarrow \qquad \frac{1}{1 + \hat{y}_t} \approx (1 - \hat{y}_t).$$

Therefore

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}}(1 + \hat{x}_t)(1 - \hat{y}_t) \approx \frac{\bar{X}}{\bar{Y}}\left(1 + \hat{x}_t - \hat{y}_t - \hat{x}_t\hat{y}_t\right)$$

So

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}}{\bar{Y}}(1 + \hat{x}_t - \hat{y}_t).$$

Since

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} = C,$$

we have

$$C = C(1 + \hat{x}_t - \hat{y}_t) \qquad \Rightarrow \qquad \hat{x}_t - \hat{y}_t = 0.$$

2.4.5 Sum

$$X_t + Y_t = C \qquad \Rightarrow \qquad \bar{X} + \bar{Y} = C.$$

$$X_t + Y_t \approx \bar{X}(1 + \hat{x}_t) + \bar{Y}(1 + \hat{y}_t) = \bar{X} + \bar{Y} + \bar{X}\hat{x}_t + \bar{Y}\hat{y}_t$$

hence

$$\bar{X}\hat{x}_t + \bar{Y}\hat{y}_t = 0.$$

2.4.6 Budget constraint

$$C_t + K_t = Y_t + (1-\delta)K_{t-1} \qquad \Rightarrow \qquad \bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K}$$

$$C_t \approx \bar{C}(1+\hat{c}_t), \qquad K_t \approx \bar{K}(1+\hat{k}_t), \qquad Y_t \approx \bar{Y}(1+\hat{y}_t), \qquad K_{t-1} \approx \bar{K}(1+\hat{k}_{t-1})$$

$$\Rightarrow \quad \bar{C}(1+\hat{c}_t) + \bar{K}(1+\hat{k}_t) = \bar{Y}(1+\hat{y}_t) + (1-\delta)\bar{K}(1+\hat{k}_{t-1})$$

$$\Rightarrow \quad \bar{C} + \bar{C}\hat{c}_t + \bar{K} + \bar{K}\hat{k}_t = \bar{Y} + \bar{Y}\hat{y}_t + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1}$$

Using

$$\bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K},$$

we obtain

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{Y}\hat{y}_t + (1-\delta)\bar{K}\hat{k}_{t-1}.$$

2.4.7 Utility FOC

$$C_t^{-\sigma} = \lambda_t \qquad \Rightarrow \qquad \bar{C}^{-\sigma} = \bar{\lambda}$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \qquad \lambda_t = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Then

$$C_t^{-\sigma} = \bar{C}^{-\sigma}e^{-\sigma\hat{c}_t} = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Since $\bar{C}^{-\sigma} = \bar{\lambda}$,

$$e^{-\sigma\hat{c}_t} = e^{\hat{\lambda}_t}.$$

Using

$$e^{-\sigma\hat{c}_t} \approx 1 - \sigma\hat{c}_t, \qquad e^{\hat{\lambda}_t} \approx 1 + \hat{\lambda}_t,$$

we get

$$1 - \sigma\hat{c}_t = 1 + \hat{\lambda}_t \qquad \Rightarrow \qquad -\sigma\hat{c}_t = \hat{\lambda}_t.$$

2.4.8 C-D production function

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha} \qquad \Rightarrow \qquad \bar{Y} = \bar{A}\bar{K}^{\alpha}\bar{N}^{1-\alpha}$$

$$\log Y_t = \log A_t + \alpha \log K_t + (1-\alpha)\log N_t$$
$$\log \bar{Y} = \log \bar{A} + \alpha \log \bar{K} + (1-\alpha)\log \bar{N}$$

Subtract:

$$\log Y_t - \log \bar{Y} = (\log A_t - \log \bar{A}) + \alpha(\log K_t - \log \bar{K}) + (1-\alpha)(\log N_t - \log \bar{N})$$

hence

$$\hat{y}_t = \hat{a}_t + \alpha\hat{k}_t + (1-\alpha)\hat{n}_t.$$

2.4.9 GDP identity

$$Y_t = C_t + I_t + G_t$$

From (5) we know

$$\hat{y}_t = \frac{\bar{C}}{\bar{Y}}\hat{c}_t + \frac{\bar{I}}{\bar{Y}}\hat{i}_t + \frac{\bar{G}}{\bar{Y}}\hat{g}_t = s_c\hat{c}_t + s_i\hat{i}_t + s_g\hat{g}_t.$$

share of GDP for each component

2.4.10 Capital law of motion

$$K_{t+1} = (1-\delta)K_t + I_t \qquad \Rightarrow \qquad \bar{K} = (1-\delta)\bar{K} + \bar{I}$$

$$\Rightarrow \quad \bar{K}(1+\hat{k}_{t+1}) = (1-\delta)\bar{K}(1+\hat{k}_t) + \bar{I}(1+\hat{i}_t)$$

$$\Rightarrow \quad \bar{K} + \bar{K}\hat{k}_{t+1} = (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_t + \bar{I} + \bar{I}\hat{i}_t$$

Using

$$\bar{K} = (1-\delta)\bar{K} + \bar{I},$$

we get

$$\bar{K}\hat{k}_{t+1} = (1-\delta)\bar{K}\hat{k}_t + \bar{I}\hat{i}_t.$$

Thus

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \frac{\bar{I}}{\bar{K}}\hat{i}_t.$$

Since

$$\bar{K} = (1-\delta)\bar{K} + \bar{I} \qquad \Rightarrow \qquad \frac{\bar{I}}{\bar{K}} = \delta,$$

we obtain

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \delta\hat{i}_t.$$

2.4.11 Euler equation

$$U'(C_t) = \beta E_t\left[U'(C_{t+1})R_{t+1}\right]$$

use CRRA utility:

$$U(C_t) = \frac{C_t^{1-\sigma}}{1-\sigma} \qquad \Rightarrow \qquad U'(C_t) = C_t^{-\sigma}$$

So

$$C_t^{-\sigma} = \beta E_t\left[C_{t+1}^{-\sigma}R_{t+1}\right].$$

At steady state:

$$\bar{C}^{-\sigma} = \beta\bar{C}^{-\sigma}\bar{R} \qquad \Rightarrow \qquad 1 = \beta\bar{R}.$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \qquad C_{t+1} = \bar{C}e^{\hat{c}_{t+1}}, \qquad R_{t+1} = \bar{R}e^{\hat{r}_{t+1}}.$$

Then

$$\bar{C}^{-\sigma}e^{-\sigma\hat{c}_t} = \beta E_t\left[\bar{C}^{-\sigma}e^{-\sigma\hat{c}_{t+1}}\bar{R}e^{\hat{r}_{t+1}}\right].$$

First-order approximation:

$$\bar{C}^{-\sigma}(1-\sigma\hat{c}_t) = \beta E_t\left[\bar{C}^{-\sigma}\bar{R}(1-\sigma\hat{c}_{t+1})(1+\hat{r}_{t+1})\right]$$

$$= \beta E_t\left[\bar{C}^{-\sigma}\bar{R}\left(1+\hat{r}_{t+1}-\sigma\hat{c}_{t+1}-\underbrace{\sigma\hat{c}_{t+1}\hat{r}_{t+1}}_{\approx 0}\right)\right]$$

Using $1 = \beta\bar{R}$:

$$1-\sigma\hat{c}_t = 1 + E_t(-\sigma\hat{c}_{t+1}+\hat{r}_{t+1}).$$

Hence

$$-\sigma\hat{c}_t = E_t(-\sigma\hat{c}_{t+1} + \hat{r}_{t+1})$$

so

$$\hat{c}_t = E_t(\hat{c}_{t+1}) - \frac{1}{\sigma}E_t(\hat{r}_{t+1}).$$

2.5 Log linearization cheat sheet for economics

$$(1)\ XY: \quad \hat{x}+\hat{y} \qquad (2)\ X/Y: \quad \hat{x}-\hat{y}$$

$$(3)\ X^{\alpha}: \quad \alpha\hat{x} \qquad (4)\ \textstyle\prod_i X_i^{\alpha_i}: \quad \sum_i \alpha_i\hat{x}_i$$

$$(5)\ \textstyle\sum_i X_i: \quad \sum_i \frac{\bar{X}_i}{\sum_j \bar{X}_j}\hat{x}_i \qquad (6)\ K' = (1-\delta)K + I: \quad \hat{k}' = (1-\delta)\hat{k} + \delta\hat{\imath}$$

$$(7)\ \pi_t = \frac{P_t}{P_{t-1}}: \quad \hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \qquad (8)\ \text{CRRA MU}: \quad \widehat{u'(c)} = -\sigma\hat{c}$$

$$(9)\ \text{CES index } X = \left[\sum_i w_i X_i^{\rho}\right]^{1/\rho} : \quad \hat{x} = \sum_i s_i\hat{x}_i$$

$$\text{with } s_i = \frac{w_i\bar{X}_i^{\rho}}{\sum_j w_j\bar{X}_j^{\rho}}$$

Lecture 3

Real Business Cycle Model

3.1 Basic RBC model: Social Planner's Problem

Preference:

$$U = E_0\left[\sum_{t=0}^{\infty}\beta^t \frac{C_t^{1-\sigma}-1}{1-\sigma}\right]$$

expectation given information at $t = 0$

Technology:

$$C_t + K_t = Z_t \boxed{K_{t-1}}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1}$$

The amount of capital is predetermined in last period.

Produced goods & depreciated capital in last period used for consumption & new capital.

In current period t, agent decides on K_t.

Z_t is the TFP which follows the process:

$$\log Z_t = (1-\psi) \underbrace{\log \bar{Z}}_{\text{steady state TFP}} + \psi \log Z_{t-1} + \varepsilon_t$$

usually normalize $\bar{Z} = 1 \Rightarrow \log \bar{Z} = 0$

Endowment:

$$\underbrace{N_t = 1}_{\text{time/labor endowment}}, \qquad \underbrace{K_{-1} > 0}_{\text{initial capital stock}}$$

Information: decision made based on all information I_t up to t.
Social planner maximize the social utility by choosing C_t, K_t:

$$\max_{\{C_t,K_t\}} U = \max_{\{C_t,K_t\}} E_0\left[\sum_{t=0}^{\infty}\beta^t \frac{C_t^{1-\sigma}-1}{1-\sigma}\right]$$

subject to a budget constraint:

$$\forall t, \qquad C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1}$$

$$N_t = 1, \qquad K_{-1} > 0$$

Set up dynamic Lagrangian equation:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma} - 1}{1-\sigma} + \lambda_t \left[Z_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t - K_t \right] \right\}$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} - \lambda_t = 0 \tag{1}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial K_t} = 0 \qquad &\Rightarrow \qquad -\lambda_t \beta^t + E_t \left[\lambda_{t+1} \beta^{t+1} \left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right] = 0 \\ &\Rightarrow \qquad -\lambda_t + \beta E_t \left[\lambda_{t+1} \left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right] = 0 \end{aligned} \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \qquad \Rightarrow \qquad C_t + K_t - Z_t K_{t-1}^{\alpha} - (1-\delta) K_{t-1} = 0$$

Transversality Condition:

$$\lim_{t \to \infty} \beta^t E_t [\lambda_t K_t] = 0$$

Plug in (1) into (2):

$$\beta E_t \left[C_{t+1}^{-\sigma} \left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right] = C_t^{-\sigma}$$

$$\Rightarrow \qquad 1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right]$$

We have a dynamic system of (K_t, C_t, Z_{t+1})

$$K_t = Z_t K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - C_t, \tag{1}$$

$$1 = \beta E_t\left[\left(\frac{C_t}{C_{t+1}}\right)^{\sigma}\left(1-\delta+\alpha Z_{t+1}K_t^{\alpha-1}\right)\right], \tag{2}$$

$$\log Z_t = (1-\psi)\log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N(0, \sigma_\varepsilon^2), \tag{3}$$

state variables $(K_{t-1},\ C_t)$, shock variable Z_t.

Three variables, three equations: #(equation) = #(variable)

3.2 What is the economic intuition behind Lagrangian multiplier?

Lagrangian multiplier is the shadow price of constrained resources. Marginal value of relaxing or tightening a constraint. For a problem,

$$V(b) = \max_x f(x) \qquad \text{subject to } g(x) \le b$$

$$\mathcal{L}(x, \lambda) = f(x) + \lambda\,(b - g(x))$$

- If constraint is binding $g(x) = b$, λ shows how much the objective function would increase if you relax the constraint slightly.——shadow value

Envelope theorem:

$$\frac{dV(b)}{db} = \frac{\partial \mathcal{L}(x,\lambda;b)}{\partial b}\bigg|_{x^*,\lambda^*} \qquad \Rightarrow \qquad V'(b) = \lambda^*(b)$$

$$V(b+\Delta) - V(b) \approx \lambda^*(b)\Delta$$

- If constraint is non-binding $g(x) < b$, $\lambda^* = 0 \quad \Rightarrow \quad V'(b) = 0$: A relaxation of constraint will not improve the objective function.

eg: RBC:

$$W_t = \max_{\{C_{t+j},K_{t+j+1}\}_{j=0}^{\infty}} E_t \sum_{j=0}^{\infty} \beta^j \frac{C_{t+j}^{1-\sigma}}{1-\sigma}$$

subject to

$$C_{t+j} + K_{t+j+1} = F(Z_{t+j}, K_{t+j}) + (1-\delta)K_{t+j} + a_t$$

a_t appear only at $j = 0$, perturb only one period's constraint

$$\mathcal{L} = E_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{t+j}^{1-\sigma}}{1-\sigma} + \lambda_{t+j}\big(F(\cdot) + (1-\delta)K_{t+j} - C_{t+j} - K_{t+j+1} + a_{t+j}\big)\right]$$

$$\frac{\partial W_t}{\partial a_t} = \frac{\partial \mathcal{L}_t}{\partial a_t}\bigg|_{\text{at optimum}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t C_t^{-\sigma} - \beta^t \lambda_t = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} = \lambda_t$$

$$\Rightarrow \qquad \frac{\partial W_t}{\partial a_t} = \lambda_t = C_t^{-\sigma} = U_C(C_t) = \frac{\partial(\text{max attainable lifetime utility } W_t)}{\partial(\text{one more unit of resources at } t)}$$

What is transversality condition (TVC)?

Boundary condition at infinity that rules out Ponzi-scheme: leaving valuable resources unspent forever.

In a typical RBC model,

$$\text{TVC:} \qquad \lim_{T\to\infty} \beta^T U_C(C_T) K_{T+1} = 0$$

The discounted marginal utility value of the capital stock carried into the far future must vanish. Otherwise you're wasting utility by not consuming some of it earlier.

At large T, if we increase C_T by ε and decrease future capital by ε:

$$\text{First-order loss in capital:} \qquad \beta^{T+1}\lambda_{T+1}\varepsilon$$

$$\text{First-order gain in utility:} \qquad \beta^T U_C \varepsilon$$

$$\Rightarrow \qquad \lim_{T\to\infty} \beta^T U_C K_{T+1} = 0$$

in far future, you can not gain by this operation

3.3 Basic RBC model: competitive market approach

divide social planner into household and firm

*Household's problem: Household's own capital K_{t-1} & one unit of labor $N_t = 1$. They lend capital and labor to firms and earn interest R_t & wage W_t.

$$\max_{\{C_t, K_t, N_t\}_{t=0}^{\infty}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1-\sigma} \right]$$

s.t.

$$C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)}$$

(s) means supply

No-ponzi Scheme condition (TVC):

$$\lim_{t\to\infty} E_0 \left[\left(\prod_{s=0}^{t} R_s^{-1} \right) K_t^{(s)} \right] = 0$$

*Firm's problem: Representative firm with tech Z_t, hire capital & labor to produce, and sells output to maximize profit. Given W_t, R_t, firm chooses $N_t^{(d)}$ & $K_{t-1}^{(d)}$.

$$\Pi_t = \max_{\{K_{t-1}^{(d)}, N_t^{(d)}\}} Z_t (K_{t-1}^{(d)})^{\alpha} (N_t^{(d)})^{1-\alpha} - W_t N_t^{(d)} - R_t K_{t-1}^{(d)}$$

(d) means demand

where tech follows AR(1):

$$\log Z_t = (1-\psi)\log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N(0, \sigma_\varepsilon^2)$$

Solve household's problem with Lagrangian Multiplier Method:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}-1}{1-\sigma} + \lambda_t \big(R_t K_{t-1}^{(s)} + W_t N_t^{(s)} + (1-\delta)K_{t-1}^{(s)} - C_t - K_t^{(s)}\big) \right]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} - \lambda_t = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \qquad \Rightarrow \qquad -\beta^t \lambda_t + \beta^{t+1} E_t[\lambda_{t+1}(R_{t+1} + 1 - \delta)] = 0$$

$$\Rightarrow \qquad \lambda_t = \beta E_t[\lambda_{t+1}(R_{t+1} + 1 - \delta)] \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \qquad \Rightarrow \qquad C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta)K_{t-1}^{(s)} \tag{3}$$

Euler equation. Plug (1) into (2):

$$\underbrace{C_t^{-\sigma}}_{\text{MU of 1 unit } C \text{ today}} = \underbrace{\beta}_{\text{discount to today}} E_t\left[\underbrace{C_{t+1}^{-\sigma}}_{\text{MU of 1 unit } C \text{ tomorrow}} \underbrace{(R_{t+1} + 1 - \delta)}_{\text{return of extra unit of } K \text{ tomorrow}} \right]$$

LHS: The utility of 1 unit of consumption today.

RHS: The expected utility of sacrificing 1 unit of consumption today, instead save as capital, and carry over to next period, get R_{t+1} (rent earning) & get $(1-\delta)$ unit of depreciated capital, consume $R_{t+1} + 1 - \delta$ in the next period at marginal utility of consumption $\lambda_{t+1} = C_{t+1}^{-\sigma}$. Finally, discount to current value by multiplying β.

$$\text{LHS} = \text{RHS} \qquad \Rightarrow \qquad \text{Intertemporal trade-off ("no arbitrage")}$$

- A TFP shock on Z_t change the level through resource constraint, and dynamics through the Euler equation.

Resource constraint:

$$C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1}$$

Solve firm's problem with Lagrangian multiplier method:

$$\frac{\partial}{\partial K_{t-1}^{(d)}} \left[Z_t (K_{t-1}^{(d)})^{\alpha} (N_t^{(d)})^{1-\alpha} - W_t N_t^{(d)} - R_t K_{t-1}^{(d)} \right] = 0$$

$$\Rightarrow \qquad \alpha Z_t (K_{t-1}^{(d)})^{\alpha-1} (N_t^{(d)})^{1-\alpha} = R_t \tag{4}$$

$$\text{MPK} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t^{(d)}} = 0 \qquad \Rightarrow \qquad (1-\alpha) Z_t (K_{t-1}^{(d)})^{\alpha} (N_t^{(d)})^{-\alpha} = W_t \tag{5}$$

$$\text{MPL} = W_t$$

Close the model: Market clearing conditions

Capital market:

$$K_{t-1}^{(d)} = K_{t-1}^{(s)} \qquad \cdots \quad \text{capital demand = supply}$$

Labor market:

$$N_t^{(d)} = N_t^{(s)} = 1 \qquad \cdots \quad \text{labor demand = supply}$$

Goods market:

$$C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1} \qquad \cdots \quad \text{Resource constraint}$$

Walras Law: if there are m markets, then we need $m-1$ number of market clearing conditions. The last market is automatically cleared.

Thus, here we need two out of three. We keep resource constraint & capital market clearing by dropping (d) & (s). We also plug in constant labor supply, demand.

How resource constraint is derived?

Competitive market $\Rightarrow \Pi_t = 0$:

$$Z_t (K_{t-1}^{(d)})^{\alpha} (N_t^{(d)})^{1-\alpha} = W_t N_t^{(d)} + R_t K_{t-1}^{(d)}$$

(3) from household's problem:

$$C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta)K_{t-1}^{(s)}$$

$$(s) = (d) \Rightarrow C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1}$$

Produced goods + depreciated capital from last period is divided into use of capital and consumption

System of equations of the competitive market version of the model:

$$\text{(i)} \quad K_t = Z_t K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - C_t \qquad \cdots \quad \text{resource constraint}$$

$$\text{(ii)} \quad 1 = E_t \left\{ \beta \left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \left[\alpha Z_{t+1} K_t^{\alpha-1} + (1-\delta) \right] \right\}$$

$$\text{(iii)} \quad \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N(0, \sigma_{\varepsilon}^2)$$

* The competitive market version is equivalent to Social planner's version. (isomorphic)

$\Rightarrow$ Same equilibrium outcome

* The competitive market version of RBC model is a decentralized version of social planner RBC.

3.4 First and second welfare theorem in dynamic equilibrium

*First welfare theorem. Provided that all product and factor markets are competitive and there are no externalities in production or consumption, dynamic competitive equilibrium are Pareto Optimal.

"no friction" world in economics

*Second welfare theorem. All Pareto optimal allocation can be decentralized as a dynamic competitive equilibrium.
1^{st} Welfare theorem in RBC: CE allocation solves the social planner's problem $\Rightarrow$ Competitive equilibria are Pareto Optimal.

2^{nd} Welfare theorem in RBC: Social planner allocation solves the competitive equilibrium $\Rightarrow$ Pareto efficient allocation is supported by CE ($\exists$ competitive prices that support it).
Both theorems rely on:

1) perfect competition (price takers)

2) No externalities, no taxes/distortions, no market power.

3) Complete contracting:no missing markets (rep agent RBC avoids this)

4) Concave utility, convex technology set (CRS + concavity)

5) TVC / no-Ponzi to rule out pathological asset plans

Question 1. If there is a exogenous TFP shock Z_t, which moves more on impact C_t or K_t

$$Y_t = C_t + i_t \qquad \Rightarrow \qquad \Delta Y = \Delta C + \Delta i$$

$$Y_t = Z_t K_{t-1}^{\alpha} \qquad \Rightarrow \qquad Z_t \uparrow \Rightarrow Y_t \uparrow$$

$$\log Z_{t+1} = (1-\psi)\log \bar{Z} + \psi \log Z_t + \varepsilon_t \qquad \Rightarrow \qquad Z_t \uparrow \Rightarrow Z_{t+1} \uparrow \Rightarrow MPK_{t+1} \uparrow \Rightarrow R_{t+1} \uparrow$$

$$i_t = K_t - (1-\delta)K_{t-1} \qquad \Rightarrow \qquad K_t \uparrow \Rightarrow i_t \uparrow$$

$$1 = \beta E_t\left[\left(\frac{C_t}{C_{t+1}}\right)^{\sigma}(R_{t+1} + 1 - \delta)\right] \qquad \Rightarrow \qquad R_{t+1} \uparrow \Rightarrow \frac{C_t}{C_{t+1}} \downarrow \Rightarrow C_{t+1} \uparrow > C_t \uparrow$$

Thus, $\quad Z_t \uparrow \Rightarrow C_t \uparrow,\ C_{t+1} \uparrow \ \&\ i_t \uparrow$
K_{t-1} is pre-determined, thus at current period t, $\Delta K_{t-1} = 0$.

$$i_t \uparrow \gg C_t \uparrow ? \qquad \text{Ambiguous for now.}$$

1) but in data, it is true
2) 1° $\quad \sigma$ captures intertemporal substitution

$$\sigma \uparrow \quad \Rightarrow \quad \text{stronger consumption smoothing} \quad \Rightarrow \quad i_t \uparrow \gg C_t \uparrow$$
more likely to hold

2° $\quad \psi$ captures shock persistence

$$\psi \uparrow \quad \Rightarrow \quad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely to hold}$$

3° $\quad \alpha$ captures MPK

$$\alpha \uparrow \quad \Rightarrow \quad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely}$$

4° $\quad \delta$ captures depreciation

$$\delta \uparrow \quad \Rightarrow \quad i_t \uparrow \gg C_t \uparrow \qquad \text{less likely}$$

5° $\quad \beta$ captures patience

$$\beta \uparrow \quad \Rightarrow \quad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely}$$

But in normal calibration, where

$$\sigma = 2, \qquad \psi = 0.95, \qquad \alpha = \frac{1}{3},$$

$$\delta = 0.025 \text{ (quarter data)}, \qquad \beta = 0.99 \text{ (quarter data)}$$

$$i_t \uparrow \gg C_t \uparrow \qquad \text{and} \qquad std(i_t) > std(C_t)$$

Question 2: What if households sell capital to firms and firms sell back to household after production?

Household's problem:

$$\max_{\{C_t,K_t,N_t\}_{t=0}^{\infty}} E_0\left[\sum_{t=0}^{\infty}\beta^t\frac{C_t^{1-\sigma}-1}{1-\sigma}\right]$$

s.t.

$$C_t+K_t^{(s)}=W_tN_t^{(s)}+R_tK_{t-1}^{(s)}$$

$$\Rightarrow \quad \begin{aligned} C_t^{-\sigma} &= \beta E_t\left[C_{t+1}^{-\sigma}R_{t+1}\right], \quad (1)\\ C_t+K_t^{(s)} &= W_tN_t^{(s)}+R_tK_{t-1}^{(s)}, \quad (2)\end{aligned}$$

Firm's problem:

$$\max_{\{K_{t-1}^{(d)},N_t^{(d)}\}}\Pi_t=\max\left\{Z_t(K_{t-1}^{(d)})^{\alpha}(N_t^{(d)})^{1-\alpha}-W_tN_t^{(d)}-R_tK_{t-1}^{(d)}+(1-\delta)K_{t-1}^{(d)}\right\}$$

$$\Rightarrow \quad \begin{aligned} \alpha Z_t(K_{t-1}^{(d)})^{\alpha-1}(N_t^{(d)})^{1-\alpha}+(1-\delta) &= R_t, \quad (3)\\ (1-\alpha)Z_t(K_{t-1}^{(d)})^{\alpha}(N_t^{(d)})^{-\alpha} &= W_t, \quad (4)\end{aligned}$$

System of equations is exactly the same as SP's problem. (isomorphic)
Here, R_t is no longer rental rate, but the gross return factor: rental + resale value.
Define the usual rental rate as

$$r_t=R_t-(1-\delta)$$

$$(3)\Rightarrow MPK=r_t$$

$$\begin{aligned}\Pi_t=Z_tK^{\alpha}N^{1-\alpha}+(1-\delta)K-RK-WN &= Z_tK^{\alpha}N^{1-\alpha}-\big(R-(1-\delta)\big)K-WN\\ &= Z_tK^{\alpha}N^{1-\alpha}-rK-WN\end{aligned}$$

$$(1)\Rightarrow \qquad C_t^{-\sigma}=\beta E_t\left[C_{t+1}^{-\sigma}(r_{t+1}+1-\delta)\right]$$

$$(2)\Rightarrow \qquad C_t+K_t=W_tN_t+r_tK_{t-1}+(1-\delta)K_{t-1}$$

$$\Rightarrow \qquad \text{Setting is totally isomorphic.}$$

3.5 Supplementary notes

3.5.1 Technology-shock responses of C, I, and K

Under a positive technology shock, K_t does *not* move on impact, because it is predetermined. Thus, on impact, both C_t and I_t rise, while K_t remains unchanged.

The more subtle question is whether C_t or I_t rises more. In the model, this is not a pure theoretical certainty independent of parameters: it depends on the parameter values.

Empirically, however, the data typically show that consumption rises by less, while investment is more volatile.Later in the course, this can be seen more clearly from the impulse response functions.

3.5.2 Definitions of return

In the baseline setup, R denotes the *capital rent*, namely the rental price.

Later, in Q2, we define

$$r = R - (1 - \delta).$$

There, R is reinterpreted as the *gross return on capital*, i.e. the total return from holding one unit of capital across periods.

Under this notation:

$$R = r + (1 - \delta).$$

Hence:

- r corresponds to the baseline R, namely the rental price ;
- R in Q2 is the gross return on capital;
- the gross return consists of two parts:

$$\text{gross return} = \text{rental price} + \text{residual value of capital}.$$

3.6 Solve the RBC model

Find the Steady State (S.S.) of the economy, drop time subscript:

$$\bar{C} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{K} \qquad \cdots \quad \text{Resource constraint}$$

$$1 = \beta\left[\alpha\bar{Z}\bar{K}^{\alpha-1} + (1-\delta)\right] \qquad \cdots \quad \text{Euler equation}$$

Two variables, $(\bar{K}, \bar{C})$ given parameters $(\bar{Z}, \beta, \alpha, \delta)$

$$C_t = C_{t+1} = \bar{C}, \qquad K_t = k_{t+1} = \bar{K}, \qquad Z_t = Z_{t+1} = \bar{Z}, \qquad N_t = 1, \ \forall t$$

$$\bar{K} = \left[\frac{\alpha\bar{Z}}{\frac{1}{\beta} - (1-\delta)}\right]^{\frac{1}{1-\alpha}}, \qquad \bar{C} = \bar{Z}\bar{K}^{\alpha} - \delta\bar{K}, \qquad \bar{Y} = \bar{Z}\bar{K}^{\alpha}$$

$$\frac{\bar{C}}{\bar{K}} = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\bar{Y}}{\bar{K}} - \delta = \frac{\frac{1}{\beta} - (1-\delta)}{\alpha} - \delta = \frac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha}$$

3.7 Log-linearization

(i) Resource constraint:

$$K_t = Z_t K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - C_t \qquad \Rightarrow \qquad \bar{K} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{C}$$

$$\bar{K}(1+\hat{k}_t) = \bar{Z}(1+\hat{z}_t)\bar{K}^{\alpha}(1+\hat{k}_{t-1})^{\alpha} + (1-\delta)\bar{K}(1+\hat{k}_{t-1}) - \bar{C}(1+\hat{c}_t)$$

Using

$$(1+\hat{k}_{t-1})^{\alpha} \approx 1 + \alpha\hat{k}_{t-1}$$

we get

$$\bar{K} + \bar{K}\hat{k}_t = \bar{Z}\bar{K}^{\alpha}\left(1 + \hat{z}_t + \alpha\hat{k}_{t-1}\right) + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C} - \bar{C}\hat{c}_t$$

Subtract steady state:

$$\bar{K}\hat{k}_t = \alpha\bar{Z}\bar{K}^{\alpha}\hat{k}_{t-1} + \bar{Z}\bar{K}^{\alpha}\hat{z}_t + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C}\hat{c}_t$$

$$\Rightarrow \qquad \hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t$$

∗ next period change in k equals

1) change in K_{t-1}, that result in change in output ($MPK = \alpha\bar{Y}/\bar{K}$) & depreciated capital $1-\delta$

2) change in tech z that result in change in output

3) change in consumption C that result in change in investment.

(ii) Euler equation:

$$1 = \beta E_t\left[\left(\frac{C_t}{C_{t+1}}\right)^{\sigma}\left(1 - \delta + \alpha Z_{t+1}K_t^{\alpha-1}\right)\right]$$

Define

$$R_{t+1} = 1 - \delta + \alpha Z_{t+1} K_t^{\alpha-1} \qquad \Rightarrow \qquad \bar{R} = \frac{1}{\beta}$$

from

$$1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} R_{t+1} \right]$$

we get

$$1 = E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \frac{R_{t+1}}{\bar{R}} \right]$$

Let

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C} e^{\hat{c}_t}}{\bar{C} e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}}, \qquad \frac{R_{t+1}}{\bar{R}} = e^{\hat{r}_{t+1}}$$

Thus

$$1 = E_t \left[e^{\sigma(\hat{c}_t - \hat{c}_{t+1}) + \hat{r}_{t+1}} \right]$$

use $e^x \approx 1 + x$ for small x:

$$1 \approx E_t \left[1 + \sigma(\hat{c}_t - \hat{c}_{t+1}) + \hat{r}_{t+1} \right]$$

$$\Rightarrow \qquad \sigma \hat{c}_t = \sigma E_t \hat{c}_{t+1} - E_t \hat{r}_{t+1} \tag{1}$$

Now derive $\hat{r}_{t+1}$. Let

$$MPK_{t+1} = \alpha Z_{t+1} K_t^{\alpha-1} \qquad \Rightarrow \qquad \overline{MPK} = \alpha \bar{Z} \bar{K}^{\alpha-1}$$

$$\bar{R} = 1 - \delta + \overline{MPK}$$

Now approximate MPK_{t+1} around steady state:

$$\alpha Z_{t+1} K_t^{\alpha-1} \approx \alpha \bar{Z}(1 + \hat{z}_{t+1}) \bar{K}^{\alpha-1} (1 + \hat{k}_t)^{\alpha-1}$$

$$= \overline{MPK} \left[1 + \hat{z}_{t+1} + (\alpha - 1)\hat{k}_t \right]$$

$$where (1+x)^a \approx 1 + ax \qquad \Rightarrow \qquad (1 + \hat{k}_t)^{\alpha-1} \approx 1 + (\alpha - 1)\hat{k}_t$$

$$R_{t+1} \approx (1-\delta) + \overline{MPK} + \overline{MPK}\left[\hat{z}_{t+1} + (\alpha-1)\hat{k}_t\right] = \bar{R} + \overline{MPK}\left[\hat{z}_{t+1} + (\alpha-1)\hat{k}_t\right]$$

$$\hat{r}_{t+1} = \log\left(\frac{R_{t+1}}{\bar{R}}\right) \approx \frac{R_{t+1}-\bar{R}}{\bar{R}} = \frac{\overline{MPK}}{\bar{R}}\left[\hat{z}_{t+1} + (\alpha-1)\hat{k}_t\right]$$

$$\frac{\overline{MPK}}{\bar{R}} = \frac{\bar{R}-(1-\delta)}{\bar{R}} = 1 - \frac{1-\delta}{\bar{R}} = 1-(1-\delta)\beta$$

Let

$$\tilde{\delta} \equiv 1-(1-\delta)\beta$$

share of gross return that comes from new output rather than keeping old capital.

Then

$$\hat{r}_{t+1} = \tilde{\delta}\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t\right] \tag{2}$$

Plug (2) into (1):

$$\sigma\hat{c}_t = \sigma E_t\hat{c}_{t+1} - \tilde{\delta}E_t\hat{z}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t$$

$$E_t(\hat{c}_{t+1}-\hat{c}_t) = \frac{\tilde{\delta}}{\sigma}\underbrace{\left(E_t\hat{z}_{t+1} - (1-\alpha)\hat{k}_t\right)}_{\text{log deviation of next period's MPK}}$$

Expected growth in consumption is proportional to expected returns.

$$E_t(\hat{c}_{t+1}-\hat{c}_t) \propto E_t\widehat{MPK}_{t+1}$$

$MPK \uparrow \Rightarrow$ want more consumption tomorrow $\Rightarrow$ save/invest more today $\Rightarrow C$ grows

$\sigma \uparrow \Rightarrow$ stronger smoothing, IES $\downarrow$ / more curvature $\Rightarrow$ households hate changing C across time

$\tilde{\delta} \uparrow \Rightarrow$ returns move more with MPK $\uparrow \Rightarrow C$ responds by more

(iii) Technology process:

$$\log Z_{t+1} = (1-\psi)\log\bar{Z} + \psi\log Z_t + \varepsilon_t$$

$$\log(\bar{Z}e^{\hat{z}_{t+1}}) = (1-\psi)\log\bar{Z} + \psi\log(\bar{Z}e^{\hat{z}_t}) + \varepsilon_t$$

$$\Rightarrow \qquad \hat{z}_{t+1} = \psi\hat{z}_t + \varepsilon_t$$

the deviation of technology from S.S. is an AR(1) process

The log-linearized system of equation:

$$\hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t$$

$$\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta = \overline{MPK} + 1 - \delta = \bar{R} = \frac{1}{\beta}$$

$$\frac{\bar{Y}}{\bar{K}} = \frac{\overline{MPK}}{\alpha} = \frac{\bar{R}-(1-\delta)}{\alpha} = \frac{\frac{1}{\beta}-(1-\delta)}{\alpha} = \frac{\tilde{\delta}}{\alpha\beta}$$

$$\frac{\bar{C}}{\bar{K}} = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\overline{MPK}}{\alpha} - \delta = \frac{\tilde{\delta}}{\alpha\beta} - \delta$$

$$\Rightarrow \qquad \begin{cases} \hat{k}_t = \dfrac{1}{\beta}\hat{k}_{t-1} + \dfrac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\dfrac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t \\ \sigma E_t\hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta}E_t\hat{z}_{t+1} = \sigma\hat{c}_t \\ \hat{z}_{t+1} = \psi\hat{z}_t + \varepsilon_t \end{cases}$$

$*$ Three equations, three variables $(\hat{k}_t, \hat{c}_{t+1}, \hat{z}_{t+1})$

3.8 Solve the system in terms of exogenous parameters & pre-determined variables

Pre-determined: $(\hat{z}_t, \hat{k}_{t-1})$

Write in matrix form:

$$\begin{pmatrix} 1 & 0 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma & -\tilde{\delta} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{k}_t \\ E_t \hat{c}_{t+1} \\ E_t \hat{z}_{t+1} \end{pmatrix} = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} & \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma & 0 \\ 0 & 0 & \psi \end{pmatrix} \begin{pmatrix} \hat{k}_{t-1} \\ \hat{c}_t \\ \hat{z}_t \end{pmatrix}$$

Because

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

is a stationary process, the dynamic property of system is pinned down by the deterministic part:

$\hat{z}_t$ is either 0 or converges to 0 since $0 < \psi < 1$

$$M\, E_t[x_t] = N x_{t-1}$$

where

$$x_{t-1} = \begin{bmatrix} \hat{k}_{t-1} & \hat{c}_t \end{bmatrix}'$$

$$M = \begin{pmatrix} 1 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma \end{pmatrix}, \qquad N = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma \end{pmatrix}$$

$$\det(M) = \sigma > 0 \qquad \Rightarrow \qquad M^{-1} \text{ exists}$$

Assume

$$M^{-1} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

Impose $MM^{-1} = I$:

$$\begin{pmatrix} 1 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \qquad b_{11} = 1, \quad b_{12} = 0, \quad b_{22} = \frac{1}{\sigma}, \quad b_{21} = -\frac{\tilde{\delta}(1-\alpha)}{\sigma}$$

$$\Rightarrow \qquad M^{-1} = \begin{pmatrix} 1 & 0 \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma} & \frac{1}{\sigma} \end{pmatrix}$$

$$\Rightarrow \qquad M^{-1}N = \begin{pmatrix} 1 & 0 \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma} & \frac{1}{\sigma} \end{pmatrix} \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma\beta} & 1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) \end{pmatrix}$$

Using

$$\delta - \frac{\tilde{\delta}}{\alpha\beta} = -\frac{\bar{C}}{\bar{K}},$$

we can write

$$M^{-1}N = \begin{pmatrix} \frac{1}{\beta} & -\frac{\bar{C}}{\bar{K}} \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma\beta} & 1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} \end{pmatrix}$$

$$E_t[x_t] = M^{-1}N x_{t-1}$$

Jacobian matrix

Thus, $M^{-1}N$ shows the property of roots.

$$T = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}}$$

$$D = \frac{1}{\beta}\left[1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}}\right] - \frac{\tilde{\delta}(1-\alpha)}{\sigma\beta}\frac{\bar{C}}{\bar{K}} = \frac{1}{\beta}$$

$$p(-1) = 1 + D + T = 2 + \frac{2}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} > 0$$

$$p(1) = 1 + D - T = -\frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} < 0$$

$$\Rightarrow \qquad \begin{cases} p(-1) > 0 \\ p(1) < 0 \end{cases}$$

$$\Rightarrow \qquad 1 \text{ stable root in } (-1, 1), \qquad 1 \text{ unstable root in } (1, +\infty)$$

Since we have one predetermined variable $\hat{k}_{t-1}$ and one jump variable $\hat{c}_t$,

$$\Rightarrow \qquad \text{the system is a saddle path equilibrium}$$

(from Blanchard and Kahn conditions)

3.9 What is Jacobian Matrix?

Partial derivative that tells how a change in today's state affects tomorrow's variables.

eg:

$$x_t = f(x_{t-1}, z_t)$$

$$J = \left.\frac{\partial f}{\partial x}\right|_{(\bar{x},\bar{z})}$$

(Notes transcribed by CMT.)

Lecture 4

Numerically Solve RBC Model

4.1 Solve LOM

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t$$

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t$$

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

State variables: $\hat{k}_{t-1}$ and shock $\hat{z}_t$.

$$\hat{k}_{t-1} \text{ is predetermined.}$$

The dynamics of the model can be described as a recursive law of motion in terms of states.

LOM

$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \end{cases}$$

and solve for $(V_{kk}, V_{kz}, V_{ck}, V_{cz})$ by the method of undetermined coefficients.

1)

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t \quad \Rightarrow \quad E_t \hat{z}_{t+1} = \psi \hat{z}_t.$$

Since $E_t(\varepsilon_t) = 0$, and $\varepsilon_t \sim iid\, N(0, \sigma^2)$,

$$E_t(\hat{z}_t) = \hat{z}_t,$$

because Z_t is already realized in period t, hence belongs to period-t's information set.

2) From

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t,$$

substitute

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t,$$

to get

$$V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t = \frac{1}{\beta}\hat{k}_{t-1} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\left(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t\right) + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t.$$

Collecting coefficients:

$$\left[\frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{ck} - V_{kk}\right] \hat{k}_{t-1} + \left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{cz} - V_{kz}\right] \hat{z}_t = 0.$$

Hence

$$\boxed{V_{kk} = \frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{ck}} \tag{1}$$

and

$$\boxed{V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{cz}} \tag{2}$$

3) From

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t,$$

substitute

$$\hat{c}_{t+1} = V_{ck}\hat{k}_t + V_{cz}\hat{z}_{t+1}, \qquad \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad E_t\hat{z}_{t+1} = \psi\hat{z}_t.$$

Then

$$\sigma E_t\left[V_{ck}\hat{k}_t + V_{cz}\hat{z}_{t+1}\right] + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta}\psi\hat{z}_t = \sigma\left(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t\right).$$

That is,

$$\sigma E_t\left[V_{ck}\left(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t\right) + V_{cz}\hat{z}_{t+1}\right] + \tilde{\delta}(1-\alpha)\left(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t\right) - \tilde{\delta}\psi\hat{z}_t = \sigma V_{ck}\hat{k}_{t-1} + \sigma V_{cz}\hat{z}_t.$$

Collecting coefficients:

$$\left[\sigma V_{ck}V_{kk} + \tilde{\delta}(1-\alpha)V_{kk} - \sigma V_{ck}\right]\hat{k}_{t-1} + \left[\sigma(V_{ck}V_{kz} + V_{cz}\psi) + \tilde{\delta}(1-\alpha)V_{kz} - \tilde{\delta}\psi - \sigma V_{cz}\right]\hat{z}_t = 0.$$

Hence

$$\boxed{\sigma V_{ck}(1 - V_{kk}) = \tilde{\delta}(1-\alpha)V_{kk}} \tag{3}$$

and

$$\boxed{\sigma V_{cz}(1-\psi) = \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]V_{kz} - \tilde{\delta}\psi} \tag{4}$$

4.2 Solve the equations for unknowns

Collect the four equations and solve for the four unknowns.

(i) (1) and (3) are two equations in (V_{kk}, V_{ck}). Eliminate V_{ck}.

From (1),

$$V_{ck} = \frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}.$$

Substitute into (3):

$$\sigma\left(\frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}\right)(1 - V_{kk}) = \tilde{\delta}(1-\alpha)V_{kk}.$$

So

$$\frac{\sigma}{\beta} - \frac{\sigma}{\beta}V_{kk} - \sigma V_{kk} + \sigma V_{kk}^2 = \tilde{\delta}(1-\alpha)\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)V_{kk}.$$

Equivalently,

$$V_{kk}^2 - \left[1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\right]V_{kk} + \frac{1}{\beta} = 0.$$

Let

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right).$$

Then the two roots are

$$V_{kk,1} = \frac{\gamma - \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0,$$

$$V_{kk,2} = \frac{\gamma + \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0.$$

We need to choose the stable root of the linear rational expectations (LRE) system.

Since

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) > 1 + \frac{1}{\beta},$$

we have

$$(1 - V_{kk,1})(1 - V_{kk,2}) = 1 + V_{kk,1}V_{kk,2} - (V_{kk,1} + V_{kk,2}) = 1 + \frac{1}{\beta} - \gamma < 0.$$

Hence

$$0 < V_{kk,1} < 1 < V_{kk,2}.$$

So $V_{kk,2}$ is the explosive root and should be discarded.

To see this, consider the shutdown-shock case $\hat{z}_t = 0$:

$$\hat{k}_t = V_{kk}\hat{k}_{t-1}.$$

Then V_{kk} is the speed of convergence:

$$|V_{kk}| < 1 \Rightarrow \text{ deviation in capital shrinks over time, economy returns to S.S.,}$$
$$|V_{kk}| > 1 \Rightarrow \text{ deviation in capital grows over time, economy explodes and violates TVC.}$$

By the Blanchard–Kahn condition, there is one predetermined state $\hat{k}_{t-1}$ and one jump variable $\hat{c}_t$, so the number of stable roots must equal the number of predetermined states. Therefore, exactly one stable root is needed.

$$\boxed{\text{Drop } V_{kk,2};\ \text{only } V_{kk,1} \text{ is admissible.}}$$

Thus

$$V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}.$$

And from (1),

$$V_{ck} = \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right).$$

(ii) (2) and (4) are two equations in (V_{kz}, V_{cz}). Solve for V_{cz} by substituting V_{kz} out.

Starting from

$$\sigma V_{cz}(1-\psi) = \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]V_{kz} - \tilde{\delta}\psi,$$

and

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)V_{cz},$$

substitute the second into the first:

$$\sigma V_{cz}(1-\psi) = \left[\sigma\frac{1}{\dfrac{\tilde{\delta}}{\alpha\beta} - \delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)V_{cz}\right] - \tilde{\delta}\psi$$

$$\Rightarrow \quad V_{cz} = \frac{\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)} \times \frac{\tilde{\delta}}{\alpha\beta}.$$

$$\Rightarrow \quad V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\tilde{\delta}}{\alpha\beta}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\cdot\frac{\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)}.$$

Now we have:

$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \\ \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t. \end{cases}$$

Where,

$$V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}, \qquad V_{ck} = \frac{\bar{K}}{\bar{C}}\left(\frac{1}{\beta} - V_{kk}\right),$$

$$V_{cz} = \frac{\sigma V_{ck} + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]\dfrac{\bar{C}}{\bar{K}}} \times \frac{\tilde{\delta}}{\alpha\beta},$$

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\bar{C}}{\bar{K}}V_{cz}.$$

Feed in quarterly parameter values

$\beta = 0.99$	matches a 4.1% annual rate: $r_q = \frac{1}{\beta} - 1 \approx 1.01\%$, $\quad r_a \approx (1+r_q)^4 - 1 \approx 4.1\%$.
$\alpha = 0.36$	matches the capital income share in the data.
$\sigma = 1$	log utility. Usually $\sigma = 2$ implies IES $= \frac{1}{\sigma} = 0.5$.
$\delta = 0.025$	capital depreciates by 2.5% per quarter (about 10% annually).
$\bar{Z} = 1$	steady-state technology level is normalized.
$\psi = 0.95$	persistence of the productivity shock, i.e. how long ε_t lasts.

Then we can compute $(V_{kk}, V_{cz}, V_{kz}, V_{ck})$.

Impulse response function (IRF) Shock ε_t and trace how variables respond.

e.g. Trace out all variables for $\varepsilon_1 = 1,\ \varepsilon_t = 0$ for $t \geq 2$, start from S.S. (one-time temporary).

4.3 IRF and half-life

Since

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t,$$

we have

$$\hat{z}_1 = \psi\hat{z}_0 + \varepsilon_1 = \varepsilon_1,$$

since we start from steady state and $\hat{z}_0 = 0$. (Recall a hat means deviation from steady state.)

Also,

$$\hat{z}_2 = \psi\hat{z}_1 + \varepsilon_2 = \psi\hat{z}_1 = \psi\varepsilon_1,$$

and in general

$$\hat{z}_t = \psi^{t-1}\varepsilon_1.$$

When $|\psi| < 1$, the shock dies out as $t \to \infty$.

For capital,

$$\hat{k}_1 = V_{kk}\hat{k}_0 + V_{kz}\hat{z}_1 = V_{kz}\varepsilon_1,$$

since $\hat{k}_0 = 0$ at steady state.

Then

$$\hat{k}_2 = V_{kk}\hat{k}_1 + V_{kz}\hat{z}_2 = V_{kk}V_{kz}\varepsilon_1 + V_{kz}\psi\varepsilon_1 = (V_{kk} + \psi)V_{kz}\varepsilon_1.$$

In general,

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\psi^{t-1}\varepsilon_1 = \sum_{i=0}^{t-1} V_{kk}^{\,i}\psi^{\,t-i-1}V_{kz}\varepsilon_1.$$

$V_{kk}^{\,i}$ and V_{kz} are constant. Each past TFP effect ψ^{t-i-1} is stored/propagated through capital at rate $V_{kk}^{\,i}$, and V_{kz} translates the TFP shock into capital.

How to measure speed of convergence? Use half-life. We have

$$\hat{k}_{t+h} = V_{kk}^{\,h}\hat{k}_t.$$

The half-life is defined by

$$\frac{\hat{k}_{t+h_{1/2}}}{\hat{k}_t} = \frac{1}{2} \quad\Rightarrow\quad V_{kk}^{\,h_{1/2}} = \frac{1}{2} \quad\Rightarrow\quad \boxed{h_{1/2}^{(k)} = \frac{\ln(1/2)}{\ln(V_{kk})}} \qquad (0 < V_{kk} < 1).$$

For the TFP shock,

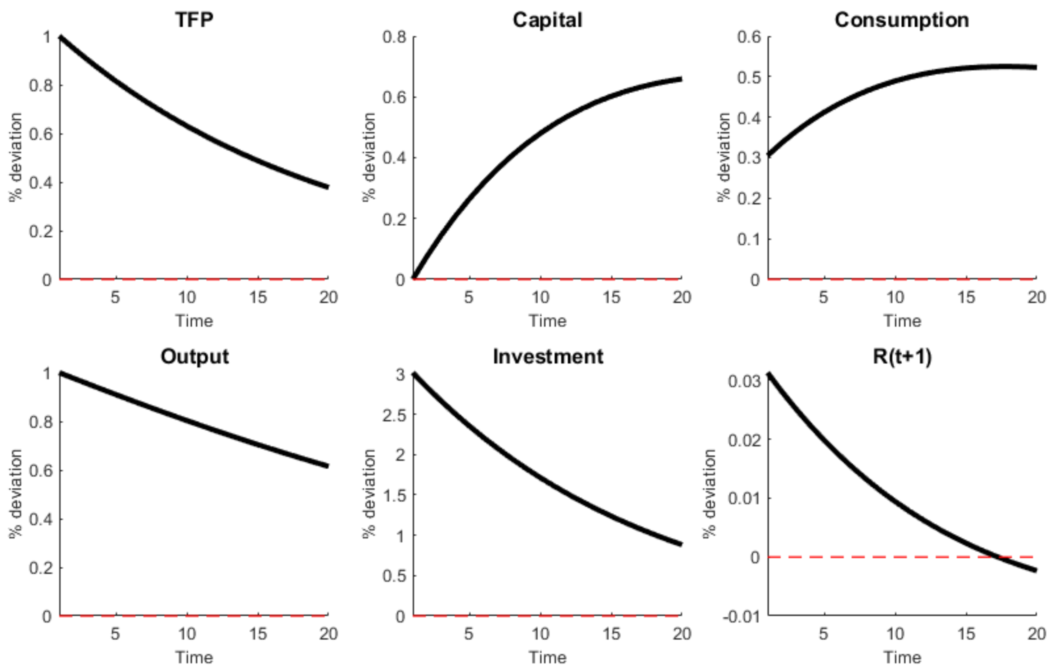
$$\boxed{h_{1/2}^{(z)} = \frac{\ln(1/2)}{\ln(\psi)}.}$$

Examples: if $V_{kk} = 0.95$,

$$h_{1/2}^{(k)} \approx 13.5 \text{ quarters} \approx 3.4 \text{ years}.$$

If $V_{kk} = 0.9$,

$$h_{1/2}^{(k)} \approx 6.6 \text{ quarters} \approx 1.6 \text{ years}.$$



Question: why is capital hump-shaped? From

$$\hat{k}_t = \sum_{i=0}^{t-1} \left(V_{kk}^{i} \psi^{\,t-i-1} \right) V_{kz} \varepsilon_1 = \begin{cases} V_{kz} \varepsilon_1 \dfrac{\psi^t - V_{kk}^t}{\psi - V_{kk}}, & \psi \neq V_{kk}, \\ V_{kz} \varepsilon_1 \, t \, \psi^t, & \psi = V_{kk}. \end{cases}$$

if $\psi = V_{kk}$, then

$$V_{kk}^{i} \psi^{\,t-i-1} = \psi^{\,t-1},$$

so

$$\hat{k}_t = \sum_{i=0}^{t-1} \psi^{\,t-1} V_{kz} \varepsilon_1 = t \psi^{\,t-1} V_{kz} \varepsilon_1.$$

Hence

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{(t+1)\psi^t}{t\psi^{t-1}} = \psi \left(1 + \frac{1}{t} \right).$$

For small t, $1 + \frac{1}{t}$ is large, so the ratio exceeds 1 and $\hat{k}_t$ keeps rising.

For large t, $1 + \frac{1}{t} \to 1$, so the ratio tends to $\psi < 1$ and $\hat{k}_t$ falls.

Peak occurs approximately when

$$\psi \left(1 + \frac{1}{t} \right) = 1 \quad \Rightarrow \quad t \approx \frac{\psi}{1 - \psi}.$$

Thus when $\psi = V_{kk}$, $\hat{k}_t$ rises at first, peaks around

$$t \approx \frac{\psi}{1 - \psi},$$

and then decays.

When $\psi \neq V_{kk}$,

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{\psi^{t+1} - V_{kk}^{t+1}}{\psi^t - V_{kk}^t} \quad \Rightarrow \quad \frac{\hat{k}_2}{\hat{k}_1} = \psi + V_{kk}.$$

We get an initial rise if $\psi + V_{kk} > 1$.
It peaks when $\hat{k}_{t+1} = \hat{k}_t$, i.e.

$$\psi^{t+1} - V_{kk}^{t+1} = \psi^t - V_{kk}^t,$$

so

$$\psi^t (1 - \psi) = V_{kk}^t (1 - V_{kk}),$$

hence

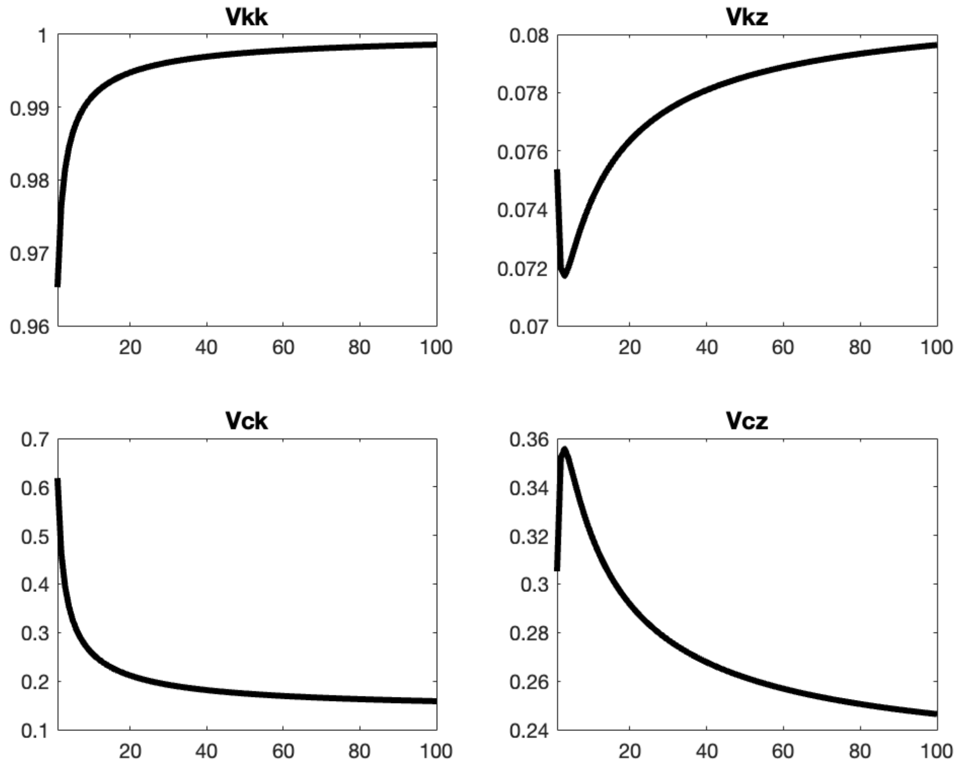
$$t^* = \frac{\ln\left(\frac{1 - V_{kk}}{1 - \psi} \right)}{\ln\left(\frac{\psi}{V_{kk}} \right)}.$$

If $t^* > 1$, the IRF rises for a while and then falls (hump-shaped). If $t^* \leq 1$, it peaks in the first period and then decays monotonically.

The effect of σ

$$\left.\begin{array}{l}\text{Supply only depends on patience } \beta \text{ and } \delta,\ \text{labor is fixed.}\\ \text{Demand: investment covers depreciation.}\end{array}\right\}\ \sigma \text{ does not have long-run effect.}$$

But does have short-run effects, governed by σ.



1.
$$\sigma \uparrow \quad \Rightarrow \quad V_{kk} \uparrow \rightarrow 1$$
Capital becomes more persistent and convergence is slower.
Households dislike shifting consumption across time more, i.e. lower IES.

2.
$$\sigma \uparrow \quad \Rightarrow \quad IES = \frac{1}{\sigma} \downarrow \quad \Rightarrow \quad V_{ck} \downarrow .$$
Marginal utility becomes more responsive to changes in consumption, so consumption reacts less for a given state incentive coming from k_{t-1}.

3.
$$\sigma \uparrow \quad \Rightarrow \quad IES \downarrow \quad \Rightarrow \quad V_{cz} \downarrow$$
Consumption's direct response to the shock shrinks.

4.
$$\sigma \uparrow \quad \Rightarrow \quad IES \downarrow \quad \Rightarrow \quad V_{cz} \downarrow \quad \Rightarrow \quad V_{kz} \uparrow$$
After a TFP shock, consumption is more reluctant to jump, so more resources show up as investment / capital accumulation.

Recall the steady state of the system

$$\bar{C} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{K},$$
$$1 = \beta\left[\alpha\bar{Z}\bar{K}^{\alpha-1} + (1-\delta)\right].$$

Thus

$$\bar{K} = \left(\frac{\alpha\bar{Z}}{\frac{1}{\beta} - 1 + \delta}\right)^{\frac{1}{1-\alpha}},$$
$$\bar{Y} = \bar{Z}\bar{K}^{\alpha}, \qquad \bar{C} = \bar{Y} - \delta\bar{K},$$

and

$$\frac{\bar{C}}{\bar{K}} = \frac{\bar{Y}}{\bar{K}} - \delta = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha}.$$

The effect of δ (depreciation) Example values in the notes:

$$\delta = 0.025: \qquad \bar{K} \approx 38, \quad \bar{Y} \approx 3.7, \quad \bar{C} \approx 2.75,$$
$$\delta = 0.1: \qquad \bar{K} \approx 6.4, \quad \bar{Y} \approx 1.95, \quad \bar{C} \approx 1.3.$$

So higher depreciation implies a lower steady-state level.
Also,

$$\delta \uparrow \Rightarrow V_{kk} \downarrow, \qquad \delta \uparrow \Rightarrow V_{kz} \uparrow.$$

With less capital stock and higher MPK, return and output both respond more proportionally.

For V_{cz}, the sign is ambiguous. The notes list several channels:

- Channel A: $\delta \uparrow \Rightarrow \tilde{\delta} \uparrow$,
- Channel B: $\delta \uparrow \Rightarrow \frac{\tilde{\delta}}{\alpha\beta} - \delta \uparrow$,
- Channel C: $\delta \uparrow \Rightarrow V_{kk} \downarrow$, $V_{ck} \uparrow$.

The final sign depends on calibration.

Complete procedure for solving RBC

1) Model setup: social planner's problem vs. competitive market (isomorphic).
2) Use dynamic programming to derive the necessary conditions(solve the model).
3) Write down the definition of equilibrium, including allocations and equilibrium conditions.
4) Find the steady state.
5) Log-linearize the dynamic system and determine local dynamic property.
6) Use the method of undetermined coefficients to solve the law of motion:

$$k_t = V_{kk}k_{t-1} + V_{kz}z_t,$$

and then solve for the other variables such as c_t.

7) Calibrate and simulate the model.

4.4 State-space approach: two-dimensional state space / saddle path analysis

Start from

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t.$$

Let $\hat{z}_t = 0$. Then

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t,$$

so

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1} - \frac{\bar{K}}{\bar{C}}\hat{k}_t.$$

Also, from

$$\sigma E_t\hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta}E_t\hat{z}_{t+1} = \sigma\hat{c}_t,$$

letting $\hat{z}_t = 0$ gives

$$\sigma(\hat{c}_t - \hat{c}_{t+1}) - \big[1-\beta(1-\delta)\big](1-\alpha)\hat{k}_t = 0$$

Rearrange:

$$\begin{cases} \hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta}-1\right)\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t, \\ \\ \hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma}\big[1-\beta(1-\delta)\big](1-\alpha)\left(\underbrace{\frac{\bar{C}}{\bar{K}}\hat{c}_t - \frac{1}{\beta}\hat{k}_{t+1}}_{=-\hat{k}_t}\right). \end{cases}$$

Steady-state loci If

$$\hat{k}_t = \hat{k}_{t-1},$$

then

$$\hat{c}_t = \left(\frac{1}{\beta}-1\right)\frac{\bar{K}}{\bar{C}}\hat{k}_{t-1}. \tag{1}$$

If

$$\hat{c}_{t+1} = \hat{c}_t,$$

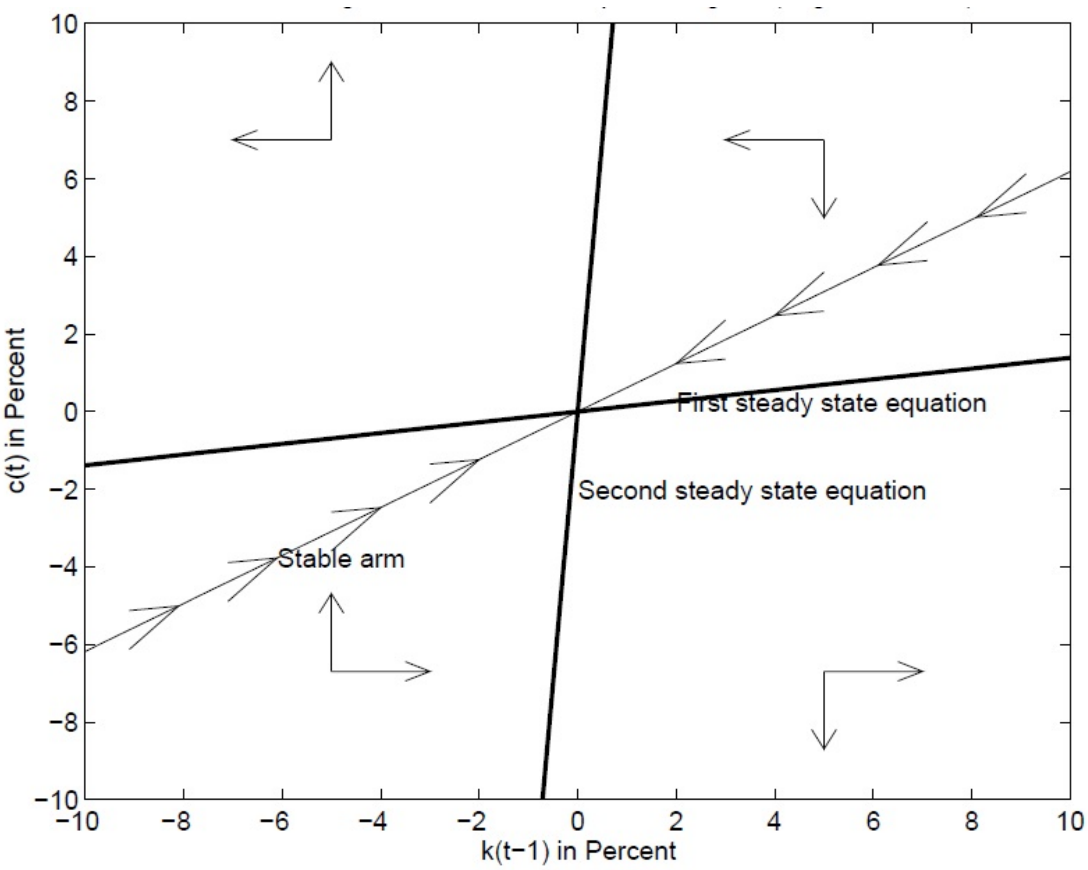
then

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1}. \tag{2}$$

The steady state is

$$\hat{k}_t = \hat{c}_t = 0.$$

The stable arm is the unique saddle path converging to the steady state.



The notes' directional interpretation:

- If $\hat{c}_t$ is above line (1), then $\hat{k}_t < \hat{k}_{t-1}$: consume too much, investment is low, so capital falls.
- If $\hat{c}_t$ is below line (1), then $\hat{k}_t > \hat{k}_{t-1}$: consume too little, save more, so capital rises.
- If $\hat{c}_t$ is above / left of line (2), then $\hat{c}_{t+1} > \hat{c}_t$: consume more next period.
- If $\hat{c}_t$ is below / right of line (2), then $\hat{c}_{t+1} < \hat{c}_t$: consume less next period.

4.5 Why these relations hold; redo in levels

Why line (1) implies $\hat{k}_t < \hat{k}_{t-1}$ If $\hat{c}_t$ is above line (1), then

$$\hat{c}_t > \left(\frac{1}{\beta} - 1\right) \frac{\bar{K}}{\bar{C}} \hat{k}_{t-1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} > 0.$$

Using

$$\hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t,$$

we get

$$\hat{k}_t < \hat{k}_{t-1}.$$

Why line (2) implies $\hat{c}_{t+1} > \hat{c}_t$ If $\hat{c}_t$ is above line (2), then

$$\hat{c}_t > \frac{\bar{K}}{\beta \bar{C}} \hat{k}_{t+1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} > 0.$$

Hence, from

$$\hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma} \big[1 - \beta(1-\delta)\big](1-\alpha) \left(\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} \right),$$

we have

$$\hat{c}_{t+1} > \hat{c}_t.$$

Stable arm The stable arm is the unique saddle path converging to steady state:

$$\hat{c}_t = V_{ck} \hat{k}_{t-1}.$$

The speed of convergence is

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} \quad \Rightarrow \quad \hat{k}_{t+h} = V_{kk}^h \hat{k}_t.$$

Nature gives $\hat{k}_{t-1}$, and households choose $\hat{c}_t$ so as to land exactly on the stable arm. Then the economy follows that arm back to steady state. Off the stable arm, the system is unstable and diverges.

Redo in levels Set $Z_t = \bar{Z}$:

$$\begin{aligned} C_t &= \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - K_t, \\ 1 &= \beta E_t \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \big(\alpha \bar{Z} K_t^{\alpha-1} + 1 - \delta \big). \end{aligned}$$

So

$$K_t - K_{t-1} = \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1} - C_t,$$

and

$$E_t \frac{C_{t+1}}{C_t} = \left\{ \beta \left[\alpha \bar{Z} \big(\bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t \big)^{\alpha-1} + 1 - \delta \right] \right\}^{1/\sigma}$$

At steady state, $\Delta K_t = 0$, so

$$C_t = \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}. \tag{1}$$

Thus:

$$\begin{aligned} &\text{if } C_t > \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}, \quad K_t < K_{t-1}, \\ &\text{if } C_t < \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}, \quad K_t > K_{t-1}. \end{aligned}$$

At steady state, $\Delta C_t = 0$, so

$$1 = \left\{ \beta \left[\alpha \bar{Z} \big(\bar{Z} K_{t+1}^{\alpha} + (1-\delta) K_{t+1} - C_t \big)^{\alpha-1} + 1 - \delta \right] \right\}^{1/\sigma},$$

which implies

$$C_t = \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - \bar{K}. \tag{2}$$

Then:

$$\begin{aligned} &\text{if } C_t > -\bar{K} + \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1}, \quad C_{t+1} > C_t, \\ &\text{if } C_t < -\bar{K} + \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1}, \quad C_{t+1} < C_t. \end{aligned}$$

Steady-state equations in levels

$$\begin{cases} C_t = \bar{Z}\, K_{t-1}^{\alpha} - \delta K_{t-1}, & (1) \\ C_t = \bar{Z}\, K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - \bar{K}. & (2) \end{cases}$$

Why is the stable arm still linear? Since

$$\hat{c}_t = V_{ck}\hat{k}_{t-1},$$

we have

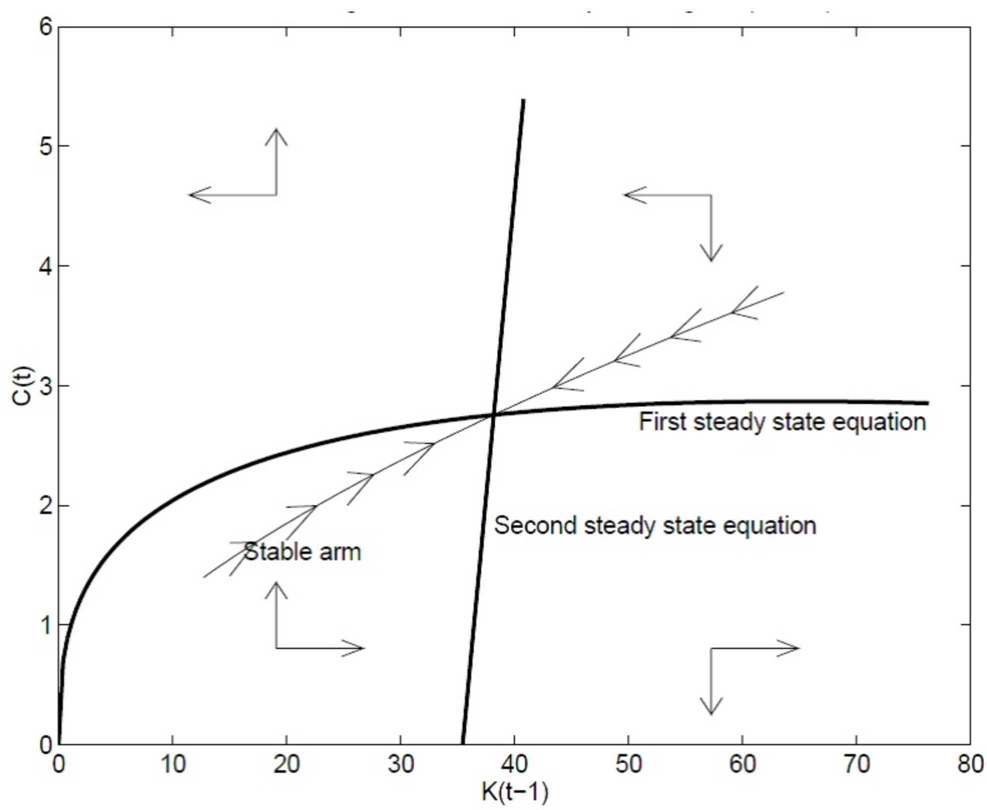
$$\frac{C_t - \bar{C}}{\bar{C}} = V_{ck}\frac{K_t - \bar{K}}{\bar{K}}.$$

Thus

$$C_t = \frac{V_{ck}\bar{C}}{\bar{K}}K_t - V_{ck}\bar{C} + \bar{C}.$$

So in levels the stable arm is linear, with slope

$$\frac{V_{ck}\bar{C}}{\bar{K}}.$$



Lecture 5

RBC with Endogenous Labor

5.1 HP filter

HP filter: decompose economic series into trend and cyclical component

$$y_t = y_t^c + y_t^T$$

$$\min_{\{y_t^T\}_{t=1}^N} \mathcal{L} = \underbrace{\sum_{t=1}^{N}(y_t - y_t^T)^2}_{(1)} + \underbrace{\lambda \sum_{t=2}^{N-1}\left[(y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T)\right]^2}_{(2)}$$

Notes:

1) penalty of calling everything trend
2) change in trend growth rate, penalty of wiggles

If λ huge, close to straight line in logs.
If $\lambda = 0$, wiggle a lot, fit to y_t.
Define

$$\Delta^2 y_t^T \equiv y_{t+1}^T - 2y_t^T + y_{t-1}^T \qquad \text{(second-order difference)}$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = 0 \quad \Rightarrow \quad \frac{\partial(1)}{\partial y_t^T} = 2(y_t^T - y_t)$$

Since

$$\Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T \qquad \text{(coefficient } +1)$$
$$\Delta^2 y_t^T = y_{t+1}^T - 2y_t^T + y_{t-1}^T \qquad \text{(coefficient } -2)$$
$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T \qquad \text{(coefficient } +1)$$

$$\begin{aligned}\frac{\partial}{\partial y_t^T}\sum_{s=2}^{N-1}(\Delta^2 y_s^T)^2 &= 2\Delta^2 y_{t-1}^T\frac{\partial \Delta^2 y_{t-1}^T}{\partial y_t^T} + 2\Delta^2 y_t^T\frac{\partial \Delta^2 y_t^T}{\partial y_t^T} + 2\Delta^2 y_{t+1}^T\frac{\partial \Delta^2 y_{t+1}^T}{\partial y_t^T} \\ &= 2\left(\Delta^2 y_{t-1}^T - 2\Delta^2 y_t^T + \Delta^2 y_{t+1}^T\right)\end{aligned}$$

Since

$$\Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T,$$

$$-2\Delta^2 y_t^T = -2(y_{t+1}^T - 2y_t^T + y_{t-1}^T) = -2y_{t+1}^T + 4y_t^T - 2y_{t-1}^T,$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T,$$

we get

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = \frac{\partial(1)}{\partial y_t^T} + \lambda \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 = 0.$$

$$\Rightarrow \quad (1+6\lambda)y_t^T - 4\lambda y_{t-1}^T - 4\lambda y_{t+1}^T + \lambda y_{t-2}^T + \lambda y_{t+2}^T = y_t, \qquad 3 \le t \le N-2.$$

At $t = 1$ (only $\Delta^2 y_2^T$ exists),

$$(1+\lambda)y_1^T - 2\lambda y_2^T + \lambda y_3^T = y_1.$$

At $t = 2$ (only $\Delta^2 y_2^T$ and $\Delta^2 y_3^T$ exist),

$$-2\lambda y_1^T + (1+5\lambda)y_2^T - 4\lambda y_3^T + \lambda y_4^T = y_2.$$

Symmetrically for $t = N$ and $t = N-1$, collect $\{y_t\}$ into the $N \times 1$ vector y and $\{y_t^T\}$ into y^T, then we have

$$Ay^T = y \quad \Rightarrow \quad y^T = A^{-1}y \quad \Rightarrow \quad y^c = y - A^{-1}y.$$

where

$$A = \begin{pmatrix} 1+\lambda & -2\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 \\ -2\lambda & 1+5\lambda & -4\lambda & \lambda & 0 & \cdots & 0 & 0 & 0 \\ \lambda & -4\lambda & 1+6\lambda & -4\lambda & \lambda & \cdots & 0 & 0 & 0 \\ 0 & \lambda & -4\lambda & 1+6\lambda & -4\lambda & \ddots & 0 & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -4\lambda & 1+6\lambda & -4\lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda & -4\lambda & 1+5\lambda & -2\lambda \\ 0 & 0 & 0 & \cdots & 0 & 0 & \lambda & -2\lambda & 1+\lambda \end{pmatrix}.$$

Let D be the $(N-2) \times N$ second-difference matrix, whose t-th row $(t = 2, \ldots, N-1)$ looks like

$$[\,0 \ \cdots \ 1 \ -2 \ 1 \ \cdots \ 0\,]$$

so that

$$(Dy^T)_{t-1} = \Delta^2 y_t^T.$$

The HP objective becomes

$$\min_{y^T} \, (y - y^T)'(y - y^T) + \lambda (Dy^T)'(Dy^T).$$

F.O.C.

$$-2(y - y^T) + 2\lambda D'Dy^T = 0 \quad \Rightarrow \quad (I + \lambda D'D)y^T = y.$$

Hence

$$A = I + \lambda D'D.$$

As a 5-diagonal pattern:

$$\text{interior diagonal: } \begin{bmatrix} 0 & \cdots & 1 & -2 & 1 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ -2 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 6$$

$$\text{first-off diagonal: } \begin{bmatrix} 0 & \cdots & 1 & -2 & 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = -4$$

$$\text{second-off diagonal: } \begin{bmatrix} 0 & \cdots & 1 & -2 & 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1$$

$$\lambda \uparrow \quad \Rightarrow \quad \text{the smoother the series } y_t^T.$$

Pros

1) Simple and transparent. (defined by one objective)

2) Produces a smooth trend with tunable smoothness λ

3) Widely used in Macro

Cons

1) End-point problem: start and end of sample is distorted.

2) λ sensitivity: moments (volatility, correlations) change a lot with λ.

3) Spurious dynamics: HP can induce artificial serial correlations and distort cross-corr between series (create cycles).

4) Leakage: low-frequency cycles leak into trend.

5) Not structural: no model interpretation.

Why $\lambda = 1600$ is the quarterly standard?
Rule of thumb:

$$\lambda = 100 \times (\text{periods per year})^2.$$

Hence

$$\lambda_{\text{quarter}} = 100 \times 4^2 = 1600, \qquad \lambda_{\text{month}} = 100 \times 12^2 = 14400, \qquad \lambda_{\text{annual}} = 100 \times 1^2 = 100.$$

5.2 Long and Plosser (1983, JPE)

Social planner's Problem Setup:

$$U = \max_{\{C_t, L_t, K_t\}} E_0 \left\{ \sum_{t=0}^{\infty} \beta^t \left[\theta \ln C_t + (1-\theta) \ln L_t \right] \right\}$$

s.t.

$$K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - C_t \qquad \text{and } K_0 \text{ is given.}$$

Resource constraint:

$$C_t + K_t = Y_t.$$

$$N_t + L_t = 1 \qquad \text{(trade-off)}$$

$$L: \text{ leisure} \qquad N: \text{ working hour.}$$

Total time is normalized to 1.

where

$$\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{i.i.d.}{\sim} N(0, \sigma^2), \qquad \psi < 1.$$

Note: capital completely turns over,

$$(1-\delta) K_{t-1} = 0 \qquad \delta = 1.$$

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big[\theta \ln C_t + (1-\theta) \ln(1-N_t) + \lambda_t \big(Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - C_t - K_t \big) \Big].$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \frac{\theta}{C_t} = \lambda_t \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad -\underbrace{\frac{1-\theta}{1-N_t}}_{\text{utility loss}} + \underbrace{\lambda_t}_{\text{MUC}_t} \underbrace{(1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}}_{\text{MPN}_t} = 0$$

$$\Rightarrow \quad \frac{1-\theta}{1-N_t} = (1-\alpha) \lambda_t Z_t K_{t-1}^{\alpha} N_t^{-\alpha} \tag{2}$$

labor-leisure condition (intratemporal)

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t \left[\lambda_{t+1} \frac{\partial}{\partial K_t} \big(Z_{t+1} K_t^{\alpha} N_{t+1}^{1-\alpha} \big) \right] = 0$$

$$\Rightarrow \quad \lambda_t = \beta E_t \big(\lambda_{t+1} \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} \big) \tag{3}$$

Euler equation (intertemporal)

TVC:

$$\lim_{T \to \infty} \beta^T U_C(C, L) K_{T+1} = 0. \tag{4}$$

Closed-form solution: constant savings rate and constant labor
Tractability: log utility + Cobb–Douglas + full depreciation

$$\Rightarrow \quad \text{constant savings rate / hours.}$$

Since

$$Y_{t+1} = Z_{t+1} K_t^{\alpha} N_{t+1}^{1-\alpha},$$

we have

$$\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} = \alpha \frac{Y_{t+1}}{K_t}.$$

Plug into (3):

$$\lambda_t = \beta E_t \left(\lambda_{t+1} \alpha \frac{Y_{t+1}}{K_t} \right).$$

Using (1):

$$\frac{\theta}{C_t} = \beta E_t \left(\frac{\theta}{C_{t+1}} \alpha \frac{Y_{t+1}}{K_t} \right)$$

$$\Rightarrow \quad \frac{1}{C_t} = \beta \alpha E_t \left[\frac{Y_{t+1}}{C_{t+1}} \frac{1}{K_t} \right]. \tag{5}$$

Guess and verify:

$C_t = (1-s)Y_t$, $\quad$ s is the savings rate, $\quad$ $K_t = sY_t$. $\quad$ (full depreciate)

Then

$$\frac{Y_{t+1}}{C_{t+1}} = \frac{1}{1-s}.$$

Plug into (5):

$$\frac{1}{(1-s)Y_t} = \alpha\beta \frac{1}{1-s} \frac{1}{sY_t}$$

$$\Rightarrow \quad s = \alpha\beta.$$

Hence

$$C_t = (1-\alpha\beta)Y_t, \qquad K_t = \alpha\beta\, Y_t.$$

$$\begin{cases} \alpha \uparrow \Rightarrow \text{ capital income share} \uparrow \Rightarrow \text{ savings rate} \uparrow, \\ \beta \uparrow \Rightarrow \text{ discount factor} \downarrow \Rightarrow \text{ savings rate} \uparrow. \end{cases}$$

Plug into (2):

$$\frac{1-\theta}{1-N_t} = (1-\alpha)\frac{\theta}{C_t}\frac{Y_t}{N_t} = (1-\alpha)\frac{\theta}{(1-\alpha\beta)Y_t}\frac{Y_t}{N_t}$$

$$\Rightarrow \quad (1-\theta)(1-\alpha\beta)N_t = (1-\alpha)\theta(1-N_t)$$

$$\Rightarrow \quad N_t = \bar{N} = \frac{\theta(1-\alpha)}{\theta(1-\alpha) + (1-\theta)(1-\alpha\beta)}.$$

Since N_t = constant $\forall t$, we have

$$\hat{n}_t = 0.$$

Also

$$K_t = \alpha\beta Y_t, \qquad C_t = (1-\alpha\beta)Y_t$$

so

$$\hat{k}_t = \hat{c}_t = \hat{y}_t.$$

Since

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1} + (1-\alpha)\hat{n}_t = \hat{z}_t + \alpha\hat{k}_{t-1},$$

we get

$$\hat{k}_t = \hat{c}_t = \hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1}.$$

And

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

TFP shock alone can generate output/consumption/investment fluctuations, but

1) labor does not fluctuate at all,

2) c, k, y move one-for-one.

Consumption is usually smoother than output; capital is usually more volatile.

3) Labor productivity moves like output in model.

$$\left(\frac{\hat{Y}}{\hat{N}}\right)_t = \hat{y}_t - \hat{n}_t = \hat{y}_t, \qquad \text{while in data,}$$

$$\sigma(Y/N) \ll \sigma(Y), \qquad \text{and } \mathrm{Corr}(Y/N, Y) = 0.55.$$

4) Prices (MRS, MRT, MPL, MPK) too-tight with output & TFP, hard to generate weak wage cyclicality & interest rate counter-cyclicality.

Labor market equilibrium
Household's problem:

$$U_{N,t} + \lambda_t W_t = 0 \quad \Rightarrow \quad U_{N,t} + U_{C,t} W_t = 0$$

labor supply
Firm's problem:

$$\max_{K_{t-1}, N_t} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

labor demand
Hence

$$W_t = Z_t \frac{Y_t}{N_t}(1-\alpha).$$

From household labor supply:

$$W_t = \frac{1-\theta}{\theta}\frac{C_t}{1-N_t}.$$

Equivalently,

$$W_t = \frac{1-\theta}{\theta}\frac{(1-\alpha\beta)Z_t K_{t-1}^{\alpha} N_t^{1-\alpha}}{1-N_t}.$$

From firm labor demand:

$$W_t = (1-\alpha)Z_t K_{t-1}^{\alpha} N_t^{-\alpha}.$$

A positive Z_t or K_{t-1} shock shifts labor supply up by the same factor as labor demand:

$$Z_t K_{t-1}^{\alpha}.$$

Hence labor supply's wealth effect is exactly offset by labor demand increase. So when Z_t rises, the real wage increases. This generates a substitution effect toward more labor, but also an income effect toward more leisure.

$$W_t = (1-\alpha)Z_t K_{t-1}^{\alpha} N_t^{-\alpha}$$

$$Z_t \uparrow \Rightarrow\ MPN_t \uparrow \Rightarrow\ W_t \uparrow \Rightarrow\ \text{labor demand curve shifts up}$$

$$Z_t \uparrow \Rightarrow\ Y_t \uparrow \Rightarrow\ C_t = (1-\alpha\beta)Y_t \uparrow \Rightarrow\ MU_{C,t} \downarrow \Rightarrow\ N_t \downarrow \qquad \text{Wealth effect}$$

$$\underbrace{\frac{1-\theta}{1-N_t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\text{marginal utility value of wage income}} = \underbrace{MU_{C,t} W_t}_{\text{intra-temporal optimality}} = \frac{\theta}{C_t}(1-\alpha)\frac{Y_t}{N_t}$$

$$= \frac{\theta(1-\alpha)}{N_t}\left(\frac{Y_t}{C_t}\right) = \frac{\theta(1-\alpha)}{1-\alpha\beta}\frac{1}{N_t}.$$

Hansen (1985): Basic idea Expand log preference to make labor respond to shocks.
Household's problem:

$$\max_{\{C_t, L_t, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) = E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$$

$$\text{s.t.} \qquad N_t + L_t = 1$$

$$C_t + K_t = W_t N_t + R_t K_{t-1} + (1-\delta) K_{t-1} \qquad \text{(budget constraint not resource constraint)}$$

$$K_t - (1-\delta) K_{t-1} = I_t \qquad \text{investment}$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big[u(C_t, 1 - N_t) + \lambda_t \big(W_t N_t + R_t K_{t-1} + (1-\delta) K_{t-1} - C_t - K_t \big) \Big]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \underbrace{U_{C,t}}_{\text{marginal utility of consumption}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N,t} + \lambda_t W_t = 0$$

$$\Rightarrow \quad - \underbrace{U_{N,t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\substack{\text{marginal utility value of} \\ \text{one more unit of wage income}}}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t [\lambda_{t+1} R_{t+1} + \lambda_{t+1}(1-\delta)] = 0$$

$$\Rightarrow \quad U_{C,t} = \beta E_t [U_{C,t+1}(R_{t+1} + 1 - \delta)]$$

Firm's problem

$$\max_{N_t, K_{t-1}} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

s.t.

$$Y_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha}, \qquad \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t.$$

FOCs:

$$\frac{\partial \Pi_t}{\partial K_{t-1}} = 0 \quad \Rightarrow \quad \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} = R_t \quad \Rightarrow \quad \alpha \frac{Y_t}{K_{t-1}} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t} = 0 \quad \Rightarrow \quad (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha} = W_t \quad \Rightarrow \quad (1-\alpha) \frac{Y_t}{N_t} = W_t.$$

Let utility function follow

$$U(C_t, 1 - N_t) = \ln C_t + A \ln(1 - N_t).$$

Then

$$U_{C,t} = \frac{1}{C_t}, \qquad U_{N,t} = -\frac{A}{1 - N_t}.$$

Hence equilibrium conditions are

$$\begin{cases} 1 = \beta E_t \left[\dfrac{C_t}{C_{t+1}} (R_{t+1} + 1 - \delta) \right], \\ \dfrac{A}{1-N_t} = \dfrac{1}{C_t} W_t, \\ W_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \\ R_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha}, \\ \Pi_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - W_t N_t - R_t K_{t-1} = 0, \\ C_t + K_t = W_t N_t + R_t K_{t-1} + (1-\delta) K_{t-1}, \\ \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Do not substitute out R_t, W_t, to make a system of equations of (C_t, N_t, K_t, Z_t). Substituting yields

$$\begin{cases} 1 = \beta E_t \left[\dfrac{C_t}{C_{t+1}} \left(\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta \right) \right], \\ A C_t = (1-N_t)(1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \\ K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1} - C_t, \\ \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Step 1: Steady state $\Rightarrow$ Step 2: Log-linearize $\Rightarrow$ Step 3: Numerically solve

5.3 Steady state

At steady state:

$$K_t = K_{t-1} = \bar{K}, \qquad C_t = C_{t+1} = \bar{C}, \qquad N_t = N_{t+1} = \bar{N}.$$

$$\bar{K} = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha} + (1-\delta)\bar{K} - \bar{C} \quad \Rightarrow \quad \bar{C} = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha} - \delta\bar{K}. \tag{1}$$

$$1 = \alpha\beta\bar{Z}\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} + (1-\delta)\beta. \tag{2}$$

$$A\bar{C} = (1-\bar{N})(1-\alpha)\bar{Z}\bar{K}^{\alpha}\bar{N}^{-\alpha}. \tag{3}$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha}\bar{N}^{1-\alpha} - \delta\bar{K} = \frac{1}{A}\Big((1-\alpha)\bar{Z}\,\bar{K}^{\alpha}\bar{N}^{-\alpha} - (1-\alpha)\bar{Z}\,\bar{K}^{\alpha}\bar{N}^{1-\alpha}\Big)$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} + \frac{1-\alpha}{A}\,\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} = \frac{1-\alpha}{A}\,\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{-\alpha} + \delta$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} = \frac{A}{1-\alpha+A}\left[\frac{1-\alpha}{A}\,\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{-\alpha} + \delta\right]$$

Using (2), after simplification one gets

$$\bar{N} = \left\{1 + \frac{A}{1-\alpha}\left[1 - \frac{\alpha\beta\delta}{1-\beta(1-\delta)}\right]\right\}^{-1}.$$

Then

$$\bar{K} = \left[\frac{1-\beta(1-\delta)}{\bar{Z}\alpha\beta}\bar{N}^{\alpha-1}\right]^{\frac{1}{\alpha-1}} = \bar{N}\left[\frac{\bar{Z}\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{1}{1-\alpha}}.$$

And

$$\bar{C} = \bar{Z}\,\bar{K}^{\alpha}\bar{N}^{1-\alpha} - \delta\bar{K} = \bar{N}^{\alpha}\cdot\left[\frac{\bar{Z}\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{\alpha}{1-\alpha}}\bar{N}^{1-\alpha}\bar{Z} - \delta\bar{K}$$

$$= \bar{Z}^{\frac{1}{1-\alpha}}\bar{N}\left[\frac{\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{\alpha}{1-\alpha}} - \delta\bar{N}\,\bar{Z}^{\frac{1}{1-\alpha}}\left[\frac{\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{1}{1-\alpha}}$$

$$= \bar{N}\,\bar{Z}^{\frac{1}{1-\alpha}}\left[\left(\frac{\alpha}{\beta^{-1}-(1-\delta)}\right)^{\frac{\alpha}{1-\alpha}} - \delta\left(\frac{\alpha}{\beta^{-1}-(1-\delta)}\right)^{\frac{1}{1-\alpha}}\right]$$

Derive the log-linearized system
(i) TFP process:

$$\log Z_t = (1-\psi)\log\bar{Z} + \psi\log Z_{t-1} + \varepsilon_t$$

Let

$$Z_t = \bar{Z}e^{\hat{z}_t}.$$

Then

$$\hat{z}_t + \log\bar{Z} = (1-\psi)\log\bar{Z} + \psi\log\bar{Z} + \psi\hat{z}_{t-1} + \varepsilon_t$$

$$\Rightarrow \quad \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

(ii)

$$K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1} - C_t.$$

Substitute

$$K_t = \bar{K}(1+\hat{k}_t), \quad Z_t = \bar{Z}(1+\hat{z}_t), \quad K_{t-1} = \bar{K}(1+\hat{k}_{t-1}), \quad N_t = \bar{N}(1+\hat{n}_t), \quad C_t = \bar{C}(1+\hat{c}_t),$$

and keep only first-order terms:

$$\bar{K} + \bar{K}\hat{k}_t = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha}(1+\hat{z}_t)(1+\alpha\hat{k}_{t-1})(1+(1-\alpha)\hat{n}_t) + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C} - \bar{C}\hat{c}_t.$$

Using the steady state condition and dropping higher-order terms:

$$\bar{K}\hat{k}_t = \alpha\bar{Y}\hat{k}_{t-1} + \bar{Y}\hat{z}_t + (1-\alpha)\bar{Y}\hat{n}_t + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C}\hat{c}_t,$$

where

$$\bar{Y} \equiv \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha}.$$

Hence

$$\hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + (1-\alpha)\frac{\bar{Y}}{\bar{K}}\hat{n}_t + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t.$$

(iii)

$$AC_t = (1-N_t)(1-\alpha)Z_t K_{t-1}^{\alpha} N_t^{-\alpha}.$$

Log-linearizing gives

$$\hat{n}_t = \left(\alpha + \frac{\bar{N}}{1-\bar{N}}\right)^{-1}\left(\hat{z}_t + \alpha\hat{k}_{t-1} - \hat{c}_t\right).$$

(iv) Euler equation:

$$1 = \beta E_t\left[\frac{C_t}{C_{t+1}}\left(\alpha Z_{t+1}K_t^{\alpha-1}N_{t+1}^{1-\alpha} + 1 - \delta\right)\right].$$

Let

$$M_{t+1} \equiv \alpha Z_{t+1}K_t^{\alpha-1}N_{t+1}^{1-\alpha}, \qquad G_{t+1} \equiv M_{t+1} + 1 - \delta.$$

Then

$$\hat{m}_{t+1} = \hat{z}_{t+1} + (\alpha-1)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}.$$

Define

$$\tilde{\delta} \equiv \frac{\bar{M}}{\bar{G}}.$$

Since

$$\bar{G} = \beta^{-1}, \qquad \bar{G} = \bar{M} + (1-\delta),$$

we have

$$\tilde{\delta} = \frac{\bar{M}}{\bar{G}} = 1 - \frac{1-\delta}{\bar{G}} = 1 - \beta(1-\delta).$$

Thus

$$G_{t+1} \approx \bar{G}(1 + \tilde{\delta}\hat{m}_{t+1}).$$

Also

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C}e^{\hat{c}_t}}{\bar{C}e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}} \approx 1 + \hat{c}_t - \hat{c}_{t+1}.$$

So Euler equation becomes

$$1 = \beta E_t\left[\bar{G}(1 + \hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1})\right]$$

Since

$$\beta\bar{G} = 1,$$

we obtain

$$E_t\left[\hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1}\right] = 0.$$

Hence

$$\hat{c}_t = E_t\hat{c}_{t+1} - \tilde{\delta}E_t\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}\right].$$

If you have CRRA,

$$\frac{C_t}{C_{t+1}} \to \left(\frac{C_t}{C_{t+1}}\right)^{\sigma} \quad \Rightarrow \quad \sigma\hat{c}_t = \sigma E_t\hat{c}_{t+1} - E_t\hat{r}_{t+1}.$$

Thus we have a system of equations:

$$\begin{cases} \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t, \\ \hat{c}_t = E_t\hat{c}_{t+1} - \tilde{\delta}E_t\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}\right], \\ \hat{n}_t = \left(\alpha + \dfrac{\bar{N}}{1-\bar{N}}\right)^{-1}(\hat{z}_t + \alpha\hat{k}_{t-1} - \hat{c}_t), \\ \hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + (1-\alpha)\frac{\bar{Y}}{\bar{K}}\hat{n}_t + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t. \end{cases}$$

Four equations, four variables:

$$(\hat{k}_{t-1}, \hat{c}_t, \hat{n}_t, \hat{z}_t).$$

$\hat{k}_{t-1}$ is predetermined variable, $\hat{z}_t$ follows a stable AR(1) process,

$(\hat{c}_t, \hat{n}_t)$ are jump variables.

Use undetermined coefficient method; guess

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \qquad \hat{n}_t = V_{nk}\hat{k}_{t-1} + V_{nz}\hat{z}_t.$$

Then

$$\hat{k}_t = V_{kz}\sum_{i=0}^{\infty}\sum_{j=0}^{i} V_{kk}^{j}\psi^{i-j}\varepsilon_{t-i}, \qquad \hat{z}_t = \sum_{s=0}^{\infty}\psi^{s}\varepsilon_{t-s},$$

and since

$$\hat{y}_t = \alpha\hat{k}_{t-1} + (1-\alpha)\hat{n}_t + \hat{z}_t,$$

we can write

$$\hat{y}_t = V_{yz}\varepsilon_t + \sum_{i=0}^{\infty}\left(V_{yk}V_{kz}\sum_{j=0}^{i} V_{kk}^{j}\psi^{i-j} + V_{yz}\psi^{i+1}\right)\varepsilon_{t-i-1}.$$

Therefore,

$$\mathrm{Var}(\hat{y}_t) = \left(V_{yz}^2 + \sum_{i=0}^{\infty} \left[V_{yk} V_{kz} \sum_{j=0}^{i} V_{kk}^j \psi^{\,i-j} + V_{yz} \psi^{i+1} \right]^2 \right) \mathrm{Var}(\varepsilon_t).$$

Given parameters' value, we can back out the explicit numerical solution.
This model works:

1) well for consumption volatility

2) bad for output volatility: requires too small TFP volatility

3) bad for hours volatility $\Rightarrow$ way too low.

4) bad for investment volatility $\Rightarrow$ far too low.

But it allows N_t to move with shocks compared to Long–Plosser. While with this set of preference and technology, it cannot generate enough labor and investment volatility relative to output.

5.4 Hansen (1985) with indivisible labor

Labor market phenomena:

1) Hours fluctuate a lot relative to productivity.

2) Micro evidence does not support huge intertemporal substitution of labor at individual level.

Target: get big aggregate hours fluctuations without assuming each person is very willing to shift leisure across time.

Key mechanism:

1) Each individual's labor is a corner-choice:

$$\tilde{n}_t \in \{0, n^*\}.$$

Either work a fixed shift or no work.

2) Lottery over jobs: assigned a job with probability π_t.

3) Complete unemployment insurance: C_t is independent of job status.

Crucial: we can write expected utility clearly at the "representative agent" level.

Thus total labor supply:

$$H_t = \bar{L}\pi_t n^*$$

where $\bar{L}$ is total labor force.

Average labor supply:

$$n_t = \frac{H_t}{\bar{L}} = \pi_t n^*.$$

Representative-agent utility: individual utility is

$$U(C_t) + v(\tilde{n}_t).$$

Given lottery,

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t v(n^*) + (1 - \pi_t)v(0).$$

Let

$$v(0) = 0, \qquad \frac{v(n^*)}{n^*} = -A < 0.$$

Then

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t(-An^*) = U(C_t) - An_t.$$

Suppose

$$U(C_t) = \ln C_t \quad \Rightarrow \quad U_t = \ln C_t - An_t.$$

Hence

$$U_{C,t} = \frac{1}{C_t}, \qquad U_{N,t} = -A, \qquad -U_{N,t} = U_{C,t}W_t$$

Micro-level: no smoothing in labor, either n^* or 0, keep small intertemporal labor substitution.

Macro-level: planner / representative agent can vary π_t a lot, aggregate hours become very responsive.

Thus, we can define the equilibrium conditions for this model:

$$\begin{cases} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} \left(\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta \right) \right], \\ K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1} - C_t, \\ AC_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \qquad \text{key difference} \qquad -U_{N,t} = U_{C,t} W_t = U_{C,t} MPL_t, \\ \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Mechanism

In the indivisible labor model, aggregate hours fluctuate mainly through the employment margin: more people work rather than each individual continuously adjusting hours.

By contrast,in the divisible labor model,

$$TFP \uparrow \Rightarrow \text{demand for labor} \uparrow \Rightarrow W_t \uparrow \Rightarrow N_t \uparrow \Rightarrow 1 - N_t \downarrow$$

$\ln(1 - N_t)$ becomes steeper $\Rightarrow$ the cost of giving up a marginal hour of leisure rises

$$\Rightarrow \quad \text{unwilling to adjust } N_t, \text{ wage moves more} \Rightarrow \text{stiff labor}$$

$$\Rightarrow \quad \text{small hour volatility.} \qquad \text{(each person chooses hours continuously)}$$

Consistency with facts

$$n_t = \frac{H_t}{\bar{L}} = \frac{H_t}{L_t} \cdot \frac{L_t}{\bar{L}} \approx n^*(1 - u_t) = n^* \pi_t,$$

where u_t is the unemployment rate.

$$\frac{H_t}{L_t} = \bar{n}_t \approx n^* \qquad \text{is empirically stable.}$$

*Cyclical movement in total hours comes from (extensive margin) employment, not from hours per worker (intensive margin).

5.5 Other RBC models

1. CRRA utility (King and Rebelo, 1999):

$$U(C_t, L_t) = \ln C_t + A\frac{(1-N_t)^{1-\chi} - 1}{1-\chi}.$$

Investment volatility ↑↑ (almost correct), but underpredicts persistence, over-predicts comovement; consumption & labor are less volatile in model than data.

change preference curvature and more volatile TFP

2. KPR utility (King, Plosser and Rebelo, 1988):

$$U(C_t, L_t) = \frac{[C_t\, v(L_t)]^{1-\sigma} - 1}{1-\sigma}.$$

Non-separable preference between C and N. $U_{C,t}$ depends on L_t, U_{L_t} depends on C_t. Relax the too-tight linkages $\Rightarrow$ improve comovement patterns.

3. GHH utility (Greenwood–Hercowitz–Huffman, 1988, AER)

$$U(C_t, N_t) = \frac{1}{1-\sigma}\left[\left(C_t - A\frac{N_t^{1+\psi}}{1+\psi}\right)^{1-\sigma} - 1\right].$$

Cancel wealth effect, labor supply responds to substitution effect (wage/productivity), $\Rightarrow$ even more volatile N_t.

Since

$$-U_{N,t} = U_{C,t}W_t \quad \Rightarrow \quad AN_t^{\psi} = W_t.$$

4. Capital utilization:

$$Y_t = Z_t(u_t K_{t-1})^{\alpha} N_t^{1-\alpha},$$
$$K_t = I_t + [1-\delta(u_t)]K_{t-1}, \qquad \delta(u) > 0$$

Extra intensive margin for capital $\Rightarrow$ higher investment volatility, higher output volatility.

Lecture 6

Money in Utility Models

6.1 Classical monetary model

1 assume perfect competition & fully flexible prices

2 limited role for money,real and nominal dichotomy:

$$N, C, Y, K, \frac{R}{P}, \frac{M}{P}$$

pinned down without monetary policy. Monetary policy only affects nominal variables.

Household's problem

$$\max_{\{C_t, N_t, B_t\}} E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t)$$

s.t.

$$P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t$$

where

B_t : one-period, nominally riskless discount bonds that mature in one period; bought in t, and pay one unit of money at $t+1$.

Q_t : price of bonds; pay Q_t units today and get 1 unit next period, or, pay 1 unit today and get $\frac{1}{Q_t}$ units,i.e.gross return next period.

T_t : lump-sum taxes (if $T_t < 0$) or dividends (if $T_t > 0$).

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big[U(C_t, N_t) + \lambda_t (B_{t-1} + W_t N_t - T_t - P_t C_t - Q_t B_t) \Big].$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad U_{C_t} - \lambda_t P_t = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N_t} + \lambda_t W_t = 0 \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \quad \Rightarrow \quad -\beta^t \lambda_t Q_t + \beta^{t+1} E_t[\lambda_{t+1}] = 0 \quad \Rightarrow \quad Q_t = \beta E_t\left[\frac{\lambda_{t+1}}{\lambda_t}\right] \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \quad \Rightarrow \quad P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t. \tag{5}$$

From the first two FOCs:

$$-\frac{U_{N_t}}{U_{C_t}} = \frac{W_t}{P_t}.$$

From the bond FOC:

$$Q_t = \beta E_t\left[\frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}}\right].$$

Suppose

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

Then

$$U_{C_t} = C_t^{-\sigma}, \qquad U_{N_t} = -N_t^{\varphi}.$$

So

$$\frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi}.$$

Hence

$$N_t = \left(\frac{W_t/P_t}{C_t^{\sigma}}\right)^{1/\varphi}.$$

$$Q_t = \beta E_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma} \frac{P_t}{P_{t+1}}\right].$$

$$\frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi} \quad \Rightarrow \quad N_t = \left(\frac{W_t/P_t}{C_t^{\sigma}}\right)^{1/\varphi} \quad \Rightarrow \quad \log N_t = \frac{1}{\varphi}\log\left(\frac{W_t}{P_t}\right) - \frac{\sigma}{\varphi}\log C_t \quad \Rightarrow \quad \frac{\partial \log N_t}{\partial \log(W_t/P_t)} = \frac{1}{\varphi}$$

$\frac{1}{\varphi}$: Frisch elasticity (how labor supply responds to real wage), $\varphi \uparrow \Rightarrow$ elasticity $\downarrow$.

Firm's problem

$$\max_{Y_t, N_t} \Pi_t = P_t Y_t - W_t N_t$$

s.t.

$$Y_t = A_t N_t^{1-\alpha}.$$

FOC:

$$\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \qquad : \text{both are real terms.} \tag{4}$$

Market clearing

$$\text{Goods market clearing:} \qquad Y_t = C_t$$

$$\text{Labor market clearing:} \qquad N_t^{(d)} = N_t^{(s)} (\text{implicitly holds})$$

$$\text{Bond market clearing:Hold by Walras Law} \qquad B_t = 0$$

Then we have the system

$$
\begin{cases}
Y_t = C_t, & \text{goods market clearing} \\
T_t = \alpha P_t Y_t, & \text{dividend} \\
Y_t = A_t N_t^{1-\alpha}, & \text{production function} \\
\dfrac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi}, & \text{labor-leisure condition} \\
\dfrac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}, & \text{real wage = MPL} \\
P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t, & \text{budget constraint} \\
Q_t = \beta E_t \left[\left(\dfrac{C_{t+1}}{C_t} \right)^{-\sigma} \dfrac{P_t}{P_{t+1}} \right], & \text{bond Euler equation} \\
\log A_t = (1-\rho_a) \log \bar{A} + \rho_a \log A_{t-1} + \varepsilon_t, & \text{productivity process}
\end{cases}
$$

No monetary policy involved.Denote the real wage as $W_t^r = \dfrac{W_t}{P_t}$.

$$
\Rightarrow \qquad \text{Real equilibrium(in units of goods):} \quad
\begin{cases}
W_t^r = C_t^{\sigma} N_t^{\varphi} & (1) \\
Y_t = A_t N_t^{1-\alpha} & (2) \\
W_t^r = (1-\alpha) A_t N_t^{-\alpha} & (3) \\
Y_t = C_t & (4)
\end{cases}
$$

Four unknowns: $W_t^r,\ C_t,\ N_t,\ Y_t$. Four equations.

At steady state Assume

$$\bar{A} = 1.$$

From (2), (3), and $Y = C$:

$$(1-\alpha)\bar{N}^{-\alpha} = \bar{C}^{\sigma} \bar{N}^{\varphi}.$$

Since

$$\bar{Y} = \bar{N}^{1-\alpha}, \qquad \bar{C} = \bar{Y},$$

we have

$$(1-\alpha)\bar{N}^{-\alpha} = \left(\bar{N}^{1-\alpha}\right)^{\sigma} \bar{N}^{\varphi} = \bar{N}^{\sigma(1-\alpha)+\varphi}.$$

Hence

$$1-\alpha = \bar{N}^{\sigma(1-\alpha)+\varphi+\alpha},$$

so

$$\bar{N} = (1-\alpha)^{\frac{1}{\sigma(1-\alpha)+\varphi+\alpha}}.$$

Thus

$$\bar{Y} = \bar{N}^{1-\alpha} = (1-\alpha)^{\frac{1-\alpha}{\sigma(1-\alpha)+\varphi+\alpha}}, \qquad \bar{C} = \bar{Y}.$$

and

$$\bar{W}^r = \bar{C}^{\sigma} \bar{N}^{\varphi} = (1-\alpha)\bar{N}^{-\alpha}.$$

Log-linearization

$$\hat{w}_t^r = \sigma \hat{c}_t + \varphi \hat{n}_t$$
$$\hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t$$
$$\hat{w}_t^r = \hat{a}_t - \alpha \hat{n}_t$$
$$\hat{y}_t = \hat{c}_t.$$

Only one shock $\hat{a}_t$, no predetermined variable. Assume

$$\hat{y}_t = \psi_{ya}\hat{a}_t, \qquad \hat{n}_t = \psi_{na}\hat{a}_t, \qquad \hat{w}_t^r = \psi_{wa}\,\hat{a}_t$$

Then

$$\sigma\hat{a}_t + \sigma(1-\alpha)\hat{n}_t + \varphi\hat{n}_t = \hat{a}_t - \alpha\hat{n}_t$$

$$\Rightarrow \quad \hat{c}_t = \hat{y}_t = \frac{1+\varphi}{\sigma(1-\alpha)+\varphi+\alpha}\,\hat{a}_t \qquad \text{coefficient} > 0$$

$$\hat{w}_t^r = \frac{\varphi+\sigma}{\sigma(1-\alpha)+\varphi+\alpha}\,\hat{a}_t \qquad \text{coefficient} > 0$$

$$\hat{n}_t = \frac{1-\sigma}{\sigma(1-\alpha)+\varphi+\alpha}\,\hat{a}_t \qquad \text{coefficient is ambiguous, depends on } \sigma$$

Income effect dominates if $\sigma > 1$: $\dfrac{\partial \hat{n}_t}{\partial \hat{a}_t} < 0.$

Substitution effect dominates if $\sigma < 1$: $\dfrac{\partial \hat{n}_t}{\partial \hat{a}_t} > 0.$

Real interest rate Bond Euler:

$$Q_t = \beta E_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma}\frac{1}{\Pi_{t+1}}\right]. \tag{*}$$

$\left(\dfrac{C_{t+1}}{C_t}\right)^{-\sigma}$: real stochastic discount factor, $\dfrac{1}{\Pi_{t+1}}$: converts $t+1$ nominal payoff into real terms.

Let

$$Q_t = e^{-i_t}, \qquad \frac{P_{t+1}}{P_t} = \Pi_{t+1} \text{ inflation}, \qquad \pi_{t+1} = \log\Pi_{t+1} = \log P_{t+1} - \log P_t, \qquad \beta = e^{-\rho}$$

Define real interest rate: $\quad r_t = i_t - E_t\pi_{t+1}$

$i_t = -\log Q_t \qquad$ log nominal interest rate

$\rho = -\log\beta \qquad$ rate of time preference

Define new deviation: $\quad \hat{i}_t = i_t - \bar{i} \qquad \hat{r}_t = r_t - \bar{r} \qquad \bar{\pi} = \log\bar{P} - \log\bar{P} = 0$

$$\begin{cases} \left(\dfrac{C_{t+1}}{C_t}\right)^{-\sigma} = e^{-\sigma\log C_{t+1}} / e^{-\sigma\log C_t} = e^{-\sigma(\log C_{t+1} - \log C_t)}, \\[2ex] \dfrac{1}{\Pi_{t+1}} = \dfrac{P_t}{P_{t+1}} = e^{\log P_t - \log P_{t+1}} = e^{-\pi_{t+1}}. \end{cases}$$

$$\Rightarrow \quad e^{-i_t} = e^{-\rho} E_t\left[e^{-\sigma(\hat{c}_{t+1}-\hat{c}_t)-\pi_{t+1}}\right] \tag{**}$$

$$-i_t = -\rho + E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}] \tag{1}$$

$$\left.\begin{aligned} -\bar{i} &= -\rho + E_t\left[-\sigma(\log \bar{C} - \log \bar{C}) - \bar{\pi}\right], \\ \bar{\pi} &= \log \bar{P} - \log \bar{P} = 0 \end{aligned}\right\} \Rightarrow \bar{i} = \rho \tag{2}$$

$$(1) + (2) \quad \Rightarrow \quad -\hat{i}_t + \bar{i} = E_t\left[-\sigma\left(\log C_{t+1} - \log C_t\right) - \pi_{t+1}\right].$$

$$\begin{cases} \hat{c}_{t+1} = \log C_{t+1} - \log \bar{C}, \\ \hat{c}_t = \log C_t - \log \bar{C} \end{cases} \quad \Rightarrow \quad \hat{c}_{t+1} - \hat{c}_t = \log C_{t+1} - \log C_t.$$

$$\Rightarrow \quad -\hat{i}_t = E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}],$$

$$\Rightarrow \quad -\hat{i}_t = -\sigma E_t \Delta \hat{c}_{t+1} - E_t[\pi_{t+1}],$$

$$\Rightarrow \quad \sigma E_t \Delta \hat{c}_{t+1} = \hat{i}_t - E_t[\pi_{t+1}], \tag{***}$$

$$\hat{\pi}_{t+1} = \pi_{t+1} - \bar{\pi} = \pi_{t+1},$$

$$\Rightarrow \quad \hat{i}_t - E_t[\hat{\pi}_{t+1}] = \sigma E_t \Delta \hat{c}_{t+1}.$$

$$\text{Since } r_t = i_t - E_t[\pi_{t+1}], \tag{3}$$

$$\bar{r} = \bar{i} - \bar{\pi} = \bar{i}, \tag{4}$$

$$(3) - (4) \;\Rightarrow\; \hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}]$$

$$\text{Plug in } (***) \;\Rightarrow\; \hat{r}_t = \sigma E_t \Delta \hat{c}_{t+1}$$

$$\hat{c}_{t+1} = \hat{y}_{t+1} = \psi_{ya} \hat{a}_{t+1},$$

$$\Rightarrow \hat{r}_t = \sigma \psi_{ya} E_t[\Delta \hat{a}_{t+1}]$$

$$\text{If the shock is transitory / mean reverting} \quad \Rightarrow \quad E_t[\Delta \hat{a}_{t+1}] < 0 \quad \Rightarrow \quad r_t < 0.$$

A positive transitory technology shock causes the current real interest rate to fall below its steady-state level.

Intuition: $a_t \uparrow$ gives a favorable temporary income shock $\Rightarrow$ households smooth consumption by saving $\Rightarrow$ bond demand rises $B_t^d \uparrow \Rightarrow$ bond price rises $Q_t \uparrow \Rightarrow$ real interest rate falls $r_t \downarrow$.

Dichotomy

- r_t pinned down by real side, money growth does not affect r_t.
- No policy is the best since only real variables enter utility.
- Does not match empirics:data say real variable move after monetary shocks and nominall variables respond slugg.

6.2 Money in utility model

- Make nominal terms affect real terms.
- Explain why agents hold money which is inferior to bond. (Money has no return but affects utility.)

Household's Problem

$$\max_{\{C_t, M_t, N_t\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, \mathbb{M}_t, N_t)$$

$$\text{s.t.} \quad P_t C_t + Q_t B_t + M_t \leq B_{t-1} + M_{t-1} + W_t N_t + T_t$$

Let $\mathbb{M}_t = \dfrac{M_t}{P_t}$, the real money balance/supply.

Financial wealth before decisions(predetermined variable):

$$A_t = B_{t-1} + M_{t-1}$$

$$\Lambda_{t,T} = Q_t \cdot Q_{t+1} \cdots Q_T$$

B.C.:

$$P_t C_t + Q_t A_{t+1} + (1 - Q_t) M_t = A_t + W_t N_t + T_t$$

Notes 1:

$$i_t : \text{ continuously compounded nominal rate}$$

$$Q_t : \text{ price today } Q_t, \text{ payoff tomorrow } 1$$

Nominal interest rate:

$$1 - Q_t = 1 - e^{-i_t} \approx i_t$$

$1 - Q_t$: the opportunity cost of holding one's financial wealth in money rather than in interest-bearing bonds; a high i_t thus means contractionary monetary policy.

Notes:

$$\Lambda_{t,T} : \text{ state-contingent discount factor from } t \text{ to } T$$

$$\frac{A_T}{P_T} : \text{ terminal real wealth}$$

No-Ponzi scheme:

$$\lim_{T \to \infty} E_t \left[\Lambda_{t,T} \frac{A_T}{P_T} \right] \geq 0, \qquad \forall t$$

Notes:

cannot end up with infinitely bad wealth

(borrow a lot $\Rightarrow$ roll-over)

TVC:

$$\lim_{T \to \infty} E_t \left[\Lambda_{t,T} \frac{A_T}{P_T} \right] = 0$$

Note:

no resources on table when world ends

All financial assets A_t yield gross nominal return $Q_t^{-1} = e^{i_t}$.

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[U(C_t, \mathbb{M}_t, N_t) + \lambda_t \big(A_t + W_t N_t + T_t - P_t C_t - Q_t A_{t+1} - (1 - Q_t) M_t \big) \right].$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial M_t} = 0 \quad \Rightarrow \quad U_{\mathbb{M}_t} \frac{1}{P_t} - \lambda_t (1 - Q_t) = 0$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad U_{C,t} = \lambda_t P_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t}.$$

Hence

$$\frac{U_{\mathbb{M}_t}}{U_{C_t}} = 1 - Q_t = 1 - e^{-i_t}. \tag{5}$$

6.3 Separable utility case (still dichotomy)

Assume

$$U(C_t, \mathbb{M}_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} + \frac{(\mathbb{M}_t)^{1-\nu}}{1-\nu} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

Neither U_{C_t} nor U_{N_t} depends on $\mathbb{M}_t \Rightarrow$ real equilibrium not affected(same as before).
But we can derive money demand:

$$U_{\mathbb{M}_t} = (\mathbb{M}_t)^{-\nu}, \qquad U_{C_t} = C_t^{-\sigma}.$$

$$\frac{U_{\mathbb{M},t}}{U_{C,t}} = 1 - e^{-i_t} \quad \Rightarrow \quad \frac{\mathbb{M}_t^{-\nu}}{C_t^{-\sigma}} = 1 - e^{-i_t}$$

$$\Rightarrow \quad \mathbb{M}_t = C_t^{\sigma/\nu} \left[1 - e^{-i_t} \right]^{-1/\nu}$$

$$\Rightarrow \quad \log M_t - \log P_t = \frac{\sigma}{\nu} \log C_t - \frac{1}{\nu} \log(1 - e^{-i_t})$$

$$\hat{\mathbb{M}}_t \approx \frac{\sigma}{\nu} \hat{c}_t - \frac{1}{\nu \left(e^{\bar{i}} - 1 \right)} \hat{i}_t.$$

math notes:

$$\text{Let } f(i) = \log(1 - e^{-i}),$$

$$f'(i) = \frac{e^{-i}}{1 - e^{-i}} = \frac{1}{e^i - 1},$$

$$f''(i) = -\frac{e^i}{(e^i - 1)^2},$$

$$\log(1 - e^{-i_t}) \approx \log(1 - e^{-\bar{i}}) + \frac{1}{e^i - 1} (i_t - \bar{i}).$$

$\log(1-e^{-\bar{i}})$ is constant, so it cancels out in derivation.

If $\sigma = \nu$, define

$$\eta \equiv \frac{1}{\nu\left(e^{\bar{i}}-1\right)},$$

then approximately

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t.$$

η: semi-elasticity of real money demand with respect to the nominal interest rate.

$\hat{m}_t - \hat{p}_t$: real money balances are determined by $\hat{y}_t$ and $\hat{i}_t$.

$\hat{y}_t$: determined by the real-side equilibrium, so monetary policy does not affect real activity in the separable case.

6.4 Inseparable utility case

Utility:

$$U(C_t, \mathbb{M}_t, N_t) = \frac{X_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

$$\text{where,}\;\; X_t = \begin{cases} \left[(1-\theta)C_t^{1-\nu} + \theta \mathbb{M}_t^{1-\nu}\right]^{\frac{1}{1-\nu}}, & \nu \neq 1, \\ C_t^{1-\theta}\mathbb{M}_t^{\theta}, & \nu = 1. \end{cases}$$

$$\text{Elasticity of substitution between } C_t \text{ and } \mathbb{M}_t \text{ is } \frac{1}{\nu}$$

$\theta \in (0,1)$ weight on money inside the CES aggregator.

Derivatives when $\nu \neq 1$

For $\nu \neq 1$,

$$X_t = \left[(1-\theta)C_t^{1-\nu} + \theta \mathbb{M}_t^{1-\nu}\right]^{\frac{1}{1-\nu}}.$$

Then

$$U_{C,t} = X_t^{-\sigma} \cdot \frac{1}{1-\nu} \cdot X_t^{\nu} \cdot (1-\theta)(1-\nu)C_t^{-\nu}$$

so

$$U_{C,t} = (1-\theta)X_t^{\nu-\sigma}C_t^{-\nu}.$$

Similarly,

$$U_{\mathbb{M},t} = X_t^{-\sigma} \cdot \frac{1}{1-\nu} \cdot X_t^{\nu} \cdot \theta(1-\nu)\mathbb{M}_t^{-\nu}$$

so

$$U_{\mathbb{M},t} = \theta X_t^{\nu-\sigma}\mathbb{M}_t^{-\nu}.$$

Also,

$$U_{N,t} = -N_t^{\varphi}.$$

Marginal rate of substitution and elasticity Define

$$X_C = (1-\theta)X^{\nu}C^{-\nu}, \qquad X_{\mathbb{M}} = \theta X^{\nu}\mathbb{M}^{-\nu}.$$

Then

$$MRS_{C,\mathbb{M}} = \frac{U_C}{U_{\mathbb{M}}} = \frac{X_C}{X_{\mathbb{M}}} = \frac{1-\theta}{\theta}\left(\frac{\mathbb{M}}{C}\right)^{\nu}.$$

Hence

$$\varepsilon = \frac{d\ln(\mathbb{M}/C)}{d\ln MRS_{C,\mathbb{M}}} = \frac{1}{\nu}.$$

Thus,

$$\nu > 1 \;\Rightarrow\; \varepsilon < 1 \quad \text{(complements)},$$
$$\nu < 1 \;\Rightarrow\; \varepsilon > 1 \quad \text{(substitutes)}.$$

FOCs Labor supply condition:

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t}.$$

Substituting the derivatives,

$$\begin{cases} \dfrac{W_t}{P_t} = \dfrac{1}{1-\theta}N_t^{\varphi}X_t^{\sigma-\nu}C_t^{\nu}, \\[2ex] Q_t = \beta E_t\left[\dfrac{U_{C,t+1}}{U_{C,t}}\dfrac{P_t}{P_{t+1}}\right] = \beta E_t\left[\left(\dfrac{C_{t+1}}{C_t}\right)^{-\nu}\left(\dfrac{X_{t+1}}{X_t}\right)^{\nu-\sigma}\dfrac{P_t}{P_{t+1}}\right]. \end{cases}$$

X_t depends on $\mathbb{M}_t$, thus money is non-neutral.

Money FOC:

$$\frac{U_{\mathbb{M},t}}{U_{C,t}} = 1 - e^{-i_t}.$$

So

$$\frac{\theta}{1-\theta}C_t^{\nu}\mathbb{M}_t^{-\nu} = 1 - e^{-i_t}.$$

Solving for real money balances,

$$\mathbb{M}_t = C_t\left(1 - e^{-i_t}\right)^{-1/\nu}\left(\frac{\theta}{1-\theta}\right)^{1/\nu}.$$

When $\nu = \sigma$ and we drop θ, the utility function reduces to the separable case, so the dichotomy holds.

When $\nu \neq \sigma$, we have the general (non-separable) case.

1) Labor supply and Euler equation are affected by the level of real balances through X_t,
2) Real balance depends on nominal interest rate.

Non-neutral monetary policy: Monetary policy $\Rightarrow$ nominal rate i_t $\Rightarrow$ real balance $\mathbb{M}$ $\Rightarrow$ labor supply and output.

Log linearize: $$\mathbb{M}_t = C_t(1-e^{-i_t})^{-1/\nu}\left(\frac{\theta}{1-\theta}\right)^{1/\nu} \;\Rightarrow\; \hat{m}_t - \hat{p}_t = \hat{c}_t - \eta\hat{i}_t \qquad \cdots (*)$$

$$\frac{W_t}{P_t} = \frac{1}{1-\theta} N_t^{\varphi} X_t^{\sigma-\nu} C_t^{\nu} \Rightarrow \hat{w}_t - \hat{p}_t = \sigma\hat{c}_t + \varphi\hat{n}_t + (\nu-\sigma)(\hat{c}_t - \hat{x}_t) \qquad \cdots (**)$$

$$X_t^{1-\nu} = (1-\theta)C_t^{1-\nu} + \theta\mathbb{M}^{1-\nu}$$

$$\text{S.S.:} \quad (1-\theta)\bar{C}^{1-\nu} = \bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu}$$

$$\Rightarrow \quad \bar{X}^{1-\nu}\hat{x}_t = (1-\theta)\bar{C}^{1-\nu}\hat{c}_t + \theta\bar{\mathbb{M}}^{1-\nu}(\hat{m}_t - \hat{p}_t).$$

Using $(1-\theta)\bar{C}^{1-\nu} = \bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu}$,

$$\bar{X}^{1-\nu}\hat{x}_t = \left[\bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu}\right]\hat{c}_t + \theta\bar{\mathbb{M}}^{1-\nu}(\hat{m}_t - \hat{p}_t).$$

$$\Rightarrow \bar{X}^{1-\nu}(\hat{c}_t - \hat{x}_t) = \theta\bar{\mathbb{M}}^{1-\nu}\left[\hat{c}_t - (\hat{m}_t - \hat{p}_t)\right].$$

$$\Rightarrow \hat{c}_t - \hat{x}_t = \underbrace{\theta\left(\frac{\bar{\mathbb{M}}}{\bar{X}}\right)^{1-\nu}}_{\chi}\left[\hat{c}_t - (\hat{m}_t - \hat{p}_t)\right]. \qquad (***)$$

Define $\left(\frac{\bar{M}}{\bar{P}}\right) \Big/ \bar{C} = k_m$

k_m : inverse consumption velocity, low k_m high velocity

Let
$$\chi = \theta\frac{\left(\frac{\bar{M}}{\bar{P}}\right)^{1-\nu}}{\bar{X}^{1-\nu}} = \theta\frac{\left(\frac{\bar{M}}{\bar{P}}\right)^{1-\nu}}{(1-\theta)\bar{C}^{1-\nu} + \theta\left(\frac{\bar{M}}{\bar{P}}\right)^{1-\nu}} = \frac{\theta\left(\left(\frac{\bar{M}}{\bar{P}}\right)/\bar{C}\right)^{1-\nu}}{1-\theta+\theta\left(\left(\frac{\bar{M}}{\bar{P}}\right)/\bar{C}\right)^{1-\nu}}$$

$$= \frac{\theta k_m^{1-\nu}}{1-\theta+\theta k_m^{1-\nu}}$$

χ : steady-state expenditure share of money inside the CES aggregator

$$\mathbb{M}_t = C_t(1-e^{-i_t})^{-1/\nu}\left(\frac{\theta}{1-\theta}\right)^{1/\nu} \Rightarrow \left(\frac{\bar{M}}{\bar{P}}\right)\Big/\bar{C} = (1-\beta)^{-1/\nu}\left(\frac{\theta}{1-\theta}\right)^{1/\nu}$$

$$\Rightarrow \quad k_m = \left[\frac{\theta}{(1-\beta)(1-\theta)}\right]^{1/\nu}$$

k_m increases with money weight θ, $\beta \uparrow \Rightarrow$ opportunity cost of holding money $\downarrow$

Plug $(***)$ into $(**)$, we have

$$\hat{w}_t - \hat{p}_t = \sigma\hat{c}_t + \varphi\hat{n}_t + (\nu-\sigma)\chi\Big[\hat{c}_t - (\hat{m}_t - \hat{p}_t)\Big]$$

Plug $(*)$ into the above equation:

$$\hat{w}_t - \hat{p}_t = \sigma\hat{c}_t + \varphi\hat{n}_t + (\nu-\sigma)\chi\left[\hat{c}_t - \hat{y}_t + \eta\hat{i}_t\right]$$

$$\Rightarrow \quad \hat{w}_t - \hat{p}_t = \sigma\hat{c}_t + \varphi\hat{n}_t + \omega\hat{i}_t$$

$$\text{where} \qquad \omega = (\nu-\sigma)\chi\eta$$

ω : like nominal interest rate and labor supply

$$\chi = \frac{\theta^{1/\nu}(1-\beta)^{1-\frac{1}{\nu}}}{(1-\theta)^{1/\nu} + \theta^{1/\nu}(1-\beta)^{1-\frac{1}{\nu}}} = \frac{\left(\frac{\theta}{1-\theta}\right)^{1/\nu}(1-\beta)^{1-\frac{1}{\nu}}}{1+\left(\frac{\theta}{1-\theta}\right)^{1/\nu}(1-\beta)^{1-\frac{1}{\nu}}}$$

$$\Rightarrow \quad \chi = \frac{k_m(1-\beta)}{1+k_m(1-\beta)}$$

$$k_m(1-\beta) = \left(\frac{\theta}{1-\theta}\right)^{1/\nu}(1-\beta)^{1-\frac{1}{\nu}}$$

$$\Rightarrow \quad \omega = \chi\eta(\nu-\sigma) = \frac{k_m(1-\beta)}{1+k_m(1-\beta)}\cdot\frac{1}{\nu\left(e^{\bar{i}}-1\right)}\cdot(\nu-\sigma)$$

$$e^{\bar{i}} = \beta^{-1} \;\Rightarrow\; e^{\bar{i}} - 1 = \frac{1-\beta}{\beta}$$

$$\Rightarrow \quad \omega = \frac{k_m\beta\,(\nu-\sigma)}{\nu\left[1+k_m(1-\beta)\right]}$$

$$\operatorname{sign}(\omega) = \operatorname{sign}\left(1-\frac{\sigma}{\nu}\right) \qquad \Rightarrow \qquad \omega > 0 \iff \nu > \sigma$$

$$\eta = \frac{1}{\nu\left(e^{\bar{i}}-1\right)}$$

$$\text{for small } \bar{i}, \qquad e^{\bar{i}} - 1 \approx \bar{i} \;\Rightarrow\; \eta \approx \frac{1}{\nu\bar{i}} \;\Rightarrow\; \nu \approx \frac{1}{\eta\bar{i}}$$

$$\text{since } \bar{i} \text{ is small,} \;\; \nu = \frac{1}{\eta\bar{i}} \gg \sigma \;\Rightarrow\; \omega > 0$$

$$\begin{cases} \hat{w}_t - \hat{p}_t = \sigma\hat{c}_t + \varphi\hat{n}_t + \omega\hat{i}_t, \\ \hat{m}_t - \hat{p}_t = \hat{y}_t - \eta\hat{i}_t, \end{cases} \qquad \hat{c}_t - \hat{x}_t = \chi\left[\hat{c}_t - (\hat{m}_t - \hat{p}_t)\right]$$

$$\hat{i}_t\uparrow,\ \eta > 0 \;\Rightarrow\; \hat{m}_t - \hat{p}_t\downarrow \Rightarrow\; \hat{x}_t\downarrow \Rightarrow\; U_{C,t}\downarrow \Rightarrow\; U_{CM} > 0$$

Money and consumption are complements.

$$\hat{i}_t\uparrow,\ \eta > 0 \;\Rightarrow\; \hat{n}_t\downarrow \quad \text{given real wage and consumption}$$

Money and labor move in the same direction: Less liquidity, $U_{C,t}\downarrow$, consumption is less valuable, and the benefit of working is lower.

Log-linearize:

$$Q_t = \beta E_t\left[\frac{U_{C,t+1}}{U_{C,t}}\frac{P_t}{P_{t+1}}\right] = \beta E_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\nu}\left(\frac{X_{t+1}}{X_t}\right)^{\nu-\sigma}\frac{P_t}{P_{t+1}}\right]$$

$$Q_t = e^{-i_t}, \qquad \beta = e^{-\rho}, \qquad \Pi_{t+1} = \frac{P_{t+1}}{P_t}, \qquad \pi_{t+1} = \log \Pi_{t+1} = \log P_{t+1} - \log P_t$$

$$\Rightarrow \quad e^{-i_t} = e^{-\rho}E_t\left[e^{-\nu(\log C_{t+1}-\log C_t)+(\nu-\sigma)(\log X_{t+1}-\log X_t)-\pi_{t+1}}\right]$$

$$\Rightarrow \quad \begin{cases} -i_t = -\rho + E_t[-\nu(\log C_{t+1}-\log C_t) + (\nu-\sigma)(\log X_{t+1}-\log X_t) - \pi_{t+1}], \\ -\bar{i} = -\rho + E_t\left[-\nu(\log \bar{C} - \log \bar{C}) + (\nu-\sigma)(\log \bar{X} - \log \bar{X}) - \bar{\pi}\right] \end{cases}$$

$$\bar{\pi} = 0$$

$$\Rightarrow \quad \hat{i}_t = i_t - \bar{i} = E_t[\nu(\hat{c}_{t+1}-\hat{c}_t) - (\nu-\sigma)(\hat{x}_{t+1}-\hat{x}_t) + \pi_{t+1}]$$

$$\Rightarrow \quad \hat{i}_t - E_t\pi_{t+1} = \nu E_t(\hat{c}_{t+1}-\hat{c}_t) - (\nu-\sigma)E_t(\hat{x}_{t+1}-\hat{x}_t)$$

$$\text{since} \qquad \hat{c}_t - \hat{x}_t = \chi\left[\hat{c}_t - (\hat{m}_t - \hat{p}_t)\right]$$

$$\hat{m}_t - \hat{p}_t = \hat{c}_t - \eta\hat{i}_t \qquad \Rightarrow \qquad \hat{c}_t - \hat{x}_t = \chi\eta\hat{i}_t \qquad \Rightarrow \qquad \hat{x}_t = \hat{c}_t - \chi\eta\hat{i}_t$$

$$\Rightarrow \quad \hat{i}_t - E_t\pi_{t+1} = \nu E_t\Delta\hat{c}_{t+1} - (\nu-\sigma)E_t\Delta\hat{c}_{t+1} + (\nu-\sigma)\chi\eta\, E_t\Delta\hat{i}_{t+1}$$

$$\Rightarrow \quad \hat{i}_t - E_t\pi_{t+1} = \sigma E_t\hat{c}_{t+1} - \sigma\hat{c}_t + \omega E_t\Delta\hat{i}_{t+1}$$
$$\Rightarrow \quad \sigma E_t\Delta\hat{c}_{t+1} = \hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}$$

$$\Rightarrow \quad \sigma E_t\hat{c}_{t+1} - \sigma\hat{c}_t = \hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}$$

$$\Rightarrow \quad \hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}\right] \qquad \cdots (*)$$

$$\text{where} \qquad \omega = (\nu-\sigma)\chi\eta, \qquad \chi = \frac{k_m(1-\beta)}{1+k_m(1-\beta)}, \qquad k_m = \frac{\bar{M}/\bar{P}}{\bar{C}}, \qquad \eta = \frac{1}{\nu(e^{\bar{i}}-1)}.$$

In $(*)$ we apply market clearing $\hat{c}_t = \hat{y}_t$

households smooth consumption $\longrightarrow$ how responsive current spending is to intertemporal incentive

$$\Rightarrow \quad \hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}\right]$$

current demand/output
= expected future output/demand − IES × expected real interest rate − forward guidance

For $\nu > \sigma \Rightarrow \omega > 0$. When $E_t \Delta \hat{i}_{t+1} \uparrow$, $\hat{c}_t \uparrow$. forward guidance

If $E_t \Delta \hat{i}_{t+1} \uparrow$, then future money holdings become less attractive, future liquidity services decline, and households shift some consumption from the future to today; hence $\hat{c}_t \uparrow$.

$$\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}, \qquad Y_t = A_t N_t^{1-\alpha}$$

$$\hat{w}_t - \hat{p}_t = \hat{a}_t - \alpha \hat{n}_t, \qquad \hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t$$

$$\Rightarrow \quad \hat{n}_t = \frac{1}{1-\alpha}(\hat{y}_t - \hat{a}_t)$$

$$\Rightarrow \quad \hat{w}_t - \hat{p}_t = \hat{y}_t - \hat{n}_t$$

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow \quad \hat{y}_t - \hat{n}_t = \sigma \hat{y}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow \quad (1-\sigma)\hat{y}_t - (1+\varphi)\hat{n}_t = \omega \hat{i}_t$$

$$\Rightarrow \quad \hat{n}_t = \frac{1-\sigma}{1+\varphi}\hat{y}_t - \frac{\omega}{1+\varphi}\hat{i}_t$$

$$\hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t$$

$$\Rightarrow \quad \hat{y}_t = \hat{a}_t + (1-\alpha)\left[\frac{1-\sigma}{1+\varphi}\hat{y}_t - \frac{\omega}{1+\varphi}\hat{i}_t\right]$$

$$\Rightarrow \quad \hat{y}_t = \frac{1+\varphi}{\alpha(1-\sigma)+\sigma+\varphi}\hat{a}_t - \frac{(1-\alpha)\omega}{\alpha(1-\sigma)+\sigma+\varphi}\hat{i}_t \qquad (1)$$

$$\Rightarrow \quad \hat{y}_t - \hat{n}_t = \sigma \hat{y}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow \quad (1-\sigma)\hat{y}_t - (1+\varphi)\hat{n}_t = \omega \hat{i}_t$$

$$\Rightarrow \quad \hat{n}_t = \frac{1-\sigma}{1+\varphi}\hat{y}_t - \frac{\omega}{1+\varphi}\hat{i}_t$$

$$\hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t$$

$$\Rightarrow \quad \hat{y}_t = \hat{a}_t + (1-\alpha)\left[\frac{1-\sigma}{1+\varphi}\hat{y}_t - \frac{\omega}{1+\varphi}\hat{i}_t\right]$$

$$\Rightarrow \quad \hat{y}_t = \frac{1+\varphi}{\alpha(1-\sigma)+\sigma+\varphi}\hat{a}_t - \frac{(1-\alpha)\omega}{\alpha(1-\sigma)+\sigma+\varphi}\hat{i}_t \qquad (1)$$

$\hat{y}_t$: interest rate $\rightarrow$ real activity $\rightarrow$ transmission coefficient

comes from money-in-the-utility non-separability.

Money is non-neutral (interest rate / monetary policy affects real output)

But the function alone cannot pin down it; we need a monetary policy rule.

Write the system of equations in terms of $(\hat{y}_t, \pi_t, \hat{i}_t)$.

$$\text{Euler equation:} \qquad \hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}\right] \tag{2}$$

Also known as IS equation

$$\text{Monetary policy rule:} \qquad \hat{i}_t = \phi_\pi \pi_t + v_t \tag{3}$$

$$\text{where} \qquad v_t = \rho_v v_{t-1} + \varepsilon_t^v \tag{4}$$

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a \tag{5}$$

Combine (1)–(5), use undetermined coefficients:

$$\pi_t = A_\pi \hat{a}_t + B_\pi v_t, \qquad \hat{i}_t = A_i \hat{a}_t + B_i v_t, \qquad \hat{y}_t = A_y \hat{a}_t + B_y v_t.$$

$$\Rightarrow \begin{cases} \pi_t = -\dfrac{\sigma(1-\rho_a)\psi_{ya}}{\phi_\pi(1+\omega\psi)(1-\theta\rho_a)}\hat{a}_t - \dfrac{1+(1-\rho_v)\omega\psi}{\phi_\pi(1+\omega\psi)(1-\theta\rho_v)}v_t, \\ \hat{i}_t = -\dfrac{\sigma(1-\rho_a)\psi_{ya}}{(1+\omega\psi)(1-\theta\rho_a)}\hat{a}_t - \dfrac{\rho_v}{\phi_n(1+\omega\psi)(1-\theta\rho_v)}v_t, \\ \hat{y}_t = \psi_{ya}\left(1 + \dfrac{\sigma(1-\rho_a)\psi_{yi}}{(1+\omega\psi)(1-\theta\rho_a)}\right)\hat{a}_t + \dfrac{\rho_v\psi_{yi}}{\phi_\pi(1+\omega\psi)(1-\theta\rho_v)}v_t. \end{cases}$$

$$\text{where} \qquad \theta = \frac{1+\omega\varphi\phi_\pi}{(1+\omega\varphi)\phi_\pi}, \qquad \psi = \frac{\alpha+\varphi}{\sigma(1-\alpha)+\alpha+\varphi}.$$

Intuition: ψ is a summary slope coming from the real-side block, capturing how strongly real activities / labor market tightness maps into wages and output in the log-linear world.

θ is a dynamic feedback object mixing the Taylor coefficient ϕ_π, the forward-guidance strength ω, and the real-side slope.
When $\omega = 0$, θ collapses to $1/\phi_\pi$, which is the usual policy-feedback scale.
When $\omega \neq 0$, θ changes because expected changes in it enter the IS equation.

Impact of the monetary policy

Conditional on an exogenous MP shock ε_t^v to v_t : $\quad \hat{y}_t = \psi_{ya}\hat{a}_t - \psi_{yi}\hat{i}_t \quad \Rightarrow \quad \frac{d\hat{y}_t}{d\hat{i}_t} = -\psi_{yi}.$

$$\text{Assume} \qquad \sigma = \phi = 1, \qquad \alpha = \frac{1}{3} \qquad \Rightarrow \qquad \psi_{yi} = \frac{\omega}{3}.$$

$$\omega = \frac{k_m \beta \left(1 - \frac{\sigma}{\nu}\right)}{1 + k_m(1-\beta)} \approx k_m \qquad \text{with } \nu \text{ large and } \beta \approx 1.$$

$$\Rightarrow \qquad \psi_{yi} \approx \frac{k_m}{3}.$$

$$\text{Recall} \qquad k_m = \frac{\bar{M}/\bar{P}}{\bar{C}},$$

how much liquidity stock households want to hold per unit of spending(cash buffer relative to spendi

Using M1, the money with highest liquidity: $\quad k_m \approx 0.3 \implies \psi_{yi} \approx 0.1.$

If i_t increases by 1% annually (0.25% quarterly), y_t decreases by 0.025% quarterly.

Using M2, the money with highest liquidity + 2nd highest liquidity: $\quad k_m \approx 3 \implies \psi_{yi} \approx 1.$

If i_t increases by 1% annually, y_t decreases by 1% annually.

* If money is quantitatively large, MIU channel becomes stronger.

Deficiency of MIU model

$$\frac{d\pi_t}{d\hat{i}_t} = \frac{d\pi_t/dv_t}{d\hat{i}_t/dv_t} = \left(1 + (1-\rho_v)\omega\psi\right)\rho_v^{-1} > 0,$$

$$\frac{d\hat{r}_t}{d\hat{i}_t} = 1 - \frac{dE_t(\pi_{t+1})/dv_t}{d\hat{i}_t/dv_t} = -(1-\rho_v)\omega\psi < 0.$$

A monetary policy shock that raises the nominal interest rate and lowers output also results in:
(i) inflation tending to increase;
(ii) the real interest rate tending to decrease

$$\left(r_t = i_t - E_t\pi_{t+1}\right).$$

This contradicts the data: contractionary monetary policy shocks raise real interest rates and lower inflation.
In the data, money is neutral in the long run once prices adjust, but non-neutral in the short run. In the model, long-run and short-run effects are the same.

Uniqueness of the nominal system

In a flexible-price (classical) economy, the real side pins down the real interest rate $\hat{r}_t$. The question is: what must monetary policy do to ensure that inflation π_t — and hence the price level p_t — is uniquely determined?
The Fisher relation is

$$\hat{i}_t = E_t[\pi_{t+1}] + \hat{r}_t, \qquad \Rightarrow \qquad E_t[\pi_{t+1}] = \hat{i}_t - \hat{r}_t.$$

Since $\hat{r}_t$ is pinned down by the real economy, if the central bank exogenously sets $\hat{i}_t$ as some stationary process, or simply sets $\hat{i}_t = 0$, then the Fisher equation pins down expected inflation only, not realized inflation.
Therefore, monetary policy must do more than fix the nominal interest rate exogenously: it must provide a rule that anchors the nominal system and uniquely determines inflation.

$$\Rightarrow \qquad E_t[\pi_{t+1}] = -\hat{r}_t \qquad \Rightarrow \qquad \pi_{t+1} = -\hat{r}_t + \zeta_{t+1}, \qquad E_t[\zeta_{t+1}] = 0.$$

Any zero-mean innovation can be added without violating Fisher equation $\Rightarrow$ indeterminacy.

With an exogenous $\hat{i}_t$, there are infinitely many inflation paths consistent with equilibrium.

$\Rightarrow$ sunspot driven (non-fundamental factors).

Since $\hat{p}_t = \pi_t + \hat{p}_{t-1}$, π_t is indetermined $\Rightarrow$ $\hat{p}_t$ is also indetermined (not unique).

Other nominal variables are also indetermined:

$$\hat{m}_t = \hat{p}_t + \hat{y}_t - \eta \hat{i}_t \qquad \Rightarrow \qquad \hat{m}_t \text{ indetermined} \qquad \text{(nominal money)}$$

$$\hat{w}_t = \hat{w}_t^r + \hat{p}_t \qquad \Rightarrow \qquad \hat{W}_t \text{ indetermined} \qquad \text{(nominal wage)}$$

At steady state,

$$P_t = \bar{P} \;\Rightarrow\; \bar{\pi} = 0 \;\Rightarrow\; \bar{i} = \bar{r} = \rho,$$

so that

$$\hat{r}_t = r_t - \rho.$$

A simple inflation-based interest rate rule (Taylor style) is

$$\hat{i}_t = \phi_\pi \pi_t, \qquad \phi_\pi \geq 0.$$

$$\hat{i}_t = E_t[\pi_{t+1}] + \hat{r}_t,$$

we obtain

$$\phi_\pi \pi_t = E_t[\pi_{t+1}] + \hat{r}_t. \tag{*}$$

This is a forward-looking difference equation in inflation.

Case 1: $\phi_\pi > 1$
Iterate forward:

$$\pi_t = \frac{1}{\phi_\pi} E_t[\pi_{t+1}] + \frac{1}{\phi_\pi} \hat{r}_t \quad \Rightarrow \quad \pi_t = \sum_{k=0}^{\infty} \phi_\pi^{-(k+1)} E_t[\hat{r}_{t+k}],$$

where the forward sum converges and yields a unique bounded solution.
Bond Euler:

$$\hat{i}_t - E_t[\pi_{t+1}] = \sigma E_t \Delta \hat{c}_{t+1}, \qquad \hat{i}_t - E_t[\pi_{t+1}] = \hat{r}_t,$$

so

$$\hat{r}_t = \sigma E_t \Delta \hat{c}_{t+1}.$$

Since $\Delta \hat{c}_{t+1} = \Delta \hat{y}_{t+1}$, and

$$\hat{y}_t = \psi_{ya} \hat{a}_t \quad \Rightarrow \quad \Delta \hat{y}_{t+1} = \psi_{ya} \Delta \hat{a}_{t+1},$$

we obtain

$$\hat{r}_t = \sigma \psi_{ya} E_t \Delta \hat{a}_{t+1}.$$

If productivity follows

$$\hat{a}_{t+1} = \rho_a \hat{a}_t + \varepsilon^a_{t+1},$$

then

$$E_t[\hat{a}_{t+1}] = \rho_a \hat{a}_t, \qquad E_t[\Delta \hat{a}_{t+1}] = E_t[\hat{a}_{t+1} - \hat{a}_t] = (\rho_a - 1)\hat{a}_t.$$

Hence,

$$\hat{r}_t = \sigma \psi_{ya} (\rho_a - 1) \hat{a}_t.$$

Guess a linear solution:

$$\pi_t = K \hat{a}_t, \qquad E_t \hat{a}_{t+1} = \rho_a \hat{a}_t.$$

Substituting into

$$\phi_\pi \pi_t = E_t \pi_{t+1} + \hat{r}_t$$

gives

$$\phi_\pi K \hat{a}_t = K \rho_a \hat{a}_t + \hat{r}_t = K \rho_a \hat{a}_t - \sigma \psi_{ya} (1 - \rho_a) \hat{a}_t,$$

so

$$\phi_\pi K - K \rho_a = -\sigma \psi_{ya} (1 - \rho_a),$$

and therefore

$$K = -\frac{\sigma \psi_{ya} (1 - \rho_a)}{\phi_\pi - \rho_a}.$$

Thus,

$$\pi_t = -\frac{\sigma \psi_{ya} (1 - \rho_a)}{\phi_\pi - \rho_a} \hat{a}_t.$$

The larger the ϕ_π, the smaller the variation of inflation caused by $\hat{a}_t$. Central bank raises it more when inflation rises. Any shock that pushes up π_t implies a larger rise in i_t; through the Fisher equation, expectations are forced to be consistent only with the saddle path.

Case 2: $\phi_\pi \leq 1$
Forward iteration breaks. Since $\phi_\pi^{-1} \geq 1$, boundedness no longer pins down a unique π_t.

To satisfy the Fisher equation,

$$\phi_\pi \pi_t = E_t[\pi_{t+1}] + \hat{r}_t.$$

The stable solution is

$$\pi_{t+1} = \phi_\pi \pi_t - \hat{r}_t + \xi_{t+1}, \qquad E_t[\xi_{t+1}] = 0,$$

with sunspot shocks. Inflation and the price level are indetermined.

If the central bank responds less than one-for-one to inflation(weak response), expected inflation can move around without being forcefully pushed back by policy. This leaves room for self-fulfilling beliefs about inflation (sunspot shocks), just as in the exogenous i_t case.

Determinacy requires the Taylor principle: raise the nominal rate more than one-for-one with inflation, that is, $\phi_\pi > 1$.

The system is

$$E_t \pi_{t+1} = \phi_\pi \pi_t - \hat{r}_t.$$

Inflation π_t is a jump variable. By the Blanchard–Kahn condition, to pin down a unique equilibrium we need

$$\#\text{ unstable roots} = \#\text{ jump variables}.$$

From

$$E_t \pi_{t+1} = \phi_\pi \pi_t,$$

the eigenvalue is ϕ_π. If $\phi_\pi > 1$, there is an unstable root, so the BK condition is satisfied. If $\phi_\pi \leq 1$, there is no unstable root to discipline the jump variable, and we get many bounded paths: sunspot indeterminacy.

Optimal Monetary Policy Assume a social planner takes the role of the central banker. Social planner's problem:

$$\max \; U\left(C_t, \frac{M_t}{P_t}, N_t\right) \qquad \text{s.t.} \qquad C_t = A_t N_t^{1-\alpha}.$$

The Lagrangian is

$$\mathcal{L} = U\left(C_t, \frac{M_t}{P_t}, N_t\right) + \lambda_t\left(A_t N_t^{1-\alpha} - C_t\right).$$

F.O.C.:

$$-\frac{U_{N,t}}{U_{C,t}} = (1-\alpha)A_t N_t^{-\alpha} = W_t = MPL_t = MRS_{N,C},$$

and

$$U_{\mathbb{M},t} = 0.$$

In competitive equilibrium,

$$\frac{U_{\mathbb{M},t}}{U_{C,t}} = 1 - e^{-i_t} \approx i_t.$$

Social and individual optima coincide iff $i_t = 0$ for all t, which is the Friedman rule.

In the long run, with the Fisher equation,

$$i = \pi + r = 0 \quad \Rightarrow \quad \pi = -r = -\rho < 0,$$

which is a moderate deflation.

In the short run, $i_t = 0$ does not guarantee determinacy. Instead,

$$E_t \pi_{t+1} = -r_t,$$

or equivalently,

$$\pi_{t+1} = -r_t + \xi_{t+1}, \qquad E_t[\xi_{t+1}] = 0,$$

so any sunspot shock can be added.

Even though money balances and other nominal terms are indetermined, real variables are pinned down by

$$MRS = MPL.$$

Thus indeterminacy does not have welfare consequences.

To avoid indeterminacy, let

$$i_t = \phi\,(r_{t-1} + \pi_t), \qquad \phi > 1.$$

Then

$$i_t = E_t \pi_{t+1} + r_t$$
$$E_t[i_{t+s}] = \phi^s i_t.$$

Since $\phi > 1$, as $s \to \infty$, i_t has to approach 0.

$$\Rightarrow \quad \pi_t = -r_{t-1}, \text{ so inflation is fully determined.}$$

Any policy setting that guarantees that current inflation moves inversely, and one-for-one, with r_{t-1} will imply $i_t = 0$ and an efficient amount of real balances.

Supplementary: IS equation under the separable case in the MIU model

$$U(C_t, \mathbb{M}_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} + \frac{(\mathbb{M}_t)^{1-\nu}}{1-\nu} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

From the bond Euler F.O.C.,

$$Q_t = \beta E_t \left[\frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right].$$

Since

$$U_{C,t} = C_t^{-\sigma},$$

we have

$$Q_t = \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right].$$

Define

$$i_t = -\log Q_t, \qquad \rho = -\log \beta, \qquad \pi_{t+1} = \log P_{t+1} - \log P_t.$$

Then

$$-i_t = -\rho + E_t[-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}].$$

At steady state,

$$-\bar{i} = -\rho + E_t\left[-\sigma(\log \bar{C} - \log \bar{C}) - \bar{\pi}\right],$$

with

$$\bar{\pi} = 0.$$

Hence

$$\bar{i} = \rho.$$

Subtracting the steady state and using hats,

$$\sigma E_t(\hat{c}_{t+1} - \hat{c}_t) = \hat{i}_t - E_t\pi_{t+1}.$$

Thus

$$\hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1}\right].$$

Since

$$\hat{y}_t = \hat{c}_t,$$

we get

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1}\right].$$

Compare to the classical monetary model: the same.

Compare to the inseparable case: $\nu = \sigma$, and the term $-\omega E_t \Delta\hat{i}_{t+1}$ disappears?

Supplement: Interpret

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}\right].$$

1. The term

$$i_t - E_t\pi_{t+1}$$

is the opportunity cost of consuming today instead of tomorrow.
Suppose you have 1 RMB.
If you buy goods today, you get

$$\frac{1}{P_t}$$

units of goods.
If you save to tomorrow and then buy goods, you get

$$\frac{1+i_t}{P_{t+1}}$$

units of goods. Therefore,

$$E_t\left[\frac{(1+i_t)/P_{t+1}}{1/P_t}\right] = E_t\left[\frac{1+i_t}{P_{t+1}/P_t}\right] = \frac{1+i_t}{1+E_t\pi_{t+1}} \approx i_t - E_t\pi_{t+1}.$$

So if

$$\hat{i}_t - E_t\pi_{t+1} \uparrow,$$

then

$$\hat{y}_t \downarrow,$$

because the opportunity cost of consuming today is higher.

2. The term

$$\omega E_t \hat{i}_{t+1}$$

is the forward-guidance term.

If tomorrow's nominal rate i_{t+1} is expected to be high, then holding money tomorrow is expensive. Households therefore want to hold fewer real balances at $t+1$:

$$\frac{M_{t+1}}{P_{t+1}} \downarrow .$$

With $\omega > 0$, there is within-period substitution between C_t and $\frac{M_{t+1}}{P_{t+1}}$, in addition to the usual intertemporal substitution of X_t.

Holding more real balances,

$$\frac{M_{t+1}}{P_{t+1}} \uparrow,$$

increases the marginal utility of consumption, $MU_{C,t+1}$, so future consumption becomes more valuable. Conversely,

$$\frac{M_{t+1}}{P_{t+1}} \downarrow$$

makes future consumption less valuable and induces households to consume more today.

In particular,

$$U_{C,t} = (1-\theta) X_t^{\nu-\sigma} C_t^{-\nu}, \qquad X_t \propto \mathbb{M}_t \quad \Rightarrow \quad U_{C\mathbb{M}} > 0.$$

Consumption is more enjoyable with more liquidity.

Supplement for computation:Log-linearized bond Euler equation Define

$$i_t = -\log Q_t \;\Rightarrow\; e^{-i_t} = Q_t, \qquad \rho = -\log \beta \;\Rightarrow\; e^{-\rho} = \beta, \qquad \pi_{t+1} = \log P_{t+1} - \log P_t,$$

and

$$r_t = i_t - E_t[\pi_{t+1}].$$

Define hats as level deviations:

$$\hat{i}_t = i_t - \bar{i}, \qquad \hat{\pi}_{t+1} = \pi_{t+1} - \bar{\pi}, \qquad \bar{\pi} = 0 \;\Rightarrow\; \hat{\pi}_{t+1} = \pi_{t+1}.$$

Also,

$$r_t - \bar{r} = i_t - \bar{i} - E_t[\pi_{t+1} - \bar{\pi}] \;\Rightarrow\; \hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}].$$

From the bond Euler equation,

$$Q_t = \beta E_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma} \frac{P_t}{P_{t+1}}\right] = \beta E_t\left[e^{-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}}\right],$$

so

$$e^{-i_t} = e^{-\rho} E_t\left[e^{-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}}\right].$$

Let

$$\hat{c}_{t+1} = \log C_{t+1} - \log \bar{C}, \qquad \hat{c}_t = \log C_t - \log \bar{C} \;\Rightarrow\; \log C_{t+1} - \log C_t = \hat{c}_{t+1} - \hat{c}_t.$$

Then

$$e^{-i_t} = e^{-\rho} E_t\left[e^{-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}}\right].$$

Taking logs,

$$-i_t = -\rho + E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}] .$$

At steady state,

$$-\bar{i} = -\rho + E_t[0 - 0] \;\Rightarrow\; \bar{i} = \rho.$$

Hence

$$-(i_t - \bar{i}) = E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t)] - E_t[\pi_{t+1} - \bar{\pi}],$$

or equivalently,

$$\hat{i}_t = \sigma E_t[\Delta\hat{c}_{t+1}] + E_t[\hat{\pi}_{t+1}], \qquad \Delta\hat{c}_{t+1} = \hat{c}_{t+1} - \hat{c}_t.$$

Therefore,

$$\sigma E_t[\Delta\hat{c}_{t+1}] = \hat{i}_t - E_t[\hat{\pi}_{t+1}] = \hat{r}_t.$$

Using $\hat{c}_t = \hat{y}_t$, we obtain

$$\sigma E_t[\Delta\hat{y}_{t+1}] = \hat{r}_t.$$

Since

$$E_t[\Delta\hat{y}_{t+1}] = E_t\hat{y}_{t+1} - \hat{y}_t, \qquad \hat{y}_t = \psi_{ya}\hat{a}_t \;\Rightarrow\; E_t[\Delta\hat{y}_{t+1}] = \psi_{ya}E_t[\Delta\hat{a}_{t+1}],$$

it follows that

$$\hat{r}_t = \sigma\psi_{ya}E_t[\Delta\hat{a}_{t+1}].$$

Let

$$\log A_{t+1} = (1-\psi)\log\bar{A} + \varphi\log A_t + \varepsilon_{t+1}, \qquad \bar{A} = 1.$$

Then

$$\log\bar{A} = \psi\log\bar{A}, \qquad \hat{a}_{t+1} = \psi\hat{a}_t + \varepsilon_{t+1}.$$

Taking expectations,

$$E_t[\hat{a}_{t+1}] = \psi\hat{a}_t + E_t[\varepsilon_{t+1}], \qquad \varepsilon_t \sim \text{i.i.d. } N(0,\sigma^2) \;\Rightarrow\; E_t[\varepsilon_{t+1}] = 0,$$

so

$$E_t[\hat{a}_{t+1}] = \psi\hat{a}_t, \qquad E_t[\hat{a}_{t+1} - \hat{a}_t] = (\psi - 1)\hat{a}_t.$$

Let

$$\Delta\hat{a}_{t+1} = \hat{a}_{t+1} - \hat{a}_t \;\Rightarrow\; E_t[\Delta\hat{a}_{t+1}] = (\psi - 1)\hat{a}_t.$$

Plugging this into $\hat{r}_t = \sigma\psi_{ya}E_t[\Delta\hat{a}_{t+1}]$, we get

$$\hat{r}_t = \sigma\psi_{ya}E_t[\Delta\hat{a}_{t+1}] = \sigma\psi_{ya}(\psi - 1)\hat{a}_t.$$

Since $\psi \in (0,1)$ and $\psi_{ya} > 0$, the coefficient on $\hat{a}_t$ is negative.

Lecture 7

Cash in Advance

Lecture 8

Basic New Keynesian

8.1 RBC to NK

RBC: Fluctuations explained by technology shocks (kydland–Prescott 1982; Prescott 1986)

Use DSGE models as tool: FOCs replacing behavioral equations, rational expectations.

1. Efficient allocation.
2. Implication: stabilization policies non-necessary.
3. Equilibrium outcome responds to exogenous shocks.
4. Limited role of monetary factors, neutrality of monetary policy: Friedman's rule

* Contradiction to:

1. Common belief: influence on output and employment.
2. Empirical challenge: non-neutrality.

⇒ Irrelevancy to policy evaluation: Normative implication different from policy practice.
Monetary policy can affect real variables or not: e.g. output, employment, consumption.

RBC	**New Keynesian**
Agents / household / firms	~
DSGE	~
Markets are frictionless and perfectly competitive	Goods market imperfection (monopolistic competition)
Price adjust instantaneously	Nominal rigidity (price, wage etc.)
Technology shocks	Other shocks and transmission mechanisms
Not really: RBC also can introduce other shocks.	

8.2 New in NK

1. Monopolistic competition: $P^m > P^c, Y^m < Y^c$, monopolistic competition inefficient response to shocks.
2. Nominal rigidities: staggered price / wage, short-run price rigidity.

3. Short-run non-neutrality of monetary policy ⇒ policy analysis & intervention.

   These elements are new to RBC but old to Keynesian (static, reduced-form).
   Empirical evidence of nominal rigidities:

- Taylor (1999): Average price adjustment frequency = 1 year.
- Bils and Klenow (2004): Heterogeneity of frequency across sectors / types of goods.
- Nakamura and Steinsson (2008): Adjusting for sales, frequency ≈ 8–11 months.
- Bewley (1998): Wages do not fall during recessions (downward nominal rigidity).

  * High welfare / social security countries face stronger downward wage rigidity.
  ? Working hour rigidity in China.

$$\text{Nominal rigidity} \Rightarrow \text{price inertia} \Rightarrow \begin{cases} \text{Monetary policy changes real balance } M/P, \\ \text{change real interest rate } r = i - \pi. \end{cases}$$

$$\Rightarrow \text{Change in aggregate demand} \Rightarrow \text{Change in employment/output.}$$

Identify monetary policy shock:

$$\begin{cases} \text{endogenous movement:} & \text{deliberate response to developments in economy } \Rightarrow \text{co-movement,} \\ \text{exogenous movement:} & \text{structural VAR residuals from policy rule.} \end{cases}$$

$$\begin{cases} \text{endogenous comovements:} & \text{high interest rate, high inflation/high output,} \\ & i_t = \phi_\pi \pi_t + \phi_y y_t + v_t, \\ & \text{policy response + exogenous component / unexpected shocks;} \\ \text{exogenous component:} & u_t \text{ identification requires: GDP and prices cannot respond} \\ & \text{contemporaneously to a monetary policy shock.} \end{cases}$$

8.3 Basic NK

Basic setup:

① DSGE structure: Households live infinitely, maximize utility subject to intertemporal budget constraint. Firms with identical technology, subject to random shocks.

② AS and AD block.

AD: log-linearized consumption Euler (relates output and price).

Let $\hat{x}_t = \log X_t - \log \bar{X}$ except i_t, r_t, π_t.

$$\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$$

$\hat{y}_t^n$: natural output under flexible prices / long-run output.

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} \underbrace{\left(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}\right)}_{\hat{r}_t} + \underbrace{v_t}_{\text{MP shock}} . \tag{1}$$

AS: New Keynesian Phillips Curve (relates price and output):

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t . \tag{2}$$

③ Policy rule:

$$\hat{i}_t = f(\Omega_t) + v_t, \qquad \Omega_t : \text{ information set / state vector.} \tag{3}$$

Four markets: goods, labor, bonds, money (cashless).

Differentiated goods: continuum of varieties $i \in [0, 1]$.

Perfect competition in labor market, no wage rigidities.

Monopolistic competition on goods market: goods prices set by firms that produce them (profit $\neq 0$).

Nominal prices are sticky: a fraction $(1 - \theta)$ of firms can reset their prices each period (Calvo pricing).

A continuum of infinitely-lived households.

A continuum of firms represented by $[0, 1]$ goods $\Rightarrow$ 1 firm produces 1 differentiated good and sets price.

8.4 Household's problem

$$\max_{C_t, N_t, B_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t U(C_t, N_t),$$

$$\text{s.t.} \quad \boxed{P_t} C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t.$$

P_t: aggregate price index of the consumption bundle, $P_t C_t = \int_0^1 P_t(i) C_t(i) \mathrm{d}i$.

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

C_t: A composite bundle of goods; continuum of varieties produced by continuum of firms.

i : variety (differentiated goods) index,
$\mathrm{d}i$: continuum of varieties produced by continuum of firms,
ε : across-variety EoS.

Solve household's problem in two steps:
1. Static optimal expenditure allocation (how to allocate across varieties?)

$$E = \min_{C_t(i)} \int_0^1 P_t(i) C_t(i) \mathrm{d}i = \text{total expenditure} \quad \text{s.t.} \quad C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

$$\mathcal{L} = \int_0^1 P_t(i) C_t(i) \mathrm{d}i - P_t \left[\left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i \right)^{\frac{\varepsilon}{\varepsilon-1}} - C_t \right].$$

$$\frac{\partial \mathcal{L}}{\partial C_t(i)} = 0 \Rightarrow P_t(i) - P_t \frac{\varepsilon}{\varepsilon-1} \left[\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i \right]^{\frac{1}{\varepsilon-1}} \frac{\varepsilon-1}{\varepsilon} C_t(i)^{-\frac{1}{\varepsilon}} = 0$$

$$\Rightarrow P_t(i) - P_t C_t^{1/\varepsilon} C_t(i)^{-1/\varepsilon} = 0$$

$$\Rightarrow C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t.$$

$$C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} = \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} C_t^{\frac{\varepsilon-1}{\varepsilon}},$$

$$\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i = C_t^{\frac{\varepsilon-1}{\varepsilon}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} \mathrm{d}i,$$

$$C_t^{\frac{\varepsilon-1}{\varepsilon}} = C_t^{\frac{\varepsilon-1}{\varepsilon}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} \mathrm{d}i,$$

$$P_t^{1-\varepsilon} = \int_0^1 P_t(i)^{1-\varepsilon} \mathrm{d}i, \qquad P_t = \left(\int_0^1 P_t(i)^{1-\varepsilon} \mathrm{d}i \right)^{\frac{1}{1-\varepsilon}}.$$

Why ε is the elasticity of substitution across varieties?

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t, \qquad C_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} C_t \Rightarrow \frac{C_t(i)}{C_t(j)} = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\varepsilon}.$$

$$\log \frac{C_t(i)}{C_t(j)} = -\varepsilon \log \frac{P_t(i)}{P_t(j)}, \qquad \frac{\mathrm{d} \log\{C_t(i)/C_t(j)\}}{\mathrm{d} \log\{P_t(i)/P_t(j)\}} = -\varepsilon.$$

$$\sigma_{ij} = -\frac{\mathrm{d} \log\{C_t(i)/C_t(j)\}}{\mathrm{d} \log\{P_t(i)/P_t(j)\}} = \varepsilon, \quad \text{elasticity of substitution between two varieties.}$$

2. Intertemporal problem

$$\max_{C_t,N_t,B_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \quad \text{s.t.} \quad P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t.$$

$$\mathcal{L} = \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left\{ U(C_t, N_t) + \lambda_t (B_{t-1} + W_t N_t + T_t - P_t C_t - Q_t B_t) \right\}.$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow U_{C,t} = P_t \lambda_t \quad (1),$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \Rightarrow U_{N,t} + \lambda_t W_t = 0 \quad (2),$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \Rightarrow \beta^{t+1} \mathbb{E}_t \lambda_{t+1} - \beta^t \lambda_t Q_t = 0 \Rightarrow \mathbb{E}_t \beta \lambda_{t+1} = \lambda_t Q_t \quad (3).$$

$$(1)\&(2) \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} \quad \text{Labor–leisure condition (intratemporal).}$$

$$(1)\&(3) \Rightarrow Q_t = \beta \mathbb{E}_t \left\{ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right\} = \beta \mathbb{E}_t \left\{ \frac{U_{C,t+1}}{U_{C,t}} \frac{1}{\Pi_{t+1}} \right\}, \quad \Pi_{t+1} = \frac{P_{t+1}}{P_t} \text{ inflation.}$$

Bond Euler equation (intertemporal).

Suppose we have a separable utility function

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}, \qquad U_{C,t} = C_t^{-\sigma}, \quad U_{N,t} = -N_t^{\varphi}.$$

$$\frac{W_t}{P_t} = N_t^{\varphi} C_t^{\sigma}, \qquad Q_t = \beta \mathbb{E}_t \left\{ \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} \right\}.$$

Log-linearize:

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t, \qquad \hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \hat{\Pi}_{t+1}).$$

$$Q_t = e^{-i_t}, \qquad \hat{\Pi}_{t+1} = \log \Pi_{t+1} - \log \bar{\Pi} = \log \Pi_{t+1} = \pi_{t+1}.$$

8.5 Firm's problem: two simultaneous problems (isomorphic)

1. Cost minimization for each producer in the continuum

$$\min_{N_t(i)} W_t N_t(i) \quad \text{s.t.} \quad Y_t(i) = A_t N_t(i)^{1-\alpha}.$$

$$\mathcal{L} = W_t N_t(i) + \mu_t \left[Y_t(i) - A_t N_t(i)^{1-\alpha} \right].$$

$$\frac{\partial \mathcal{L}}{\partial N_t(i)} = W_t - \mu_t (1-\alpha) A_t N_t(i)^{-\alpha} = 0 \Rightarrow W_t = \mu_t (1-\alpha) A_t N_t(i)^{-\alpha}.$$

μ_t is the Lagrangian multiplier: shadow value of relaxing the production requirement by one unit.

$$\Rightarrow \mu_t \text{ is the nominal marginal cost: } \mu_t = MC_t, \qquad \mu_t = MC_t = \frac{W_t}{MPL_t(i)}.$$

W_t: 1 labor's cost; $MPL_t(i)$: 1 labor's output; $\frac{W_t}{MPL_t(i)}$: cost per unit of output.

2. Profit maximization by choosing price.

$$\max_{P_t(i)} P_t(i)Y_t(i) - W_t N_t(i) = \max_{P_t(i)} P_t(i)Y_t(i) - W_t \left(\frac{Y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}} \qquad \text{production function}$$

$$\frac{\partial Profit}{\partial P_t(i)} = Y_t(i) + P_t(i)\frac{\partial Y_t(i)}{\partial P_t(i)} - W_t \frac{1}{1-\alpha}\left(\frac{Y_t(i)}{A_t}\right)^{\frac{\alpha}{1-\alpha}} \frac{\partial Y_t(i)}{\partial P_t(i)} \frac{1}{A_t} = 0. \qquad (*)$$

$$MC_t = \frac{\partial}{\partial Y_t(i)}\left[W_t \left(\frac{Y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}}\right] = W_t \frac{1}{1-\alpha}\left(\frac{Y_t(i)}{A_t}\right)^{\frac{\alpha}{1-\alpha}} \frac{1}{A_t}$$

$$Y_t(i) + [P_t(i) - MC_t]\frac{\partial Y_t(i)}{\partial P_t(i)} = 0.$$

Plug in:

$$Y_t(i) - \varepsilon\,[P_t(i) - MC_t]\frac{Y_t(i)}{P_t(i)} = 0.$$

Divide both sides by $Y_t(i)$:

$$1 - \varepsilon \frac{P_t(i) - MC_t}{P_t(i)} = 0.$$

$$1 - \varepsilon + \varepsilon \frac{MC_t}{P_t(i)} = 0.$$

Therefore,

$$P_t(i) = \frac{\varepsilon}{\varepsilon - 1} MC_t.$$

Variety price equals to a constant markup μ over the marginal cost

where $\mu = \frac{\varepsilon}{\varepsilon - 1}$; monopolistic competition pricing.

When $\varepsilon \to \infty \Rightarrow \mu \to 1$, perfect competition, $P_t(i) = MC_t$.
In general ε is finite, $\mu > 1$, $P_t(i) > MC_t \Rightarrow$ inefficiency, low employment and output.

Cost minimization 1 and profit maximization 2 are dual problems:

1. tells how expensive it is to produce at each output level.
2. tells the price-taking that cost schedule and demand as given.

Nesting the two:

$$C(Y_t(i)) = \min_{N_t(i)} W_t N_t(i) \quad \text{s.t.} \quad Y_t(i) = A_t N_t(i)^{1-\alpha}.$$

Then solve

$$\max_{P_t(i)} \{P_t(i)Y_t(i) - C(Y_t(i))\} \quad \text{s.t.} \quad Y_t(i) = \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} C_t.$$

When $\alpha = 0$, marginal cost connects the two problems:

$$MC_t = \frac{\partial C(Y_t(i))}{\partial Y_t(i)}.$$

8.6 Sticky-price equilibrium

Calvo (1983) pricing: Each period a measure $(1-\theta)$ fraction of the continuum of firms reset their prices, the other θ keeps price unchanged.

$$\Rightarrow \text{Average duration of a price is } \frac{1}{1-\theta}.$$

$$\theta: \text{ index of price stickiness, pinned down by data.}$$

Thus

$$P_t = \left[\theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P_t^*)^{1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}.$$

Log-linearize:

$$\begin{aligned}
P_t^{1-\varepsilon} &= \theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P_t^*)^{1-\varepsilon},\\
\bar{P}^{1-\varepsilon}(1+\hat{p}_t)^{1-\varepsilon} &= \theta\bar{P}^{1-\varepsilon}(1+\hat{p}_{t-1})^{1-\varepsilon} + (1-\theta)\bar{P}^{1-\varepsilon}(1+\hat{p}_t^*)^{1-\varepsilon},\\
\bar{P}^{1-\varepsilon}(1-\varepsilon)\hat{p}_t &= \theta\bar{P}^{1-\varepsilon}(1-\varepsilon)\hat{p}_{t-1} + (1-\theta)\bar{P}^{1-\varepsilon}(1-\varepsilon)\hat{p}_t^*,\\
\hat{p}_t &= \theta\hat{p}_{t-1} + (1-\theta)\hat{p}_t^*,\\
\hat{p}_t - \hat{p}_{t-1} &= (1-\theta)(\hat{p}_t^* - \hat{p}_{t-1}).
\end{aligned}$$

$$\begin{aligned}
\hat{\pi}_t &= \pi_t - \bar{\pi} = \pi_t = \log P_t - \log P_{t-1} = \log P_t - \log\bar{P} - (\log P_{t-1} - \log\bar{P})\\
&= \hat{p}_t - \hat{p}_{t-1}.
\end{aligned}$$

$$\Rightarrow \hat{\pi}_t = (1-\theta)(\hat{p}_t^* - \hat{p}_{t-1}).$$

* Positive inflation arises iff firms that adjust prices in t choose to charge above the price level prevailed in $t-1$.

$\theta \uparrow \Rightarrow$ more sticky firms $\Rightarrow$ inflation is lower given the price difference.

Due to sticky price, firms must set price taking into account the risk that they will not be able to change price in the future.

$$\begin{aligned}
V_t(i) = P_t(i)Y_t(i) - cost_t(i) &+ \theta\mathbb{E}_t\{Q_{t,t+1}[P_t(i)Y_{t+1}(i) - cost_{t+1}(i)]\}\\
&+ \theta^2\mathbb{E}_t\{Q_{t,t+2}[P_t(i)Y_{t+2}(i) - cost_{t+2}(i)]\} + \cdots.
\end{aligned}$$

Optimal pricing under sticky price: Choose P_t^* to maximize the expected PV of profits while this price remains effective, $P_t^* = P_t^*(i)$.

$$V_t(i) = \max_{P_t^*}\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\left[Q_{t,t+k}(P_t^*Y_{t+k|t} - \psi_{t+k}(Y_{t+k|t}))\right] \quad \text{conditional on the firm can reset at } t.$$

$$\text{s.t.}\quad Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\varepsilon} C_{t+k}.$$

$$Q_{t,t+k} = \beta^k\left(\frac{C_{t+k}}{C_t}\right)^{-\sigma}\left(\frac{P_t}{P_{t+k}}\right) \quad \text{stochastic discount factor for nominal payoffs.}$$

$\psi_{t+k}(\cdot)$ is the cost function.
$Y_{t+k|t}$ is output in period $t+k$ for a firm that last reset price at t.

When $\theta = 0$, collapse to with only $k=0$ term $\Rightarrow P_t^* = \frac{\varepsilon}{\varepsilon-1}\psi_t$.

* $1-\theta$ determines how many firms reset in the cross section.
θ^k matters for how long a chosen reset price survives in the time dimension.
FOC:

$$\frac{\partial V_t(i)}{\partial P_t^*} = 0 \Rightarrow \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[Y_{t+k|t} + P_t^* \frac{\partial Y_{t+k|t}}{\partial P_t^*} - \frac{\partial \psi_{t+k}}{\partial Y_{t+k|t}} \frac{\partial Y_{t+k|t}}{\partial P_t^*} \right] \right\} = 0,$$

$$\frac{\partial Y_{t+k|t}}{\partial P_t^*} = -\varepsilon \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon-1} \frac{C_{t+k}}{P_{t+k}} = -\varepsilon \frac{Y_{t+k|t}}{P_t^*}.$$

Define

$$\varphi_{t+k|t} = \frac{\partial \psi_{t+k}}{\partial Y_{t+k|t}} \quad \text{as nominal marginal cost.}$$

$$\mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[Y_{t+k|t} + (P_t^* - \varphi_{t+k|t})(-\varepsilon) \frac{Y_{t+k|t}}{P_t^*} \right] \right\} = 0,$$

$$\mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t} \left[1 - \varepsilon + \varepsilon \frac{\varphi_{t+k|t}}{P_t^*} \right] \right\} = 0.$$

$$P_t^* = \underbrace{\frac{\varepsilon}{\varepsilon - 1}}_{\text{markup } \mu} \cdot \underbrace{\frac{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t} \varphi_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t}}}_{\text{weighted}}$$

Weighted average of future marginal cost(*).

$$\frac{\theta^k Q_{t,t+k} Y_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t}} : \text{weight/how nominal MC of } t+k \text{ matters.}$$

When $\theta \to 0$, $P_t^* \to \frac{\varepsilon}{\varepsilon-1} \varphi_t$.

Weights are larger when:
1) $\theta \uparrow$: price is more sticky;
2) $Q_{t,t+k} \uparrow$: future profit is more valuable in utility terms;
3) $Y_{t+k|t} \uparrow$: firms expect to sell more in that period.

Why $Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t}\right)^{-\sigma} \left(\frac{P_t}{P_{t+k}}\right)$?

$$Q_t = \beta \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right] \quad \text{one-period nominal discount factor.}$$

$$Q_t = Q_{t,t+1},$$

$$Q_{t+1} = \beta \left[\left(\frac{C_{t+2}}{C_{t+1}} \right)^{-\sigma} \frac{P_{t+1}}{P_{t+2}} \right],$$

$$Q_{t,t+2} = Q_{t,t+1} Q_{t+1,t+2} = \beta^2 \left(\frac{C_{t+2}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+2}},$$

$$Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}}.$$

β^k: impatience.
$\left(\frac{C_{t+k}}{C_t}\right)^{-\sigma}$: marginal utility of future consumption.
$\frac{P_t}{P_{t+k}}$: inflation erosion of nominal payoff; converts nominal payoff into real payoff.

$$Q_{t,t+k} = \beta^k \widetilde{Q}_{t,t+k}, \qquad \widetilde{Q}_{t,t+k} = \left(\frac{C_{t+k}}{C_t}\right)^{-\sigma} \left(\frac{P_t}{P_{t+k}}\right).$$

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\varepsilon} C_{t+k} = (P_t^*)^{-\varepsilon} P_{t+k}^{\varepsilon} Y_{t+k}, \quad (Y_t = C_t, \forall t).$$

$$(*) \Rightarrow P_t^* = \mu \frac{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k} \varphi_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k}}.$$

$$P_t^* \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k} = \mu \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k} \varphi_{t+k|t}.$$

Let

$$A_t \equiv \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k}, \qquad a_{t+k} \equiv (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k}.$$

$$\widehat{LHS} = \hat{p}_t^* + \mathbb{E}_t \hat{A}_t.$$

$$\hat{A}_t = \sum_{k=0}^{\infty} \frac{\bar{a}_{t+k}}{\bar{A}} \hat{a}_{t+k},$$

$$\frac{\bar{a}_{t+k}}{\bar{A}} = \frac{(\beta\theta)^k \bar{\widetilde{Q}} \bar{P}^{\varepsilon} \bar{Y}}{\sum_{k=0}^{\infty} (\beta\theta)^k \bar{\widetilde{Q}} \bar{P}^{\varepsilon} \bar{Y}} = \frac{(\beta\theta)^k \bar{\widetilde{Q}} \bar{P}^{\varepsilon} \bar{Y}}{\frac{1}{1-\beta\theta} \bar{\widetilde{Q}} \bar{P}^{\varepsilon} \bar{Y}} = (1-\beta\theta)(\beta\theta)^k,$$

$$\hat{a}_{t+k} = \hat{\tilde{q}}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k}.$$

$$\widehat{LHS} = \hat{p}_t^* + (1-\beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\hat{\tilde{q}}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k}).$$

$$B_t \equiv \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \widetilde{Q}_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k} \varphi_{t+k|t}, \qquad \widehat{RHS} = \mathbb{E}_t \hat{B}_t.$$

$$\hat{B}_t = \sum_{k=0}^{\infty} \frac{\bar{b}_{t+k}}{\bar{B}} \hat{b}_{t+k}, \qquad \frac{\bar{b}_{t+k}}{\bar{B}} = (1-\beta\theta)(\beta\theta)^k,$$

$$\hat{b}_{t+k} = \hat{\tilde{q}}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k} + \hat{\varphi}_{t+k|t}.$$

$$\widehat{RHS} = (1-\beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\hat{\tilde{q}}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k} + \hat{\varphi}_{t+k|t}).$$

Combine $\widehat{LHS} = \widehat{RHS}$:

$$\begin{aligned} \hat{p}_t^* &= (1-\beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \hat{\varphi}_{t+k|t} \\ &= (1-\beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\widehat{mc}_{t+k|t}^{\,r} + \hat{p}_{t+k}). \end{aligned}$$

nominal MC = real MC × price

$$\varphi_{t+k|t} = MC^r_{t+k|t} \cdot P_{t+k} \Rightarrow \hat{\psi}_{t+k|t} = \widehat{mc}^{\,r}_{t+k|t} + \hat{p}_{t+k}.$$

1. Weights $\sum_{k=0}^{\infty}(1-\beta\theta)(\beta\theta)^k = 1$: weighted average over expected future nominal marginal cost.

2. Aggregate price level denotes willingness to maintain (in expectation) the relative price unchanged.

$$\text{Suppose } \widehat{mc}^{\,r}_{t+k|t} = 0 \text{ for all } k, \quad \text{real marginal cost is constant.}$$

$$\hat{p}^*_t = (1-\beta\theta)\mathbb{E}_t \sum_{k=0}^{\infty}(\beta\theta)^k \hat{p}_{t+k}.$$

* Reset price rises when expected future aggregate prices rise. Front-load future inflation to today's reset price. Expected inflation $\Rightarrow \hat{p}^*_t \uparrow$.

3. Real MC $\widehat{mc}^{\,r}_{t+k|t}$ denotes desire to change the expected relative price to avoid any gap that may emerge between expected and desired markup. Monopolistic competitive firm's desired markup rule: $P = \mu MC$.

$$\text{Thus, firm wants its real price } \frac{P^*_t}{P_{t+k}} \text{ to be high enough relative to real MC.}$$

$$\hat{p}^*_t - \hat{p}_{t+k} = \widehat{mc}^{\,r}_{t+k|t} + \text{desired markup} \Rightarrow \widehat{mc}^{\,r}_{t+k|t} \uparrow \Rightarrow \hat{p}^*_t - \hat{p}_{t+k} \uparrow.$$

$\Rightarrow$ Expected future real marginal cost is higher: the firm wants a higher relative price.

8.7 Equilibrium

1. Goods market clearing:

$$Y_t(i) = C_t(i), \quad \forall i \in [0,1].$$

Define

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i\right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

Get $Y_t = C_t$. Log-linearize: $\hat{y}_t = \hat{c}_t$. Since

$$\hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}),$$

$$\Rightarrow \hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}).$$

2. Labor market clearing:

$$N_t = \int_0^1 N_t(i)\mathrm{d}i.$$

$$Y_t(i) = A_t N_t(i)^{1-\alpha} \Rightarrow N_t = \int_0^1 \left(\frac{Y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}} \mathrm{d}i,$$

$$Y_t(i) = \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} Y_t \Rightarrow N_t = \left(\frac{Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} \int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} \mathrm{d}i,$$

$$\Rightarrow N_t^{1-\alpha} = \frac{Y_t}{A_t}\left[\int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} \mathrm{d}i\right]^{1-\alpha}.$$

8.8 Price dispersion

Let

$$D_t = \left[\int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} \mathrm{d}i\right]^{1-\alpha} = \exp(d_t).$$

$Y_t = A_t N_t^{1-\alpha} D_t^{-1}$, D_t measures how uneven prices are across firms.

When all firms charge the same price: $\dfrac{P_t(i)}{P_t} = 1,\ \forall i \Rightarrow D_t = 1 \Rightarrow d_t = 0.$

Reduced to $Y_t = A_t N_t^{1-\alpha}$: frictionless benchmark under flexible prices, no price distortion. But under Calvo pricing, some firms use old price, some firms use new price $\Rightarrow D_t > 1$.

$$Y_t(i) = \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} Y_t.$$

Firms with low relative prices sell too much; firms with high relative prices sell too little.

$$d_t = (1-\alpha)\log\left[\int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} \mathrm{d}i\right].$$

Define $p_t(i) = \log P_t(i)$, $p_t = \log P_t$, $P_t(i)/P_t = e^{p_t(i)-p_t}$. Let $\chi_i = p_t(i) - p_t$.

$$\Rightarrow d_t = (1-\alpha)\log\left[\int_0^1 \exp\left(-\frac{\varepsilon}{1-\alpha}\chi_i\right) \mathrm{d}i\right].$$

Second-order Taylor expansion around zero-inflation / zero-dispersion steady state. Why zero inflation $\bar{\Pi} = 1 \Leftrightarrow$ zero dispersion $P_t(i)/P_t = 1 \Leftrightarrow \bar{d} = 0$?

$$P_t = \left[\theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P_t^*)^{1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}, \quad \text{to have } P_t = P_{t-1} = \bar{P}, \text{ need } P_t^* = \bar{P}.$$

The firms that reset price set the same price as others.

$$\text{Since } e^z = e^0 + \frac{(e^z)|_{z=0}}{1!}(z-0) + \frac{(e^z)|_{z=0}}{2!}(z-0)^2 + o(z^2).$$

$$e^z \approx 1 + z + \frac{z^2}{2} \quad \text{for small } z.$$

$$e^{-\frac{\varepsilon}{1-\alpha}\chi_i} \approx 1 - \frac{\varepsilon}{1-\alpha}\chi_i + \frac{1}{2}\left(\frac{\varepsilon}{1-\alpha}\right)^2\chi_i^2.$$

$$\int_0^1 e^{-\frac{\varepsilon}{1-\alpha}\chi_i}\mathrm{d}i \approx 1 - \frac{\varepsilon}{1-\alpha}\int_0^1 \chi_i \mathrm{d}i + \frac{1}{2}\left(\frac{\varepsilon}{1-\alpha}\right)^2\int_0^1 \chi_i^2 \mathrm{d}i,$$

$$\chi_i = p_t(i) - p_t \Rightarrow \int_0^1 \chi_i \mathrm{d}i = \int_0^1 p_t(i)\mathrm{d}i - \int_0^1 p_t \mathrm{d}i = p_t - p_t = 0,$$

$$\int_0^1 e^{-\frac{\varepsilon}{1-\alpha}\chi_i}\mathrm{d}i \approx 1 + \frac{1}{2}\left(\frac{\varepsilon}{1-\alpha}\right)^2\int_0^1 \chi_i^2 \mathrm{d}i.$$

$$\int_0^1 \chi_i^2 \mathrm{d}i \approx \mathrm{Var}_i\{p_t(i) - p_t\} \quad \text{cross-sectional second moment.}$$

Take logs:

$$d_t = (1-\alpha)\log\left[1 + \frac{1}{2}\left(\frac{\varepsilon}{1-\alpha}\right)^2 Var_i\{p_t(i) - p_t\}\right],$$

since $\log(1+u) \approx u$,

$$d_t \approx (1-\alpha)\frac{1}{2}\left(\frac{\varepsilon}{1-\alpha}\right)^2 Var_i\{p_t(i) - p_t\},$$

$$d_t \approx \frac{\varepsilon^2}{2(1-\alpha)} Var_i\{p_t(i) - p_t\},$$

$$d_t \propto Var_i\{p_t(i)\}.$$

Thus, in the neighborhood of $\hat{\pi}_t = 0$ or $\bar{\Pi} = 1$, d_t is 0 up to first-order approximation:

$$\hat{d}_t \approx 0.$$

Thus

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1} \quad \text{log linearization}$$

$$\hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t - \hat{d}_t \Rightarrow \hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t, \qquad \hat{n}_t = \frac{1}{1-\alpha}(\hat{y}_t - \hat{a}_t).$$

8.9 Average real marginal cost

Average firm:

$$MC_t^r = \frac{W_t}{P_t}\frac{1}{MPN_t} = \frac{W_t}{P_t}\frac{1}{(1-\alpha)A_t N_t^{-\alpha}}.$$

$$\begin{aligned}\widehat{mc}_t^r &= \hat{w}_t - \hat{p}_t - \widehat{mpn}_t = \hat{w}_t - \hat{p}_t - (\hat{a}_t - \alpha\hat{n}_t)\\ &= \hat{w}_t - \hat{p}_t - \left(\hat{a}_t - \frac{\alpha}{1-\alpha}\hat{y}_t + \frac{\alpha}{1-\alpha}\hat{a}_t\right)\\ &= \hat{w}_t - \hat{p}_t - \frac{1}{1-\alpha}(\hat{a}_t - \alpha\hat{y}_t).\end{aligned}$$

For firms that set price at t and remains at $t+k$:

$$\begin{aligned}\widehat{mc}_{t+k|t}^r &= \hat{w}_{t+k} - \hat{p}_{t+k} - \widehat{mpn}_{t+k|t}\\ &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1-\alpha}\left(\hat{a}_{t+k} - \alpha\hat{y}_{t+k|t}\right)\\ &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1-\alpha}\left(\hat{a}_{t+k} - \alpha\hat{y}_{t+k} + \alpha\hat{y}_{t+k} - \alpha\hat{y}_{t+k|t}\right)\\ &= \widehat{mc}_{t+k}^r - \frac{\alpha}{1-\alpha}\left(\hat{y}_{t+k} - \hat{y}_{t+k|t}\right).\end{aligned}$$

Since

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\varepsilon} Y_{t+k} \Rightarrow \hat{y}_{t+k} - \hat{y}_{t+k|t} = \varepsilon(\hat{p}_t^* - \hat{p}_{t+k}).$$

$$\Rightarrow \widehat{mc}_{t+k|t}^r = \widehat{mc}_{t+k}^r - \frac{\alpha\varepsilon}{1-\alpha}(\hat{p}_t^* - \hat{p}_{t+k}).$$

With CRS: $\alpha = 0 \Rightarrow \widehat{mc}_{t+k|t}^r = \widehat{mc}_{t+k}^r$. The real marginal cost is common across firms regardless whether they can reset price at t.

Since

$$\widehat{mc}_{t+k|t}^r = \widehat{mc}_{t+k}^r - \frac{\alpha\varepsilon}{1-\alpha}(\hat{p}_t^* - \hat{p}_{t+k}),$$

we have

$$\hat{p}_t^* = (1-\beta\theta)E_t\sum_{k=0}^{\infty}(\beta\theta)^k\left[\Theta\widehat{mc}_{t+k}^r + \hat{p}_{t+k}\right], \qquad \Theta = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon} \leq 1.$$

Derivation:

$$\begin{aligned}\hat{p}_t^* &= (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k\left[\widehat{mc}_{t+k}^r - \frac{\alpha\varepsilon}{1-\alpha}(\hat{p}_t^* - \hat{p}_{t+k}) + \hat{p}_{t+k}\right]\right\},\\ \hat{p}_t^* &= -\frac{\alpha\varepsilon}{1-\alpha}\hat{p}_t^* + (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k\left[\widehat{mc}_{t+k}^r + \frac{1-\alpha+\alpha\varepsilon}{1-\alpha}\hat{p}_{t+k}\right]\right\},\\ \frac{1-\alpha+\alpha\varepsilon}{1-\alpha}\hat{p}_t^* &= (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k\left[\widehat{mc}_{t+k}^r + \frac{1-\alpha+\alpha\varepsilon}{1-\alpha}\hat{p}_{t+k}\right]\right\},\\ \hat{p}_t^* &= (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k[\Theta\widehat{mc}_{t+k}^r + \hat{p}_{t+k}]\right\}.\end{aligned}$$

$$\begin{aligned}
\hat{p}_t^* &= (1-\beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + \beta\theta\mathbb{E}_t\hat{p}_{t+1}^* \quad (*),\\
\beta\theta\hat{p}_{t+1}^* &= \beta\theta(1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k[\Theta\widehat{mc}_{t+1+k}^r + \hat{p}_{t+1+k}]\right\}\\
&= (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^{k+1}[\Theta\widehat{mc}_{t+(k+1)}^r + \hat{p}_{t+(k+1)}]\right\}.
\end{aligned}$$

Let $k' = k+1, \forall k$:

$$\beta\theta\hat{p}_{t+1}^* = (1-\beta\theta)\mathbb{E}_t\left\{\sum_{k'=1}^{\infty}(\beta\theta)^{k'}[\Theta\widehat{mc}_{t+k'}^r + \hat{p}_{t+k'}]\right\}.$$

k' is any index; we can use k. (*) just take out the term of $k = 0$. Combine (*) with log-linearized price index:

$$\hat{p}_t = \theta\hat{p}_{t-1} + (1-\theta)\hat{p}_t^*.$$

$$\hat{p}_t = \theta\hat{p}_{t-1} + (1-\theta)(1-\beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + (1-\theta)\beta\theta\mathbb{E}_t\hat{p}_{t+1}^*,$$

$$\text{since } \hat{p}_{t+1}^* = \frac{\mathbb{E}_t\hat{p}_{t+1} - \theta\hat{p}_t}{1-\theta},$$

$$\hat{p}_t = \theta\hat{p}_{t-1} + (1-\theta)(1-\beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + (1-\theta)\beta\theta\frac{\mathbb{E}_t\hat{p}_{t+1} - \theta\hat{p}_t}{1-\theta}.$$

$$\begin{aligned}
\hat{p}_t &= \theta\hat{p}_{t-1} + (1-\theta)(1-\beta\theta)\Theta\widehat{mc}_t^r + (1-\theta-\beta\theta+\beta\theta^2)\hat{p}_t + \beta\theta\mathbb{E}_t\hat{p}_{t+1} - \beta\theta^2\hat{p}_t,\\
\hat{p}_t - \theta\hat{p}_{t-1} - (1-\theta)\hat{p}_t &= (1-\theta)(1-\beta\theta)\Theta\widehat{mc}_t^r + \beta\theta(\mathbb{E}_t\hat{p}_{t+1} - \hat{p}_t),\\
\theta(\hat{p}_t - \hat{p}_{t-1}) &= (1-\theta)(1-\beta\theta)\Theta\widehat{mc}_t^r + \beta\theta\mathbb{E}_t\hat{\pi}_{t+1},\\
\hat{\pi}_t &= \underbrace{\frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta}_{\lambda}\,\widehat{mc}_t^r + \beta\mathbb{E}_t\hat{\pi}_{t+1}.
\end{aligned}$$

$$\Rightarrow \hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \lambda\widehat{mc}_t^r \quad \text{iterate forward}$$

$$\Rightarrow \hat{\pi}_t = \lambda\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t\widehat{mc}_{t+k}^r.$$

λ is the slope of NKPC w.r.t. real marginal cost.
$\theta \uparrow$ sticky price $\Rightarrow \lambda \downarrow \Rightarrow$ flatter Phillips curve.
$\Theta \downarrow$ stronger decreasing returns $\Rightarrow \lambda \downarrow$.

Inflation results from aggregate consequences of purposeful price-setting decisions by firms, which adjust their prices in light of current and anticipated cost conditions.

8.10 Relate marginal cost with output gap

$$\begin{aligned}
\widehat{mc}_t^r &= \hat{w}_t - \hat{p}_t - \frac{1}{1-\alpha}(\hat{a}_t - \alpha\hat{y}_t),\\
\hat{w}_t - \hat{p}_t &= \sigma\hat{c}_t + \varphi\hat{n}_t \Rightarrow \hat{w}_t - \hat{p}_t = \sigma\hat{y}_t + \varphi\hat{n}_t,\\
\Rightarrow \widehat{mc}_t^r &= \frac{\varphi+\alpha+\sigma-\alpha\sigma}{1-\alpha}\hat{y}_t - \frac{\varphi+1}{1-\alpha}\hat{a}_t.
\end{aligned}$$

Real marginal cost rises with output, falls with productivity.

1. Labor becomes harder to expand due to household requires more compensation.
2. Decreasing return to scale: labor becomes less productive at the margin.

Under flexible price:

$$P_t(i) = \frac{\varepsilon}{\varepsilon - 1}\frac{W_t}{A_t F_N, t(N_t(i))} = \mu MC_t = P_t,$$

$$MC_t^r = \frac{MC_t}{P_t} = \frac{1}{\mu} \Rightarrow \widehat{mc}_t^r = 0, \quad \text{real marginal cost has no variation.}$$

Define the output under flexible price as natural output (long-run output) Y_t^n.

$$\Rightarrow \text{log deviation } \hat{y}_t^n \Rightarrow \hat{y}_t^n = \frac{\varphi + 1}{\varphi + \alpha + \sigma - \alpha\sigma}\hat{a}_t.$$

Define output gap as $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$.

$$\begin{aligned}
\text{Then } \widehat{mc}_t^r &= \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha}(\hat{y}_t^n + \tilde{y}_t) - \frac{\varphi+1}{1-\alpha}\hat{a}_t \\
&= \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha}\frac{\varphi + 1}{\varphi + \alpha + \sigma - \alpha\sigma}\hat{a}_t + \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha}\tilde{y}_t - \frac{\varphi+1}{1-\alpha}\hat{a}_t \\
&= \frac{\varphi + \alpha + \sigma(1-\alpha)}{1-\alpha}\tilde{y}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)\tilde{y}_t.
\end{aligned}$$

NKPC:

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \lambda \widehat{mc}_t^r \quad (\hat{\pi}_{t+1} = \pi_{t+1})$$

$$\Rightarrow \hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t, \quad \kappa = \lambda\left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right), \quad \lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta, \quad \Theta = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon}.$$

When demand pushes output above natural output, firms face tighter resource constraints, and the marginal cost rises.

1. σ: demand / consumption smoothing channel.

$$Y_t > Y_t^n \Rightarrow C_t > C_t^n \Rightarrow MU \text{ of leisure} \uparrow \Rightarrow \text{tends to work less, but to keep } N \Rightarrow \frac{W}{P} \uparrow \Rightarrow MC^r \uparrow.$$

$$\begin{aligned}
&\sigma \uparrow \Rightarrow \text{less willing to substitute consumption intertemporally} \\
&\Rightarrow \text{stronger intertemporal } C, N \text{ substitution} \\
&\Rightarrow \text{stronger wage pressure} \Rightarrow \text{stronger MC pressure.}
\end{aligned}$$

2. φ: labor market channel.

$$\varphi \uparrow \Rightarrow \text{households dislike work} \Rightarrow W \text{ increases by more} \Rightarrow MC^r \text{ increases by more.}$$

3. α: diminishing-returns-to-scale channel.

$$\alpha \uparrow \Rightarrow \text{to increase output by 1 unit} \Rightarrow \alpha \uparrow \text{ units increase in labor} \Rightarrow W/P \uparrow \Rightarrow MC^r \uparrow.$$

8.11 Dynamic IS curve

$$
\begin{aligned}
\hat{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}\} - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}\right), \\
\hat{y}_t^n + \tilde{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}^n + \tilde{y}_{t+1}\} - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}\right), \\
\tilde{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}^n - \hat{y}_t^n\} + \mathbb{E}_t\{\tilde{y}_{t+1}\} - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}).
\end{aligned}
$$

Define natural interest rate as:

$$
\hat{r}_t^n = \sigma\mathbb{E}_t\{\hat{y}_{t+1}^n - \hat{y}_t^n\} = \frac{\sigma(1+\varphi)}{\varphi+\alpha+\sigma(1-\alpha)}\mathbb{E}_t\{\Delta\hat{a}_{t+1}\} = \sigma\psi_{ya}^n\mathbb{E}_t\{\Delta\hat{a}_{t+1}\}.
$$

$$
\Rightarrow \tilde{y}_t = -\frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.
$$

Define

$$
\hat{r}_t = \hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}.
$$

$$
\Rightarrow \tilde{y}_t = -\frac{1}{\sigma}(\hat{r}_t - \hat{r}_t^n) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.
$$

Assume $\lim_{T\to\infty}\mathbb{E}_t\{\tilde{y}_{t+T}\} = 0$ and iterate forward:

$$
\Rightarrow \tilde{y}_t = -\frac{1}{\sigma}\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\hat{r}_{t+k} - \hat{r}_{t+k}^n)\right\}.
$$

Output gap is proportional to the sum of current and anticipated deviations between the real interest rate and natural real interest rate.

NKPC:

$$
\hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t \quad \text{iterate forward}
$$

$$
\Rightarrow \hat{\pi}_t = \kappa\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t\{\tilde{y}_{t+k}\}, \quad \text{where } \kappa = \lambda\left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right).
$$

Inflation today is the discounted sum of the expected future output gap.

Basic New Keynesian non-policy block:

$$
\begin{cases}
\hat{\pi}_t = \kappa\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t\{\tilde{y}_{t+k}\}, & \text{NKPC: determines inflation given output gap path,} \\
\tilde{y}_t = -\frac{1}{\sigma}\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\hat{r}_{t+k} - \hat{r}_{t+k}^n)\right\}, & \text{DIS: determines output gap given real interest rate gap path.}
\end{cases}
$$

To close, we need more equations to pin down how nominal interest rate i / $\hat{i}_t$ evolves (monetary policy).

$$
\begin{cases}
\tilde{y}_t = -\frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n\right) + \mathbb{E}_t\{\tilde{y}_{t+1}\} & \dashrightarrow \text{DIS (Demand)} \\
\hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t & \dashrightarrow \text{NKPC (Supply)}
\end{cases}
$$

8.12 Dynamic monetary policy rule

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + v_t, \quad v_t \text{ is exogenous with zero mean.}$$

$$\phi_\pi > 0, \quad \phi_y > 0, \quad \text{assume S.S. } \bar{\Pi} = 0.$$

DIS:

$$\tilde{y}_t = -\frac{1}{\sigma}[\phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - (\hat{r}_t^n - v_t)] + \mathbb{E}_t\{\tilde{y}_{t+1}\} \quad (***)$$

$$\Rightarrow \left(1 + \frac{\phi_y}{\sigma}\right)\tilde{y}_t + \frac{\phi_\pi}{\sigma}\hat{\pi}_t = \mathbb{E}_t\{\tilde{y}_{t+1}\} + \frac{1}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \frac{1}{\sigma}(\hat{r}_t^n - v_t).$$

$$-\kappa\tilde{y}_t + \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\}.$$

Write in matrix form in system of $[\tilde{y}_t, \hat{\pi}_t]'$:

$$\underbrace{\begin{bmatrix} 1+\phi_y/\sigma & \phi_\pi/\sigma \\ -\kappa & 1 \end{bmatrix}}_{M} \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} (\hat{r}_t^n - v_t).$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = M^{-1} \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + M^{-1} \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} (\hat{r}_t^n - v_t).$$

$$M^{-1} = \frac{1}{|M|} \begin{bmatrix} 1 & -\phi_\pi/\sigma \\ \kappa & 1+\phi_y/\sigma \end{bmatrix} = \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma+\phi_y \end{bmatrix}, \quad \Omega = \frac{1}{\sigma+\phi_y+\kappa\phi_\pi}.$$

$$A_T = \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma+\phi_y \end{bmatrix} \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} = \begin{bmatrix} \sigma & 1-\beta\phi_\pi \\ \kappa\sigma & \kappa+\beta(\sigma+\phi_y) \end{bmatrix} \Omega,$$

$$B_T = \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma+\phi_y \end{bmatrix} \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega.$$

$$\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + B_T(\hat{r}_t^n - v_t).$$

Both $\tilde{y}_t$ and $\hat{\pi}_t$ are not predetermined $\stackrel{BK}{\Longrightarrow}$ both eigenvalues of A_T are within the unit circle.

For a 2×2 matrix A_T, both eigenvalues lie inside unit circle iff Jury/Schur condition holds:

$$\begin{cases} 1-\det(A_T) > 0 \quad (D < 1), \\ 1-\operatorname{tr}(A_T)+\det(A_T) > 0 \quad (p(1) > 0), \\ 1+\operatorname{tr}(A_T)+\det(A_T) > 0 \quad (p(-1) > 0). \end{cases}$$

$$\det(A_T) = \Omega^2[\sigma(\kappa+\beta(\sigma+\phi_y)) - (1-\beta\phi_\pi)\kappa\sigma] = \Omega^2\sigma\beta(\sigma+\phi_y+\kappa\phi_\pi) = \frac{\beta\sigma}{\sigma+\phi_y+\kappa\phi_\pi},$$

$$\operatorname{tr}(A_T) = \Omega[\sigma+\kappa+\beta(\sigma+\phi_y)] = \frac{\sigma+\kappa+\beta(\sigma+\phi_y)}{\sigma+\phi_y+\kappa\phi_\pi}.$$

$$1^\circ \quad 1-\det(A_T) = 1 - \frac{\beta\sigma}{\sigma+\phi_y+\kappa\phi_\pi} = \frac{(1-\beta)\sigma+\phi_y+\kappa\phi_\pi}{\sigma+\phi_y+\kappa\phi_\pi} > 0 \quad \text{satisfy},$$

$\beta < 1, \sigma > 0, \phi_y > 0, \kappa > 0, \phi_\pi > 0.$

$$2^\circ \quad 1 - \mathrm{tr}(A_T) + \det(A_T) = 1 - \frac{\sigma + \kappa + \beta(\sigma + \phi_y)}{\sigma + \phi_y + \kappa\phi_\pi} + \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi}$$
$$= \frac{\kappa(\phi_\pi - 1) + (1-\beta)\phi_y}{\sigma + \phi_y + \kappa\phi_\pi} > 0$$

$$\Rightarrow \kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \quad \text{satisfy if.}$$

$$3^\circ \quad 1 + \mathrm{tr}(A_T) + \det(A_T) = \frac{(\sigma + \phi_y + \kappa\phi_\pi) + (\sigma + \kappa + \beta(\sigma + \phi_y)) + \beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} > 0 \quad \text{satisfy.}$$

The system is stable iff: $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0.$

If $\phi_y = 0 \Rightarrow \phi_\pi > 1$, consistent with the Taylor principle: when inflation rises by 1%, the central bank needs to raise the nominal interest rate by $> 1\%$.

$$r_t = i_t - \mathbb{E}_t\pi_{t+1} \qquad \text{Fisher equation.}$$

$$\mathbb{E}_t\pi_{t+1} \uparrow 1\%, \quad i_t \uparrow > 1\% \Rightarrow r_t \uparrow \Rightarrow \mathbb{E}_t\pi_{t+1} \downarrow.$$

8.13 Monetary policy shock

Assume v_t follows AR(1):

$$v_t = \rho_v v_{t-1} + \varepsilon_t^v, \qquad \rho_v \in [0,1].$$

$$\varepsilon_t^v > 0: \text{ contractionary MP shock,} \qquad \varepsilon_t^v < 0: \text{ expansionary MP shock.}$$

For simplicity, shut down technology shock:

$$\hat{a}_t = 0 \Rightarrow \hat{y}_t^n = 0, \quad \hat{r}_t^n = 0, \quad \forall t.$$

Guess

$$\tilde{y}_t = \psi_{yv} v_t, \qquad \hat{\pi}_t = \psi_{\pi v} v_t.$$

Plug into (***):

$$\begin{aligned}
\psi_{yv} v_t &= -\frac{1}{\sigma}(\phi_\pi \psi_{\pi v} v_t + \phi_y \psi_{yv} v_t + v_t - \rho_v \psi_{\pi v} v_t) + \rho_v \psi_{yv} v_t, \\
-\sigma \psi_{yv} &= \phi_\pi \psi_{\pi v} + \phi_y \psi_{yv} + 1 - \rho_v \psi_{\pi v} - \sigma \rho_v \psi_{yv}, \\
[\sigma(1-\rho_v) + \phi_y]\psi_{yv} &= (\rho_v - \phi_\pi)\psi_{\pi v} - 1 \quad (1).
\end{aligned}$$

From NKPC:

$$\begin{aligned}
\hat{\pi}_t &= \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t, \\
\psi_{\pi v} v_t &= \beta \rho_v \psi_{\pi v} v_t + \kappa \psi_{yv} v_t, \\
\psi_{yv} &= \frac{1-\beta\rho_v}{\kappa} \psi_{\pi v} \quad (2).
\end{aligned}$$

Plug (2) into (1):

$$\psi_{\pi v} = -\frac{\kappa}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y] + \kappa(\phi_\pi - \rho_v)}.$$

$$\hat{\pi}_t = \underbrace{-\kappa \Lambda_\nu}_{<0} v_t$$

$$\Lambda_v = \frac{1}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y] + \kappa(\phi_\pi - \rho_v)} > 0.$$

where

$$\kappa = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)\lambda = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)\frac{(1-\theta)(1-\beta\theta)}{\theta}\frac{1-\alpha}{1-\alpha+\alpha\varepsilon} > 0.$$

$$\tilde{y}_t = \underbrace{-(1-\beta\rho_v)\Lambda_v}_{<0} v_t$$

$$\begin{aligned}
\hat{r}_t &= \hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} = \phi_\pi \psi_{\pi v} v_t + \phi_y \psi_{yv} v_t + v_t + \kappa \Lambda_v \rho_v v_t \\
&= \underbrace{\sigma(1-\rho_v)(1-\beta\rho_v)\Lambda_v}_{>0} v_t.
\end{aligned}$$

$$\hat{i}_t = \underbrace{\left[\sigma(1-\rho_v)(1-\beta\rho_v) - \kappa\rho_v\right]\Lambda_v}_{\text{coefficient on } v_t} v_t.$$

ambiguous; depends on ρ_v. if $\rho_v \uparrow$, $\hat{i}_t \downarrow$.

If monetary policy is more persistent, $\hat{i}_t \downarrow$ real interest rate decreases.

$$\text{Therefore} \begin{cases} v_t = \rho_v v_{t-1} + \varepsilon_t^v, & \rho_v \in [0,1], \\ \hat{\pi}_t = -\kappa \Lambda_v v_t, \\ \tilde{y}_t = -(1-\beta\rho_v)\Lambda_v v_t, \\ \hat{r}_t = \sigma(1-\rho_v)(1-\beta\rho_v)\Lambda_v v_t, \\ \hat{i}_t = \left[\sigma(1-\rho_v)(1-\beta\rho_v) - \kappa\rho_v\right]\Lambda_v v_t. \end{cases}$$

Calibration: $\beta = 0.99$ (quarterly data), $\sigma = 1$ (log utility), $\varphi = 1$, $\alpha = 1/3$, $\varepsilon = 6$, $\theta = 2/3$, $\bar{y} = 4$.
With $\phi_\pi = 1.5$, $\phi_y = 0.5$, $\rho_v = 0.5$, and shock $\varepsilon_t^v = 25$bp: $\hat{\pi}_t < 0$, $\tilde{y}_t < 0$, $\hat{y}_t < 0$, $\hat{i}_t > 0$, $\hat{m}_t < 0$.

8.14 Technology shock

Assume an AR(1) process of $\hat{a}_t$:

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \qquad \rho_a \in [0,1].$$

$$\hat{r}_t^n = \sigma \mathbb{E}_t\{\hat{y}_{t+1}^n - \hat{y}_t^n\} = \sigma\psi_{ya}^n \mathbb{E}_t\{\Delta\hat{a}_{t+1}\} = -\sigma\psi_{ya}^n(1-\rho_a)\hat{a}_t.$$

Setting $v_t = 0, \forall t$. Assume:

$$\tilde{y}_t = \psi_{yr}\hat{r}_t^n, \qquad \hat{\pi}_t = \psi_{\pi r}\hat{r}_t^n.$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + B_T(\hat{r}_t^n - v_t).$$

Opposite sign to v_t:

$$\psi_{yr} = -\psi_{yv}, \qquad \psi_{\pi r} = -\psi_{\pi v}.$$

$$\tilde{y}_t = (1-\beta\rho_a)\Lambda_a\hat{r}_t^n = -\sigma\psi_{ya}^n(1-\rho_a)(1-\beta\rho_a)\Lambda_a\hat{a}_t,$$
$$\hat{\pi}_t = \kappa\Lambda_a\hat{r}_t^n = -\sigma\psi_{ya}^n(1-\rho_a)\kappa\Lambda_a\hat{a}_t,$$

where

$$\Lambda_a = \frac{1}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y] + \kappa(\phi_\pi - \rho_a)} > 0.$$

$$\begin{aligned} \hat{y}_t &= \hat{y}_t^n + \tilde{y}_t = \psi_{ya}^n\hat{a}_t - \sigma\psi_{ya}^n(1-\rho_a)(1-\beta\rho_a)\Lambda_a\hat{a}_t \\ &= \psi_{ya}^n\left[1 - \underbrace{\sigma(1-\rho_a)(1-\beta\rho_a)\Lambda_a}_{(*)}\right]\hat{a}_t. \end{aligned}$$

$$\begin{aligned} 1-(*) &= 1 - \sigma(1-\rho_a)(1-\beta\rho_a)\frac{1}{\sigma(1-\rho_a)(1-\beta\rho_a) + \phi_y(1-\beta\rho_a) + \kappa(\phi_\pi-\rho_a)} \\ &= \frac{\sigma(1-\rho_a)(1-\beta\rho_a) + \phi_y(1-\beta\rho_a) + \kappa(\phi_\pi-\rho_a) - \sigma(1-\rho_a)(1-\beta\rho_a)}{\sigma(1-\rho_a)(1-\beta\rho_a) + \phi_y(1-\beta\rho_a) + \kappa(\phi_\pi-\rho_a)} \\ &= \frac{\phi_y(1-\beta\rho_a) + \kappa(\phi_\pi-\rho_a)}{\sigma(1-\rho_a)(1-\beta\rho_a) + \phi_y(1-\beta\rho_a) + \kappa(\phi_\pi-\rho_a)}. \end{aligned}$$

$$\Rightarrow \hat{y}_t = \frac{\phi_y(1-\beta\rho_a)+\kappa(\phi_\pi-\rho_a)}{\sigma(1-\rho_a)(1-\beta\rho_a)+\phi_y(1-\beta\rho_a)+\kappa(\phi_\pi-\rho_a)}\psi^n_{ya}\hat{a}_t.$$

$$\begin{aligned}\phi_y(1-\beta\rho_a)+\kappa(\phi_\pi-\rho_a) &= \kappa(\phi_\pi-1)+(1-\beta)\phi_y+\kappa-\rho_a\kappa+\beta\phi_y-\beta\rho_a\phi_y\\ &= \underbrace{\kappa(\phi_\pi-1)+(1-\beta)\phi_y}_{\text{stable condition}}+\underbrace{(1-\rho_a)\kappa}_{0\le\rho_a<1}+\underbrace{(1-\rho_a)\beta\phi_y}_{0\le\rho_a<1}>0.\end{aligned}$$

y_t moves positively with a_t.

$$(1-\alpha)\hat{n}_t = \hat{y}_t-\hat{a}_t = \left[(\psi^n_{ya}-1)-\sigma\psi^n_{ya}(1-\rho_a)(1-\beta\rho_a)\Lambda_a\right]\hat{a}_t.$$

In calibration: $\sigma=1 \Rightarrow \psi^n_{ya}=\frac{1+\varphi}{\varphi+\alpha+\sigma(1-\alpha)}=1 \Rightarrow n_t$ declines when a_t increases.

8.15 Equilibrium with exogenous money supply: $\Delta\hat{m}_t$

$$\frac{M_t}{P_t}=L(i_t,Y_t).$$

Close the model with money demand equation instead of Taylor rule. Define log real money balance:

$$\widehat{\mathbb{M}}_t=\hat{m}_t-\hat{p}_t \Rightarrow \widehat{\mathbb{M}}_t=\hat{y}_t-\eta\hat{i}_t.$$

$$\Rightarrow \tilde{y}_t-\eta\hat{i}_t=\widehat{\mathbb{M}}_t-\hat{y}^n_t \Rightarrow \hat{i}_t=\frac{1}{\eta}\left(\tilde{y}_t+\hat{y}^n_t-\widehat{\mathbb{M}}_t\right).$$

Output $\uparrow\Rightarrow$ money demand $\uparrow\Rightarrow i\uparrow$. Real money supply $\uparrow\Rightarrow i\downarrow$.

$$DIS \Rightarrow (1+\sigma\eta)\tilde{y}_t=\sigma\eta\mathbb{E}_t\{\tilde{y}_{t+1}\}+\widehat{\mathbb{M}}_t+\eta\mathbb{E}_t\{\hat{\pi}_{t+1}\}+\eta\hat{r}^n_t-\hat{y}^n_t.$$

From $\widehat{\mathbb{M}}_t$ definition:

$$\widehat{\mathbb{M}}_t-\widehat{\mathbb{M}}_{t-1}=\Delta\hat{m}_t-\hat{\pi}_t \Rightarrow \widehat{\mathbb{M}}_{t-1}=\widehat{\mathbb{M}}_t+\hat{\pi}_t-\Delta\hat{m}_t.$$

$$A_{M,0}\begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\\ \widehat{\mathbb{M}}_{t-1}\end{bmatrix}=A_{M,1}\begin{bmatrix}\mathbb{E}_t\{\tilde{y}_{t+1}\}\\ \mathbb{E}_t\{\hat{\pi}_{t+1}\}\\ \widehat{\mathbb{M}}_t\end{bmatrix}+B_M\begin{bmatrix}\hat{r}^n_t\\ \hat{y}^n_t\\ \Delta\hat{m}_t\end{bmatrix}.$$

$\widehat{\mathbb{M}}_{t-1}$ is predetermined and $\tilde{y}_t$, $\hat{\pi}_t$ are not predetermined. Stationary solution will exist iff

$$A_M=A^{-1}_{M,0}A_{M,1} \text{ has two eigenvalues inside the unit circle.}$$

Assume the monetary policy shock follows AR(1):

$$\Delta\hat{m}_t=\rho_m\Delta\hat{m}_{t-1}+\varepsilon^m_t, \qquad \rho_m\in[0,1].$$

Let

$$\hat{r}^n_t=\hat{y}^n_t=0, \quad \varepsilon^m_t=0.25, \quad t=0, \quad \varepsilon^m_t=0, \quad \forall t\neq 0.$$

An expansionary monetary policy shock:

$$\Delta\hat{m}_t \uparrow \Rightarrow \tilde{y}_t \uparrow,\ \hat{\pi}_t \uparrow,\ \widehat{\mathbb{M}}_t \uparrow, \qquad \hat{i}_t \uparrow\text{: no liquidity effect } (\sigma = 1), \qquad \hat{r}_t \downarrow.$$

$$\hat{a}_t \uparrow \Rightarrow \hat{r}^n_t \downarrow,\ \hat{y}_t \uparrow,\ \hat{\pi}_t \downarrow,\ \hat{n}_t \downarrow,\ \hat{r}_t \uparrow,\ \hat{i}_t \approx 0, \qquad \hat{y}^n_t \uparrow\uparrow,\ \tilde{y}_t \downarrow.$$

Liquidity effect: when real money balance $\uparrow$, $\widehat{\mathbb{M}}_t \uparrow \Rightarrow$ households hold too much money than they want.

$$\Rightarrow \hat{i}_t \downarrow \text{ to induce households to hold that money.}$$

Why $\sigma = 1$ liquidity effect is gone?

$$\hat{i}_t = \frac{1}{\eta}\left(\tilde{y}_t + \hat{y}^n_t - \widehat{\mathbb{M}}_t\right) \quad \text{with } \hat{y}^n_t = 0.$$

DIS:

$$\tilde{y}_t = -\frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t\hat{\pi}_{t+1}) + \mathbb{E}_t\tilde{y}_{t+1}.$$

Set $\hat{y}^n_t = 0$:

$$\hat{i}_t = \frac{1}{\eta}(\tilde{y}_t - \widehat{\mathbb{M}}_t).$$

Liquidity effect: $\Delta\hat{m}_t > 0$, $\hat{i}_t < 0$, requires $\widehat{\mathbb{M}}_t > \tilde{y}_t$. Real balance must rise by more than the output gap.

$$DIS \Rightarrow \hat{i}_t = \underbrace{\mathbb{E}_t\hat{\pi}_{t+1}}_{>0} + \sigma \underbrace{\left[\mathbb{E}_t\hat{y}_{t+1} - \hat{y}_t\right]}_{<0}.$$

$$\hat{i}_t < 0 \Leftrightarrow \sigma(\tilde{y}_t - \mathbb{E}_t\tilde{y}_{t+1}) > \mathbb{E}_t\hat{\pi}_{t+1}.$$

After expansionary money shock, $\tilde{y}_t \uparrow \Rightarrow \tilde{y}_t > \mathbb{E}_t\tilde{y}_{t+1} \Rightarrow \mathbb{E}_t\tilde{y}_{t+1} - \tilde{y}_t < 0$.

When $\sigma = 1$, the negative term is not large enough to dominate the positive term.

Liquidity effect disappears whenever:

$$\mathbb{E}_t\hat{\pi}_{t+1} > \tilde{y}_t - \mathbb{E}_t\tilde{y}_{t+1}.$$

Thus, $\sigma \gg 1$, $\rho_m \ll 1$ liquidity effect presents; money shock not persistent $\Rightarrow$ expected inflation $\downarrow$.

Lecture 9

New Keynesian Policy Design

9.1 Efficient allocation and distortions

Monopolistic competition + flexible prices $\Rightarrow$ efficient allocation, with a subsidy to correct distortions of monopolistic competition.

Sticky prices + policy that fully stabilizes the price level $\Rightarrow$ efficient allocation.

9.2 Efficient allocation under social planner's problem

$$\max_{\{C_t,N_t\}} U(C_t,N_t)$$

$$\text{s.t.} \qquad C_t(i) = A_t F(N_t(i)), \quad \text{for all } i \in [0,1],$$

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i\right)^{\frac{\varepsilon}{\varepsilon-1}}, \qquad N_t = \int_0^1 N_t(i)\mathrm{d}i.$$

$$C_t(i) = C_t, \qquad \forall i \in [0,1], \tag{1}$$

$$N_t(i) = N_t, \qquad \forall i \in [0,1], \tag{2}$$

$$-\frac{U_{N,t}}{U_{C,t}} = MPN_t = (1-\alpha)A_t N_t^{-\alpha}, \tag{3}$$

nominal rigidity $\Rightarrow \begin{cases} C_t(i) \neq C_t, \\ N_t(i) \neq N_t, \end{cases}$

nominal rigidity or monopolistic competition $\Rightarrow$ (3) does not hold.

(1)&(2) symmetric allocation $\Rightarrow$ $MPN_t = MPN_t(i)$.

Optimality conditions can be met due to:

i. All goods enter utility symmetrically.
   For planner, weight are the same even distribution $C_t(i)$.

ii. Identical technology of production.
   For planner, technology for i is the same, even distribution $N_t(i)$.

iii. Concavity of utility.
   For planner, there is an optimal allocation of $C_t(i)$ and $N_t(i)$.

$$\mathrm{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}}, \qquad \mathrm{MRT}_{C,N} = \frac{\mathrm{d}C_t}{\mathrm{d}N_t} = F_N(N_t)A_t \quad \text{output transfered one-for-one to consumption.}$$

$$\Rightarrow \mathrm{MRS}_{C,N} = \mathrm{MRT}_{C,N} = \mathrm{MPN}.$$

9.3 Suboptimality in NK

Two sources of distortion:

1) Monopolistic competitive firms $\Rightarrow$ market power.

2) Sticky price.

Let's study (1) separately. Assume flexible price. With isoelastic demand function, optimal pricing:

$$P_t = \mu \frac{W_t}{\mathrm{MPN}_t}, \qquad \mu = \frac{\varepsilon}{\varepsilon - 1} \quad \text{markup.}$$

$$\Rightarrow \mathrm{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mathrm{MPN}_t}{\mu}.$$

$\mu > 1 \Rightarrow$ low employment and output, and optimal condition (3) is violated.

Let τ denotes the employment subsidy; subsidy financed by lump-sum tax, no distortions.

$$P_t = \mu \frac{(1-\tau)W_t}{\mathrm{MPN}_t} \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mathrm{MPN}_t}{\mu(1-\tau)}.$$

$$\mathrm{MPN}_t = \frac{\mathrm{MPN}_t}{\mu(1-\tau)} \Rightarrow \mu(1-\tau) = 1 \Rightarrow \tau = \frac{1}{\varepsilon}.$$

The optimal allocation can be achieved by giving employment subsidy of $\tau = \frac{1}{\varepsilon}$.

Now study (2) price stickiness.

9.3.1 Average markup varies over time

$$\mu_t = \frac{P_t}{(1-\tau)W_t/\mathrm{MPN}_t} = \frac{\mu P_t}{W_t/\mathrm{MPN}_t} \Rightarrow \frac{W_t}{P_t} = \frac{\mu}{\mu_t}\mathrm{MPN}_t.$$

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mu}{\mu_t}\mathrm{MPN}_t.$$

If $\mu_t \neq \mu$, optimal allocation condition (3) is violated.

9.3.2 Staggered price setting: sticky price

$$P_t(i) \neq P_t(j), \qquad C_t(i) \neq C_t(j), \qquad N_t(i) \neq N_t(j).$$

Optimal conditions (1) and (2) are violated.

9.4 Optimal monetary policy in the basic NK

Assumptions:

1. An optimal subsidy to offset distortion of firms' market power.
2. Economy starts without relative price distortion:

$$P_{-1}(i) = P_{-1}, \qquad \text{for all } i \in [0,1].$$

Optimal policy should achieve:

1. $P_t = P_{t-1}$, $\forall t = 0, 1, 2, \ldots$, so that conditions (1) and (2) hold.
2. $\mu_t = \mu$, $\forall t$, so that condition (3) holds.

$$\Leftrightarrow \tilde{y}_t = \hat{y}_t - \hat{y}_t^n = 0, \quad \forall t; \qquad \hat{\pi}_t = 0, \quad \forall t.$$

DIS:

$$\tilde{y}_t = -\frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n\right) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^n, \qquad \forall t.$$

1. Stabilizing output: Not freezing output at constant.

$$\hat{y}_t = \hat{y}_t^n, \qquad \forall t.$$

Eliminating gap, not eliminating fluctuations.

$\hat{y}_t^n$ is subject to technology shock.

2. Stabilizing price: divine coincidence.

If central bank achieves price stability, then it automatically gets $\tilde{y}_t = 0$.

Stabilizing inflation $\Rightarrow$ stabilizing output gap.

9.4.1 How to implement the optimal policy?

Non-policy block:

$$\text{DIS:} \quad \tilde{y}_t = -\frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n\right) + \mathbb{E}_t\{\tilde{y}_{t+1}\},$$

$$\text{NKPC:} \quad \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t.$$

When $\hat{\pi}_t = 0, \ \forall t \Rightarrow \tilde{y}_t = 0, \ \forall t.$ divine coincidence.

Policy block:

Case a: An exogenous interest rate rule

$$\hat{i}_t = \hat{r}_t^n, \qquad \forall t.$$

$$\Rightarrow \begin{cases} \boxed{\tilde{y}_t} = \dfrac{1}{\sigma}E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t \end{cases}$$

Substitute the first equation into the second:

$$\begin{aligned}\hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\left(\frac{1}{\sigma}E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\}\right)\\ &= \kappa E_t\{\tilde{y}_{t+1}\} + \left(\beta + \frac{\kappa}{\sigma}\right)E_t\{\hat{\pi}_{t+1}\}.\end{aligned}$$

$$\Rightarrow \begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = A_0\begin{bmatrix}E_t\{\tilde{y}_{t+1}\}\\ E_t\{\hat{\pi}_{t+1}\}\end{bmatrix}, \qquad \text{where } A_0 = \begin{bmatrix}1 & \frac{1}{\sigma}\\ \kappa & \beta + \frac{\kappa}{\sigma}\end{bmatrix}.$$

$\tilde{y}_t = \hat{\pi}_t = 0$ is one solution associated with optimal policy, but not unique.

$\tilde{y}_t$ and $\hat{\pi}_t$ are neither predetermined.

$$\det(A_0) = \beta, \qquad \operatorname{tr}(A_0) = 1 + \beta + \frac{\kappa}{\sigma}.$$

$$P(1) = 1 - T + D = 1 - 1 - \beta - \frac{\kappa}{\sigma} + \beta = -\frac{\kappa}{\sigma} < 0.$$

$$P(-1) = 1 + T + D = 1 + 1 + \beta + \frac{\kappa}{\sigma} + \beta = 2 + 2\beta + \frac{\kappa}{\sigma} > 0.$$

There is one root between $(-1, 1)$ and another one outside the unit circle.

jump variables $>$ # unstable roots $\Rightarrow$ system is indeterminate.

No guarantee on the realization of $\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t$.

Case b: A Taylor-type interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi\hat{\pi}_t + \phi_y\tilde{y}_t, \qquad \phi_\pi \geq 0, \quad \phi_y \geq 0, \quad \forall t.$$

$$\begin{cases}\tilde{y}_t = -\dfrac{\phi_\pi}{\sigma}\hat{\pi}_t - \dfrac{\phi_y}{\sigma}\tilde{y}_t + \dfrac{1}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \mathbb{E}_t\{\tilde{y}_{t+1}\},\\ \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t.\end{cases}$$

$$\begin{cases}\tilde{y}_t = \dfrac{1-\beta\phi_\pi}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\tilde{y}_{t+1}\},\\ \hat{\pi}_t = \dfrac{\kappa+\beta\sigma+\beta\phi_y}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma\kappa}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\tilde{y}_{t+1}\}.\end{cases}$$

$$\begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = A_T\begin{bmatrix}\mathbb{E}_t\{\tilde{y}_{t+1}\}\\ \mathbb{E}_t\{\hat{\pi}_{t+1}\}\end{bmatrix},$$

$$A_T = \Omega\begin{bmatrix}\sigma & 1-\beta\phi_\pi\\ \kappa\sigma & \kappa+\beta(\sigma+\phi_y)\end{bmatrix}, \qquad \Omega = \frac{1}{\sigma+\phi_y+\kappa\phi_\pi}.$$

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t \quad \text{is always a solution.}$$

To guarantee uniqueness, we need two stable roots

notice that $x_t = A_T E_t x_{t+1}$, but BK is defined on $E_t x_{t+1} = A_T^{-1} x_t$,

2 stable roots in $A_T \Longleftrightarrow$ 2 unstable roots in A_T^{-1}.

$$\mathrm{tr}(A_T) = \Omega[\sigma + \kappa + \beta(\sigma + \phi_y)], \qquad \det(A_T) = \Omega\beta\sigma.$$

To have stable result, need $D < 1, \quad P(1) > 0, \quad P(-1) > 0.$

$$\begin{aligned} P(1) = 1 - \mathrm{tr}(A_T) + \det(A_T) &= 1 - \Omega[\sigma + \kappa + \beta\phi_y] > 0 \\ &\Rightarrow \sigma + \kappa + \beta\phi_y < \sigma + \phi_y + \kappa\phi_\pi \\ &\Rightarrow \kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \qquad (*) \end{aligned}$$

$$P(-1) = 1 + \mathrm{tr}(A_T) + \det(A_T) = 1 + \Omega[\sigma + \kappa + \beta\phi_y + \beta\sigma] > 0 \quad \text{for sure.}$$

$$\Delta = \mathrm{Tr}(A_T)^2 - 4\det(A_T) = \Omega^2[\sigma + \kappa + \beta(\sigma + \phi_y)]^2 - 4\Omega\beta\sigma, \qquad \text{sign does not matter.}$$

$$\begin{aligned} D = \det(A_T) = \Omega\beta\sigma &= \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} = \frac{1}{\frac{1}{\beta} + \frac{\phi_y + \kappa\phi_\pi}{\beta\sigma}} \\ &< \frac{1}{1 + \frac{\kappa - \kappa\phi_\pi + \kappa\phi_\pi}{\beta\sigma}} = \frac{1}{1 + \frac{\kappa}{\beta\sigma}} < 1. \end{aligned}$$

If $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$ is satisfied, $\quad \tilde{y}_t = \hat{\pi}_t = 0 \quad$ and $\quad \hat{i}_t = \hat{r}_t^n$ hold$\forall t$.

If $\phi_y = 0 \Rightarrow \kappa(\phi_\pi - 1) > 0 \Rightarrow \phi_\pi > 1$ Taylor principle.

If $\phi_y > 0$, output gap helps, lower bound on ϕ_π can be relaxed a little.

But $1-\beta$ is usually small, inflation is the main stabilizing force.

Case c: A forward-looking interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \phi_y \mathbb{E}_t\{\tilde{y}_{t+1}\}.$$

$$\begin{cases} \tilde{y}_t = \dfrac{1-\phi_\pi}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma - \phi_y}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t. \qquad (*) \end{cases}$$

$$\begin{cases} \tilde{y}_t = \dfrac{1-\phi_\pi}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma - \phi_y}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}, \qquad (**) \\ \hat{\pi}_t = \dfrac{\kappa(1-\phi_\pi) + \beta\sigma}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\kappa(\sigma - \phi_y)}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}. \qquad (***) \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_F \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix},$$

$$A_F = \begin{bmatrix} \frac{\sigma - \phi_y}{\sigma} & \frac{1-\phi_\pi}{\sigma} \\ \frac{\kappa(\sigma - \phi_y)}{\sigma} & \beta + \frac{\kappa(1-\phi_\pi)}{\sigma} \end{bmatrix}.$$

$$\mathrm{tr}(A_F) = \beta + \frac{\kappa(1-\phi_\pi) + \sigma - \phi_y}{\sigma}, \qquad \det(A_F) = \beta\frac{\sigma - \phi_y}{\sigma}.$$

$$\begin{aligned}P(1) &= 1 - \mathrm{tr}(A_F) + \det(A_F)\\ &= 1 - \beta - \frac{\kappa(1-\phi_\pi) + (\sigma - \phi_y)}{\sigma} + \beta\frac{\sigma - \phi_y}{\sigma} > 0\\ &\Rightarrow \kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \qquad (1) \quad \text{policy cannot be too passive.}\end{aligned}$$

$$\begin{aligned}P(-1) &= 1 + \mathrm{tr}(A_F) + \det(A_F)\\ &= 1 + \beta + \frac{\kappa}{\sigma}(1-\phi_\pi) + 1 - \frac{1}{\sigma}\phi_y + \beta - \frac{\beta}{\sigma}\phi_y > 0\\ &\Rightarrow \kappa(\phi_\pi - 1) + (1+\beta)\phi_y < 2\sigma(1+\beta) \qquad (2) \quad \text{policy cannot be too aggressive.}\end{aligned}$$

$$D < 1 \Rightarrow \beta\left(1 - \frac{\phi_y}{\sigma}\right) < 1, \quad \text{since } \beta \in (0,1) \text{ and } \phi_y \geq 0, \text{ so this holds.}$$

$$\Rightarrow \text{Under (1) and (2), we have 2 stable roots in } A_F, \text{ 2 unstable roots in } A_F^{-1}.$$

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t, \quad \text{is the unique solution.}$$

1) If $\phi_\pi > 1$, the coefficient on $\mathbb{E}_t\{\hat{\pi}_{t+1}\}$ is negative.

$$\mathbb{E}_t\{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

If $\phi_\pi < 1$, then

$$\mathbb{E}_t\{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow,$$

self-fulfilling inflation; policy cannot be too passive.

2) If policy is too aggressive so that condition (2) is violated,

$$P(-1) < 0, \quad \text{one eigenvalue crosses } -1, \quad \text{system unstable.}$$

But all optimal policy listed are not realistic, since policy rate must be adjusted one-for-one with the natural interest rate.

But $\hat{r}_t^n$ is usually unobservable, because it depends on the true model of the economy,

parameter values and the realized shocks. $\Rightarrow$ Simple rules are needed!

Due to divine coincidence, unobservable output gap is not the main issue, since $E_t\hat{\pi}_{t+1} = 0 \Rightarrow \tilde{y}_t = 0$.

$$\kappa(\phi_\pi - 1) + (1-\beta)\phi_y = 0 \quad \Longleftrightarrow \quad \phi_\pi = 1 - \frac{1-\beta}{\kappa}\phi_y.$$

$$\kappa(\phi_\pi - 1) + (1+\beta)\phi_y = 2\sigma(1+\beta) \quad \Longleftrightarrow \quad \phi_\pi = 1 + \frac{2\sigma(1+\beta)}{\kappa} - \frac{1+\beta}{\kappa}\phi_y.$$

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Figure 9.1: Determinacy region

9.5 Simple monetary policy rule

9.5.1 Rotemberg and Woodford (1999)

Second-order approximation to the utility losses due to deviations from efficient allocation:

$$W = \frac{1}{2}\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\kappa} \hat{\pi}_t^2 \right].$$

Per-period loss:

$$L = \frac{1}{2} \left[\left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \mathrm{Var}(\tilde{y}_t) + \frac{\varepsilon}{\kappa} \mathrm{Var}(\hat{\pi}_t) \right].$$

$\sigma \uparrow,\ \varphi \uparrow,\ \alpha \uparrow \Rightarrow L \uparrow.$ Curvature amplifies the effect (consumption, labor, technology).

$\varepsilon \uparrow,\ \frac{1}{\kappa} \uparrow \Rightarrow L \uparrow.$ Amplify losses from price dispersion; $\frac{1}{\lambda}$ amplifies the degree of price dispersion.

9.5.2 Taylor-type interest rate rule

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t, \qquad \phi_\pi > 0, \quad \phi_y > 0.$$

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + \phi_y \hat{y}_t^n.$$

Plug in:

$$\begin{cases} \text{DIS:} \quad (\sigma + \phi_y)\tilde{y}_t + \phi_\pi \hat{\pi}_t = \sigma \mathbb{E}_t\{\tilde{y}_{t+1}\} + \mathbb{E}_t\{\hat{\pi}_{t+1}\} + (\hat{r}_t^n - \phi_y \hat{y}_t^n), \\ \qquad \text{NKPC:} \quad \hat{\pi}_t = \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t. \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + B_T(\hat{r}_t^n - \phi_y \hat{y}_t^n),$$

$$A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ \kappa\sigma & \kappa + \beta(\sigma + \phi_y) \end{bmatrix}, \quad B_T = \Omega \begin{bmatrix} 1 \\ \kappa \end{bmatrix}, \quad \Omega = \frac{1}{\sigma + \phi_y + \kappa\phi_\pi}.$$

Assume

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \qquad \hat{y}_t^n = \psi_{ya}^n \hat{a}_t, \qquad \psi_{ya}^n = \frac{1+\phi}{\sigma + \phi + \alpha(1-\sigma)} > 0.$$

$$\Rightarrow v_t = \phi_y \hat{y}_t^n = \phi_y \psi_{ya}^n \hat{a}_t.$$

$$\hat{r}_t^n = \sigma \mathbb{E}_t\{\Delta \hat{y}_{t+1}^n\} = -\sigma \psi_{ya}^n (1 - \rho_a)\hat{a}_t.$$

$$\Delta \hat{y}_{t+1}^n = \hat{y}_{t+1}^n - \hat{y}_t^n = \psi_{ya}^n(\hat{a}_{t+1} - \hat{a}_t), \qquad \mathbb{E}_t \Delta \hat{y}_{t+1}^n = \psi_{ya}^n(\rho_a - 1)\hat{a}_t.$$

$$\hat{r}_t^n - v_t = -\psi_{ya}^n[\sigma(1-\rho_a) + \phi_y]\hat{a}_t.$$

$$B_T(\hat{r}_t^n - v_t) = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega \left\{ -\psi_{ya}^n \left[\sigma(1-\rho_a) + \phi_y \right] \right\} \hat{a}_t.$$

Shock loading: $\quad f(\phi_y) = \dfrac{\sigma(1-\rho_a) + \phi_y}{\sigma + \phi_y + \kappa\phi_\pi}, \qquad f'(\phi_y) = \dfrac{\sigma\rho_a + \kappa\phi_\pi}{(\sigma + \phi_y + \kappa\phi_\pi)^2} > 0.$

Shock term gets larger if the central bank responds more aggressively to output.

$\Rightarrow$ output gap and inflation variance $\uparrow$, welfare $\downarrow$.

$\phi_\pi \to +\infty \Rightarrow f(\phi_y) \to 0 \Rightarrow$ the exogenous driving force hitting the system goes to 0.

$\Rightarrow$ Aggressive inflation targeting gets arbitrarily close to the optimum.

9.6 A constant money growth rule: Friedman (1960)

$$\Delta\hat{m}_t = 0, \qquad \forall t.$$

$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta\hat{i}_t - \xi_t,$ $\quad$ ξ_t is an exogenous money demand shock: $\Delta\xi_t = \rho_\xi \Delta\xi_{t-1} + \varepsilon_t^\xi$.

Let

$\widehat{\mathbb{M}}_t^+ = \widehat{\mathbb{M}}_t + \xi_t,$ $\quad$ denotes real balance adjusted by exogenous component of money demand.

$$\widehat{\mathbb{M}}_t^+ = \tilde{y}_t + \hat{y}_t^n - \eta\hat{i}_t.$$

$$\Rightarrow \hat{i}_t = \frac{1}{\eta}\left(\tilde{y}_t + \hat{y}_t^n - \widehat{\mathbb{M}}_t^+\right).$$

$$\widehat{\mathbb{M}}_t^+ - \widehat{\mathbb{M}}_{t-1}^+ = \hat{m}_t - \hat{m}_{t-1} - \hat{p}_t + \hat{p}_{t-1} + \xi_t - \xi_{t-1} = 0 - \hat{\pi}_t + \Delta\xi_t.$$

$$\Rightarrow \widehat{\mathbb{M}}_{t-1}^+ = \widehat{\mathbb{M}}_t^+ + \hat{\pi}_t - \Delta\xi_t.$$

$$A_{M,0}\begin{bmatrix}\tilde{y}_t \\ \hat{\pi}_t \\ \widehat{\mathbb{M}}_{t-1}^+\end{bmatrix} = A_{M,1}\begin{bmatrix}\mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \\ \widehat{\mathbb{M}}_t^+\end{bmatrix} + B_M\begin{bmatrix}\hat{r}_t^n \\ \hat{y}_t^n \\ \Delta\xi_t\end{bmatrix}.$$

The degree of stability of money demand is a key element in determining the desirability of a money growth rule.

Lecture 10

Discretion vs. Commitment

Myopic optimiser
Optimize period by period.

discretion: period-by-period optimisation without committing to future actions.

Dynamic optimiser
Optimize the sum of present and future utilities.

commitment: central bank can credibly announce and follow a future path. Backward induction.

Recap: Optimal Monetary Policy is strict inflation targeting.

$$\hat{\pi}_t = 0 \Rightarrow \hat{y}_t = \hat{y}_t^n = \hat{y}_t^e.$$

However, possible and optimal but not feasible since central bank can not fully stabilize inflation.

Facing short-run tradeoff due to supply shock.

10.1 NK with imperfection at the supply side

Under flexible price:

$y_t^n \neq y_t^e$ (natural output $\neq$ efficient output) $\Rightarrow$ short-run tradeoff between quantity and prices.

Move oppositely facing supply shock.

Assume at steady state, the inefficiency associated with the flexible price is not there:

$$\bar{y} = \bar{y}^n = \bar{y}^e,$$

while exists short-run deviations:

$$\hat{y}_t \neq \hat{y}_t^n \neq \hat{y}_t^e.$$

Welfare losses are:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2\right).$$

Define welfare relevant output gap as

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^e = \log y_t - \log \bar{y} - \left(\log y_t^e - \log \bar{y}^e\right) = \log y_t - \log y_t^e,$$
$$\hat{\pi}_t = \log P_t - \log P_{t-1}, \qquad \alpha_x = \frac{\kappa}{\varepsilon}.$$

$$\left.\begin{aligned} \tilde{y}_t &= \hat{y}_t - \hat{y}_t^n = \hat{x}_t + (\hat{y}_t^e - \hat{y}_t^n), \\ \hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t \end{aligned}\right\} \Rightarrow \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t,$$

where

$$v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

- $E_t\{\hat{\pi}_{t+1}\}$: taken as given when choosing current-period policy.
- u_t: exogenous cost-push shock.

A nonzero u_t makes it impossible to attain simultaneously zero inflation and an efficient level of output. For example, u_t may come from exogenous changes in desired wage markups, price markups, labor income taxes, minimum wages, etc.

Assume u_t follows:

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u, \qquad \rho_u \in [0, 1).$$

$$\text{NKPC:} \quad \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t,$$

$$\text{DIS:} \qquad \hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

$\hat{r}_t^e$ is the efficient real interest rate that supports the efficient allocation.

10.2 Optimal policy under discretion (period-by-period optimization)

$$\min_{\hat{x}_t, \hat{\pi}_t} \quad \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \qquad \text{s.t.} \quad \hat{\pi}_t = \kappa \hat{x}_t + v_t,$$

where

$$v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

$E_t\{\hat{\pi}_{t+1}\}$: independent of current policy; $\quad u_t$: exogenous.

$$\mathcal{L} = \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 - \lambda(\hat{\pi}_t - \kappa \hat{x}_t - v_t).$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow 2\hat{\pi}_t = \lambda,$$

$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow 2\alpha_x \hat{x}_t = -\lambda\kappa \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t, \qquad \forall t \quad \text{targeting rule.}$$

To dampen the rise in inflation due to cost-push shock,

$$\hat{x}_t < 0 \Rightarrow \hat{y}_t < \hat{y}_t^e.$$

$\Rightarrow$ drive output below efficiency level; negative output gap $\Rightarrow$ policy trade-off.

Plug into NKPC:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t = \beta E_t\{\hat{\pi}_{t+1}\} - \frac{\kappa^2}{\alpha_x}\hat{\pi}_t + u_t,$$

$$\Rightarrow \quad \hat{\pi}_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} E_t\{\hat{\pi}_{t+1}\} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t.$$

Iterate forward:

$$\begin{cases} \hat{\pi}_t = \alpha_x \psi u_t, \\ \hat{x}_t = -\kappa \psi u_t, \end{cases} \qquad \psi = \frac{1}{\kappa^2 + \alpha_x(1 - \beta\rho_u)}.$$

Central bank lets output gap and inflation deviate from targets in proportion to the current value of the cost-push shock:

$$\hat{\pi}_t = \alpha_x \psi u_t, \qquad \hat{x}_t = -\kappa \psi u_t.$$

$$\psi_{0.5} = \frac{1}{\kappa^2 + \alpha_x\left(1 - \frac{\beta}{2}\right)} > \psi_0 = \frac{1}{\kappa^2 + \alpha_x}.$$

Optimal to partly accommodate inflation pressure from the cost-push shock by letting inflation rise.

Without any policy response, $\kappa = 0$:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

Guess $\hat{\pi}_t = A u_t$ and $E_t u_{t+1} = \rho_u u_t$:

$$A u_t = \beta A \rho_u u_t + u_t \Rightarrow A = \frac{1}{1 - \beta\rho_u}.$$

Without output gap response:

$$\hat{\pi}_t = \frac{1}{1 - \beta\rho_u} u_t > \frac{\alpha_x}{\kappa^2 + \alpha_x(1 - \beta\rho_u)} u_t.$$

$$\hat{x}_t = -\kappa\psi u_t \Rightarrow \kappa\hat{x}_t = -\kappa^2\psi u_t < 0 \quad \text{when } u_t > 0,$$
$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} \boxed{-\kappa^2\psi u_t} + u_t.$$

u_t pushes inflation up — direct effect.
Induced recessionary output gap pushes inflation down.
Using

$$E_t\{\hat{\pi}_{t+1}\} = \alpha_x \psi \rho_u u_t,$$
$$\hat{\pi}_t = \left[\beta\alpha_x\psi\rho_u - \kappa^2\psi + 1\right] u_t = \alpha_x\psi u_t.$$

$$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \Rightarrow \hat{p}_t = \hat{p}_{t-1} + \sum_{j=0}^{t} \hat{\pi}_j.$$

Since $\hat{\pi}_j = \alpha_x \psi u_j$, and $u_j \to 0$ as $j \to \infty$, $\Rightarrow \hat{\pi}_j \to 0$ as $j \to \infty$.

$$\Rightarrow \lim_{t\to\infty} \hat{p}_t = \hat{p}_{-1} + \sum_{j=0}^{\infty} \hat{\pi}_j > \hat{p}_{-1}, \qquad \text{level price stay permanently higher.}$$

10.3 Optimal discretionary policy implementation

$$\hat{\pi}_t = \alpha_x \psi u_t, \qquad \hat{x}_t = -\kappa\psi u_t. \qquad \text{(Target rule)}$$

$$\hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

Plug in:

$$\begin{aligned}
-\kappa\psi u_t &= -\frac{1}{\sigma}\left(\hat{i}_t - \alpha_x \psi \rho_u u_t - \hat{r}_t^e\right) - \kappa\psi\rho_u u_t, \\
\Rightarrow \quad \hat{i}_t &= \underbrace{\hat{r}_t^e}_{\text{efficiency rate}} + \underbrace{\psi_i u_t}_{\text{response from cost-push shock}}, \qquad \psi_i = \psi\left[\kappa\sigma(1-\rho_u) + \alpha_x\rho_u\right].
\end{aligned}$$

$\kappa\sigma(1-\rho_u)$: need to affect current output gap through IS curve. $\alpha_x\rho_u$: forward-looking part.

$u_t \uparrow$ cost-push shock $\Rightarrow$ inflation $\hat{\pi}_t \uparrow \Rightarrow$ cool demand through $\hat{i}_t \uparrow > \hat{r}_t^e \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow$. But output losses are costly $\Rightarrow$ only fights $\hat{\pi}_t \uparrow$ partially.

10.3.1 DIS

$$\hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

Calculation:

$$C_t^{-\sigma} = \beta(1+i_t)E_t\left[\frac{C_{t+1}^{-\sigma}}{1+\pi_{t+1}}\right].$$

At steady state, $\beta(1+\bar{i}) = 1$ and $\bar{\pi} = 0$:

$$1 = \beta(1+\bar{r}), \qquad 1+\bar{r} = \frac{1}{\beta}, \qquad \rho = -\log\beta \qquad \rho: \text{ steady-state real rate.}$$

Log-linearization:

$$\hat{c}_t = E_t\{\hat{c}_{t+1}\} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right],$$

$$\hat{y}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right] + E_t\{\hat{y}_{t+1}\}.$$

$$\hat{x}_t + \hat{y}_t^e = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right] + E_t\{\hat{x}_{t+1}\} + E_t\{\hat{y}_{t+1}^e\},$$

$$\hat{x}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \sigma E_t\{\Delta\hat{y}_{t+1}^e\}\right] + E_t\{\hat{x}_{t+1}\},$$

$$\hat{x}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right] + E_t\{\hat{x}_{t+1}\} \qquad \text{where } \hat{r}_t^e = \sigma E_t\{\Delta\hat{y}_{t+1}^e\}.$$

Consider an inflation targeting rule:

$$\hat{i}_t = \hat{r}_t^e + \phi_\pi\hat{\pi}_t, \qquad \phi_\pi = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u.$$

$$\hat{i}_t = \hat{r}_t^e + \psi_i u_t, \qquad \hat{\pi}_t = \alpha_x\psi u_t \Rightarrow u_t = \frac{\hat{\pi}_t}{\alpha_x\psi}$$
$$\Rightarrow \hat{i}_t = \hat{r}_t^e + \frac{\psi_i}{\alpha_x\psi}\hat{\pi}_t.$$

$$\frac{\psi_i}{\alpha_x\psi} = \frac{\kappa\sigma(1-\rho_u) + \alpha_x\rho_u}{\alpha_x} = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u = \phi_\pi.$$

Since a determinate equilibrium exists iff $\phi_\pi > 1$:

$$\phi_\pi > 1 \Rightarrow \frac{\kappa\sigma}{\alpha_x} > 1 \quad \text{may not be satisfied.}$$

A rule that decentralizes the desired discretionary equilibrium:

$$\begin{aligned}\hat{i}_t &= \hat{r}_t^e + \psi_i u_t + \phi_\pi(\hat{\pi}_t - \alpha_x\psi u_t)\\ &= \hat{r}_t^e + \Theta_i u_t + \phi_\pi\hat{\pi}_t,\end{aligned} \qquad \text{(Instrument rule)}$$

where

$$\Theta_i = \psi\left[\kappa\sigma(1-\rho_u) - \alpha_x(\phi_\pi - \rho_u)\right].$$

Now we have a unique equilibrium, and it is the desired discretionary equilibrium.

1. Under desired equilibrium:

$$\hat{\pi}_t = \alpha_x \psi u_t, \qquad \hat{i}_t = \hat{r}_t^e + \psi_i u_t.$$

   We do not change the intended discretionary allocation.

2. Central bank reacts more aggressively:

$$\hat{i}_t \uparrow \quad \text{when} \quad \hat{\pi}_t > \alpha_x \psi u_t.$$

   If $\phi_\pi > 1$, nominal rate rises more than one-for-one with inflation deviations; real rate rises, demand falls, inflation falls, eliminating self-fulfilling inflation paths.

Target rule: tell central bank the relationship between target variables to achieve.

E.g. $\hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t$, This tells tradeoff, but still infeasible, since y_t^e is unobservable.

Instrument rule: tell central bank how to set its instrument.

E.g. $\hat{i}_t = \hat{r}_t^e + \Theta_i u_t + \phi_\pi \hat{\pi}_t$, $\hat{r}_t^e$ and u_t are hard to observe and measure.

10.4 Optimal policy under commitment

Central bank able to commit with full credibility to a policy plan.

$$\min_{\{\hat{x}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2\right)$$

$$\text{s.t.} \quad \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \qquad u_t = \rho_u u_{t-1} + \varepsilon_t^u.$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) + \gamma_t(\hat{\pi}_t - \kappa \hat{x}_t - \beta E_t\{\hat{\pi}_{t+1}\} - u_t)\right].$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1} = 0, \qquad \gamma_{-1} = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow \alpha_x \hat{x}_t - \kappa \gamma_t = 0 \Rightarrow \gamma_t = \frac{\alpha_x}{\kappa}\hat{x}_t, \qquad \forall t.$$

Constraint in period -1 not effective for optimal plan in period 0.

$$\Rightarrow \begin{cases} \hat{x}_0 = -\dfrac{\kappa}{\alpha_x}\hat{\pi}_0, \\ \hat{x}_t = \hat{x}_{t-1} - \dfrac{\kappa}{\alpha_x}\hat{\pi}_t, \qquad t \geq 1. \end{cases} \qquad (*) \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t.$$

$$\hat{x}_1 = \hat{x}_0 - \frac{\kappa}{\alpha_x}\hat{\pi}_1 = -\frac{\kappa}{\alpha_x}\left(\hat{\pi}_0 + \hat{\pi}_1\right),$$
$$\hat{x}_2 = \hat{x}_1 - \frac{\kappa}{\alpha_x}\hat{\pi}_2 = -\frac{\kappa}{\alpha_x}\left(\hat{\pi}_0 + \hat{\pi}_1 + \hat{\pi}_2\right).$$

$$\Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\sum_{s=0}^{t}\hat{\pi}_s.$$

$$\because \quad \hat{\pi}_s = \log P_s - \log P_{s-1}, \qquad \sum_{s=0}^{t}\hat{\pi}_s = \sum_{s=0}^{t}(\log P_s - \log P_{s-1}) = \log P_t - \log P_{-1}.$$

$$\Rightarrow \quad \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t. \qquad \check{p}_t = \log P_t - \log P_{-1}, \qquad \check{p}_t: \text{ price-level targeting.}$$

While under discretion, we have purely static: $\hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t$.
Now today's policy depends on past policy through $\hat{x}_{t-1}$: history dependent policy making.
Then

$$\begin{aligned}
\hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\hat{x}_t + u_t,\\
\check{p}_t - \check{p}_{t-1} &= \beta E_t\{\check{p}_{t+1} - \check{p}_t\} - \frac{\kappa^2}{\alpha_x}\check{p}_t + u_t,\\
\left(1+\beta+\frac{\kappa^2}{\alpha_x}\right)\check{p}_t &= \beta E_t\{\check{p}_{t+1}\} + \check{p}_{t-1} + u_t.
\end{aligned}$$

Let

$$a = \frac{\alpha_x}{(1+\beta)\alpha_x + \kappa^2}.$$

Then

$$\check{p}_t = a\check{p}_{t-1} + a\beta E_t\{\check{p}_{t+1}\} + au_t. \qquad \forall t$$

Assume

$$\begin{aligned}
\check{p}_t &= \delta\check{p}_{t-1} + \eta u_t, \qquad \text{undetermined coefficients.}\\
\check{p}_t &= \frac{a}{\delta}(\check{p}_t - \eta u_t) + a\beta\delta\check{p}_t + a\beta\eta\rho_u u_t + au_t,\\
\left(1-\frac{a}{\delta}-a\beta\delta\right)\check{p}_t &+ \left(\frac{a\eta}{\delta} - a\beta\eta\rho_u - a\right)u_t = 0.
\end{aligned}$$

$$\begin{cases} 1-\frac{a}{\delta}-a\beta\delta = 0,\\ \frac{a}{\delta}\eta - a\beta\eta\rho_u - a = 0, \end{cases} \Rightarrow \begin{cases} \delta = \frac{1-\sqrt{1-4\beta a^2}}{2\beta a} \in (0,1),\\ \eta = \frac{\delta}{1-\delta\beta\rho_u}. \end{cases}$$

Thus

$$\check{p}_t = \delta\check{p}_{t-1} + \frac{\delta}{1-\delta\beta\rho_u}u_t,$$

where

$$\delta = \frac{1-\sqrt{1-4\beta a^2}}{2\beta a} \in (0,1).$$

$$\hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t \Rightarrow \hat{x}_t = \delta\hat{x}_{t-1} - \frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}u_t. \tag{**}$$

$$\hat{x}_0 = -\frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}u_0.$$

$$\hat{\pi}_t = \kappa\hat{x}_t + \kappa\sum_{k=1}^{\infty}\beta^{k-1}E_t\{\hat{x}_{t+k}\} + u_t.$$

In response to u_t, central bank may lower future output gap with credible promises. Given $\hat{x}_t$, current $\hat{\pi}_t$ may decline less. Since cost function is convex, damping of deviations in period of shock improves welfare.

Assume transitory shock ($\rho_u = 0$):

$$\begin{aligned}
DIS: \hat{x}_t &= -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\} \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} + \sigma\left[E_t\{\hat{x}_{t+1}\} - \hat{x}_t\right], \\
\hat{x}_t &= -\frac{\kappa}{\alpha_x}\check{p}_t, \\
E_t\{\hat{x}_{t+1}\} - \hat{x}_t &= -\frac{\kappa}{\alpha_x}\left(E_t\{\check{p}_{t+1}\} - \check{p}_t\right), \\
E_t\{\hat{\pi}_{t+1}\} &= E_t\{\check{p}_{t+1} - \check{p}_t\}, \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} - \sigma\frac{\kappa}{\alpha_x}E_t\{\hat{\pi}_{t+1}\}, \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + \left(1 - \frac{\sigma\kappa}{\alpha_x}\right)E_t\{\hat{\pi}_{t+1}\}.
\end{aligned} \tag{1}$$

Since $\check{p}_t = \delta\check{p}_{t-1} + \delta u_t$ when $\rho_u = 0$,

$$E_t\{\check{p}_{t+1}\} = \delta\check{p}_t, \qquad E_t\{\hat{\pi}_{t+1}\} = \delta\check{p}_t - \check{p}_t = -(1-\delta)\check{p}_t.$$

Thus

$$\hat{i}_t = \hat{r}_t^e - (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\check{p}_t$$

Iterate backward:

$$\check{p}_t = \delta u_t + \delta^2 u_{t-1} + \cdots = \sum_{k=0}^{t}\delta^{k+1}u_{t-k}.$$

$$\Rightarrow \quad \hat{i}_t = \hat{r}_t^e - (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\sum_{k=0}^{t}\delta^{k+1}u_{t-k}.$$

Equilibrium nominal rate consistent with optimal allocation.
To make an implementable rule that makes the allocation unique:

$$\hat{i}_t = \hat{r}_t^e - \left[\phi_p + (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\right]\sum_{k=0}^{t}\delta^{k+1}u_{t-k} + \phi_p\check{p}_t, \qquad \text{ensure determinacy, } \phi_p > 0.$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e - (\phi_p + A)\check{p}_t + \phi_p\check{p}_t, \qquad A = (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right).$$

Along the intended equilibrium, the rule collapses back to

$$\hat{i}_t^* = \hat{r}_t^{e^*} - (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\check{p}_t^*.$$

Without a term with $\check{p}_t$, the rule does not actively push back against the deviation.
Define deviation from the target equilibrium:

$$\tilde{p}_t = \check{p}_t - \check{p}_t^*, \qquad \tilde{x}_t = \hat{x}_t - \hat{x}_t^*, \qquad \tilde{\pi}_t = \hat{\pi}_t - \hat{\pi}_t^*, \qquad \tilde{i}_t = \hat{i}_t - \hat{i}_t^*.$$

$$\begin{cases} \text{DIS:} & \tilde{x}_t = E_t\tilde{x}_{t+1} - \frac{1}{\sigma}\left(\phi_p\tilde{p}_t - E_t\tilde{\pi}_{t+1}\right), \\ \text{NKPC:} & \tilde{\pi}_t = \kappa\tilde{x}_t + \beta E_t\tilde{\pi}_{t+1}, \\ \text{Price identity:} & \tilde{\pi}_t = \tilde{p}_t - \tilde{p}_{t-1}, \\ \text{Guess:} & \tilde{p}_t = \lambda\tilde{p}_{t-1}. \end{cases} \Longrightarrow \begin{cases} \tilde{x}_t = \frac{1+\phi_p-\lambda}{\sigma(\lambda-1)}\tilde{p}_t, \\ \tilde{x}_t = \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]\tilde{p}_t, \\ \tilde{\pi}_t = \left(1 - \frac{1}{\lambda}\right)\tilde{p}_t, E_t\tilde{\pi}_{t+1} = (\lambda-1)\tilde{p}_t \\ E_t\tilde{x}_{t+1} = \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]E_t\tilde{p}_{t+1} \\ \qquad = \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]\lambda\tilde{p}_t = \lambda\tilde{x}_t. \end{cases}$$

Equating the two expressions for $\tilde{x}_t$:

$$\frac{1+\phi_p-\lambda}{\sigma(\lambda-1)} = \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right].$$

$$H(\lambda) \equiv \sigma(\lambda-1)^2(1-\beta\lambda) - \kappa\lambda(1+\phi_p-\lambda) = 0.$$

When $\phi_p = 0$:
$H(1) = 0 \to \tilde{p}_t = \tilde{p}_{t-1} \to \tilde{p}_t$ is constant and can hold at any level deviation.
When $\phi_p > 0$:
$H(1) = \sigma(1-1)^2(1-\beta) - \kappa(1+\phi_p-1) = -\kappa\phi_p < 0 \quad \to \lambda = 1$ is not a root.
Proof that there are 1 stable root and 2 unstable roots which is what we want:

(i) For $\lambda \in (0,1)$,

$$H(\lambda) = 0 \Longleftrightarrow \frac{\kappa}{\sigma}\frac{\lambda(1+\phi_p-\lambda)}{(1-\lambda)^2(1-\beta\lambda)} = 1.$$

Then

$$LHS(0) = 0, \qquad LHS(1) = +\infty \quad \Longrightarrow \quad \exists\, \lambda_1 \in (0,1).$$

(ii) For $\lambda \in (-\infty, 0]$,

$$(\lambda-1)^2 > 0, \qquad 1-\beta\lambda > 0, \qquad -\kappa\lambda(1+\phi_p-\lambda) \geq 0.$$

$$H(\lambda) > 0 \quad \Longrightarrow \quad \text{there is no negative real root.}$$

Therefore, there are either two positive real roots or conjugate complex roots.

$$\lambda_1\lambda_2\lambda_3 = \frac{1}{\beta} > 1 \qquad \lambda_1 \in (0,1) \Rightarrow \lambda_2\lambda_3 = \frac{1}{\beta\lambda_1} > 1.$$

Thus the remaining roots are unstable:

$$\begin{cases} |\lambda_2| = |\lambda_3| > 1, & \text{if they are complex conjugate roots,} \\ \lambda_2 > 1,\ \lambda_3 > 1, & \text{if they are unstable real roots.} \end{cases}$$

Economic intuition:

(i) $\phi_p > 0$:

$$\check{p}_t \uparrow \Longrightarrow \hat{i}_t \uparrow \Longrightarrow \text{reduced demand} \Longrightarrow \text{negative output gap} \xrightarrow{\text{NKPC}} \hat{\pi}_t \downarrow.$$

Hence the belief that prices will drift up is no longer self-fulfilling; monetary policy offsets the deviation.

(ii) $\phi_p = 0$:

The central bank does not react to the endogenous price-level drift,beliefs may survive as alternative equilibria.

10.5 Distorted steady state

Permanent supply-side distortion:

$$\bar{y}^n < \bar{y}^e.$$

Real imperfections generate a permanent gap between natural and efficient levels of output:

$$-\frac{U_N}{U_C} = MPN(1-\phi), \qquad \phi = 1 - \frac{1}{\mu}.$$

Output is inefficiently low.

The welfare losses:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t \right], \qquad \text{where } \Lambda = \phi \frac{\lambda}{\varepsilon} > 0.$$

Deviation of the welfare-relevant output gap:

$$\hat{x}_t = \log X_t - \log \bar{X} = (\log Y_t - \log Y_t^e) - (\log \bar{Y} - \log \bar{Y}^e) = (\log Y_t - \log Y^e) - (\log \bar{Y}^n - \log \bar{Y}^e).$$

$$\log \bar{X} = \log \bar{Y} - \log \bar{Y}^e = \log \bar{Y}^n - \log \bar{Y}^e < 0.$$

NKPC constraint:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \qquad \text{where } u_t = \kappa(\hat{y}_t^e - \hat{y}_t^n) \text{ is cost-push shock.}$$

10.5.1 Optimal discretionary policy

$$\min_{\hat{x}_t, \hat{\pi}_t} \frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t \qquad \text{s.t.} \quad \hat{\pi}_t = \kappa \hat{x}_t + v_t, \quad v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

F.O.C.:

$$\hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\hat{\pi}_t.$$

$\frac{\Lambda}{\alpha_x}$: central bank always wants a more expansionary policy.
Plug into the constraint.

$$\hat{\pi}_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} E_t\{\hat{\pi}_{t+1}\} + \frac{\kappa \Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t.$$

Guess $\hat{\pi}_t = A + B u_t$:

$$\Rightarrow A + B u_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} \left(A + B \rho_u u_t \right) + \frac{\kappa \Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t.$$

$$A = \frac{\kappa \Lambda}{\kappa^2 + \alpha_x (1-\beta)}, \qquad B = \alpha_x \psi, \qquad \psi = \frac{1}{\kappa^2 + \alpha_x(1 - \beta \rho_u)}.$$

Thus

$$\Rightarrow \hat{\pi}_t^* = \frac{\Lambda \kappa}{\kappa^2 + \alpha_x (1-\beta)} + \alpha_x \psi u_t.$$

$\frac{\Lambda \kappa}{\kappa^2 + \alpha_x(1-\beta)}$: constant > 0; inflation bias from distorted S.S.

$\alpha_x \psi u_t$: usual response.
Changed average level of inflation / output gap; shock response no change.

Inefficient low natural level of output $\Rightarrow$
(i)positive average inflation, because CB's incentive to push output above its natural S.S. level.
(ii)increases the incentive to keep inflation higher than 0 as Λ increases (inflation bias).

10.5.2 Optimal policy under commitment

$$\mathcal{L} = E_t \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t + \gamma_t(\hat{\pi}_t - \kappa \hat{x}_t - \beta E_t\{\hat{\pi}_{t+1}\} - u_t) \right].$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1} = 0, \tag{1}$$
$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow \alpha_x \hat{x}_t - \kappa \gamma_t - \Lambda = 0, \tag{2}$$

$$\gamma_{-1} = 0$$

$$(1) \Rightarrow \gamma_t = \frac{\alpha_x \hat{x}_t - \Lambda}{\kappa}, \qquad \gamma_{t-1} = \frac{\alpha_x \hat{x}_{t-1} - \Lambda}{\kappa}, \quad \text{plug into (2)} \Rightarrow \hat{x}_t = \hat{x}_{t-1} - \frac{\kappa}{\alpha_x}\hat{\pi}_t.$$

Since $\hat{\pi}_t = \check{p}_t - \check{p}_{t-1}$, by iterating forward:

$$\left.\begin{aligned} \hat{x}_t &= -\frac{\kappa}{\alpha_x}\check{p}_t + \text{constant} \\ (1) \Rightarrow \hat{x}_t &= \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\check{p}_t \end{aligned}\right\} \Rightarrow \text{constant} = \frac{\Lambda}{\alpha_x}$$

Plug into NKPC:
$$\left.\begin{aligned} \hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \\ \hat{\pi}_t &= \check{p}_t - \check{p}_{t-1}, \\ \hat{\pi}_{t+1} &= \check{p}_{t+1} - \check{p}_t. \end{aligned}\right\} \Rightarrow \check{p}_t = a\check{p}_{t-1} + a\beta E_t\{\check{p}_{t+1}\} + \frac{a\kappa\Lambda}{\alpha_x} + a u_t.$$

$$\text{where } a = \frac{\alpha_x}{\kappa^2 + \alpha_x(1+\beta)}.$$

Solve with undetermined coefficients:

$$\check{p}_t = A\check{p}_{t-1} + Bu_t + C.$$

$$A = \delta = \frac{1 - \sqrt{1 - 4\beta a^2}}{2\beta a} \in (0,1), \qquad B = \frac{\delta}{1 - \delta\beta\rho_u}, \qquad C = \frac{\delta\kappa\Lambda}{(1-\delta\beta)\alpha_x}.$$

Thus
$$\check{p}_t = \delta\check{p}_{t-1} + \frac{\delta}{1 - \delta\beta\rho_u}u_t + \frac{\delta\kappa\Lambda}{(1-\delta\beta)\alpha_x}. \tag{3}$$

$\frac{\delta}{1-\delta\beta\rho_u}u_t$: shock response, not affected.
$\frac{\delta\kappa\Lambda}{1-\delta\beta}$: stabilization bias; shifts price-level path upward by a constant

$\Rightarrow$ price level converges to a constant and average inflation is 0.

$$\lim_{T\to\infty} P_T = P_{-1} + \frac{\delta\kappa\Lambda}{(1-\delta\beta)(1-\delta)\alpha_x}.$$

Zero average inflation, but positive upward bias above P_{-1}.

$$\hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\check{p}_t.$$

Plug in (3):

$$\hat{x}_t = \underbrace{\delta\hat{x}_{t-1}}_{\text{history dependence}} \underbrace{- \frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}u_t}_{\text{same shock response}} + \underbrace{\frac{\Lambda}{\alpha_x}\left[1-\delta\left(1+\frac{\kappa^2}{(1-\delta\beta)\alpha_x}\right)\right]}_{\text{constant S.S. deviation}}.$$

Commitment leads to:

1. output can be raised above its natural level with welfare improvement;
2. more subdued effects on inflation;
3. commitment avoids the inflation bias under discretionary policy.